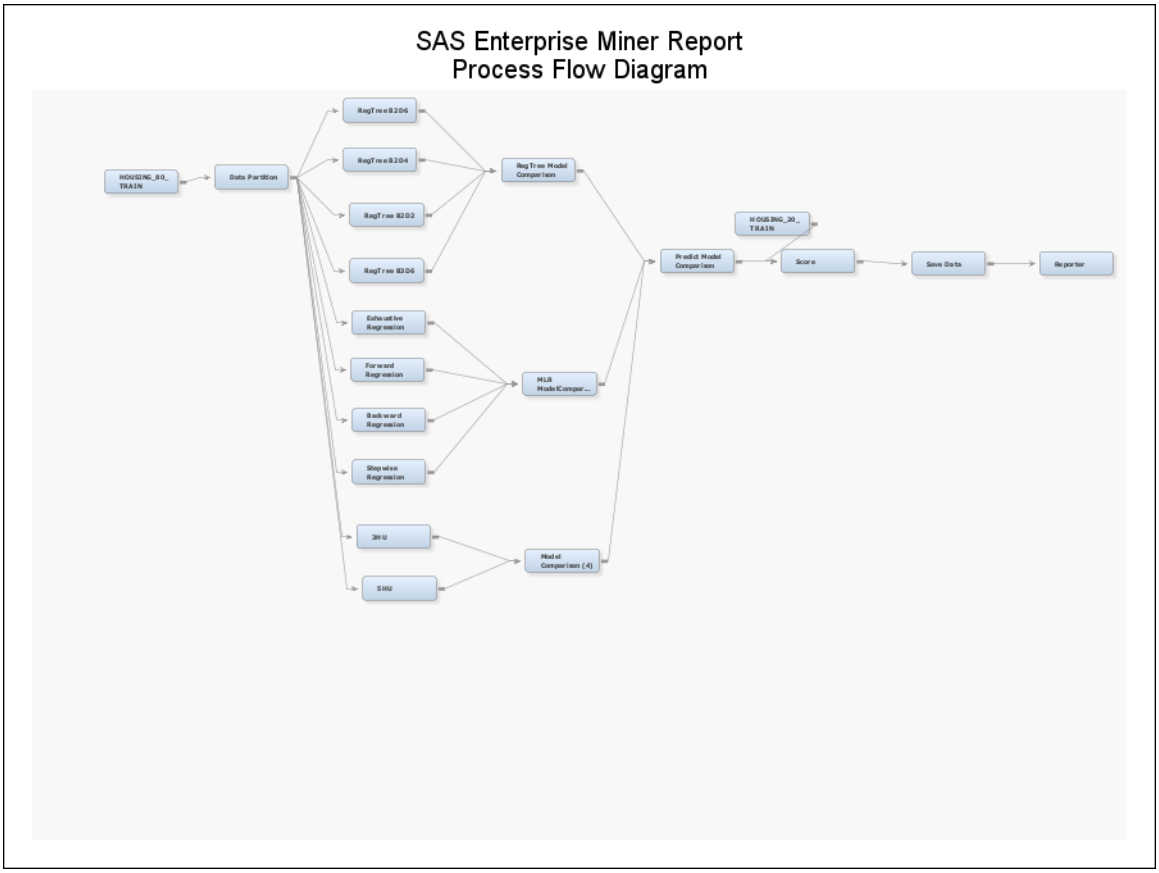
SAS Enterprise Miner Report

User = smupp
Date = 23:09:38 December 11
Project = HU.PREDICTIVEANALYSIS
Diagram = HU.Prediction

Start Node = Report
Node label = Reporter
Nodes = PATH
Showall = N

Format = PDF
Style = LISTING



SAS Enterprise Miner Report

Node=HOUSING_80_TRAIN
Summary

Node id = Ids2
Node label = HOUSING_80_TRAIN
Meta path = Ids2
Notes =

Node=HOUSING_80_TRAIN
Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	DataSource		DsCreatedBy	smupp		NBytes	1508352	.
ApplyIntervalLevelLowerLimit	Y		DsId	housingtrain1		NCols	11	.
ApplyMaxClassLevels	Y		DsModifiedBy	smupp		NObs	16512	.
ApplyMaxPercentMissing	Y		DsModifyDate	2049495415.9		NewTable		
CMeta	WORK.M3IA26CI		DsSampleName			NewVariableRole	REJECT	
ComputeStatistics	N		DsSampleSize			OutputType	VIEW	
DBPassThrough	Y		DsSampleSizeType			Role	RAW	TRAIN
Data	HUDATA.HOUSING_80_TRAIN		DsScope	LOCAL		Sample	D	
DataSelection	DATASOURCE		IdentifyEmptyColumns	Y		SampleSizeObs	10000	
DataSource	housingtrain1		IntervalLowerLimit	20		SampleSizePercent	20	
DataSourceRole	RAW		Library	HUDATA		SampleSizeType	PERCENT	
Description			MaxClassLevels	20		Scope	LOCAL	
DropMapVariables	Y		MaxPercentMissing	50		Segment		
DsCreateDate	2049495415.8		MetaAdvisor	BASIC		Table	HOUSING_80_TRAIN	

Node=HOUSING_80_TRAIN
Data Attributes

Attribute	Value	Attribute	Value	Attribute	Value
Data Name	HOUSING_80_TRAIN	Date Created	11Dec2024:00:14:55	Data Size	1508352
Data Type	DATA	Date Modified	11Dec2024:00:14:55	Role	RAW
Data Label		Number Rows	16512	Segment	
Engine	BASE	Number Columns	11	Data Library	HUDATA

Node=HOUSING_80_TRAIN
Variables List

Name	Label	Role	Level	Type	Length	Format	Creator
dataobs	Observation Number	INPUT	INTERVAL	N	8		
households		INPUT	INTERVAL	N	8	BEST12.0	
housing_median_age		INPUT	INTERVAL	N	8	BEST12.0	
latitude		INPUT	INTERVAL	N	8	BEST12.0	
longitude		INPUT	INTERVAL	N	8	BEST12.0	
median_house_value		TARGET	INTERVAL	N	8	BEST12.0	
median_income		INPUT	INTERVAL	N	8	BEST12.0	
ocean_proximity		INPUT	NOMINAL	C	8	$8.	
population		INPUT	INTERVAL	N	8	BEST12.0	
total_bedrooms		INPUT	INTERVAL	N	8	BEST12.0	
total_rooms		INPUT	INTERVAL	N	8	BEST12.0	

SAS Enterprise Miner Report

Node=Data Partition
Summary

Node id = Part
Node label = Data Partition
Meta path = Ids2 => Part
Notes =

Node=Data Partition
Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	Partition		Method	DEFAULT		TestPct	30	
ClassDistribution	Y		OutputType	DATA		TrainPct	40	
IntervalDistribution	Y		RandomSeed	12345		ValidatePct	30	

Node=Data Partition
Variable Summary

Role	Level	Frequency Count	Name
TARGET	INTERVAL	1	median_house_value
INPUT	INTERVAL	9	_dataobs_ households housing_median_age latitude longitude median_income population total_bedrooms total_rooms
INPUT	NOMINAL	1	ocean_proximity

SAS Enterprise Miner Report

Node=5HU
Summary

Node id = Neural2
Node label = 5HU
Meta path = Ids2 => Part => Neural2
Notes =

Node=5HU
Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	NeuralNetwork		Hidden	5	3	Prelim	Y	
AbsConvValue	-1.34078E154	-7.237006E75	HiddenActivation	DEFAULT		PrelimMaxTime	1 HOUR	
AbsFTime	1		HiddenBias	Y		PrelimMaxiter	10	
AbsFValue	0		HiddenCombFunction	DEFAULT		PrelimOutest		
AbsGTime	1		HiddenUnits	N		PreliminaryRuns	5	
AbsGValue	0.00001		InitialDs			RandDist	NORMAL	
AbsXTime	1		InitialSeed	12345		RandLoc	0	
AbsXValue	1E-8		InputStandardization	STD		RandScale	0.1	
Accelerate	1.2		Learn	0.1		Residuals	Y	
AddHidden	Y		MaxLearn	50		Standardizations	N	
CodefileNoRes			MaxMomentum	1.75		SuppressOutput	N	
CodefileRes			Maxiter	50		TargetActivation	DEFAULT	
ConvDefaults	Y		Maxtime	4 HOURS		TargetBias	Y	
Decelerate	0.5		MinLearn	0.00001		TargetCombFunction	DEFAULT	
DirectConnection	N		ModelSelectionCriterion	PROFIT/LOSS		TargetError	DEFAULT	
FConvTime	1		Momentum	0		Tilt	0	
FConvValue	0		NetworkArchitecture	MLP		TrainCode		
GConvTime	1		Outest			TrainingTechnique	DEFAULT	
GConvValue	1E-6		Outfit			UseEstimates	N	

Node=5HU
Variable Summary

Role	Level	Frequency Count	Name
TARGET	INTERVAL	1	median_house_value
INPUT	INTERVAL	8	households housing_median_age latitude longitude median_income population total_bedrooms total_rooms
INPUT	NOMINAL	1	ocean_proximity

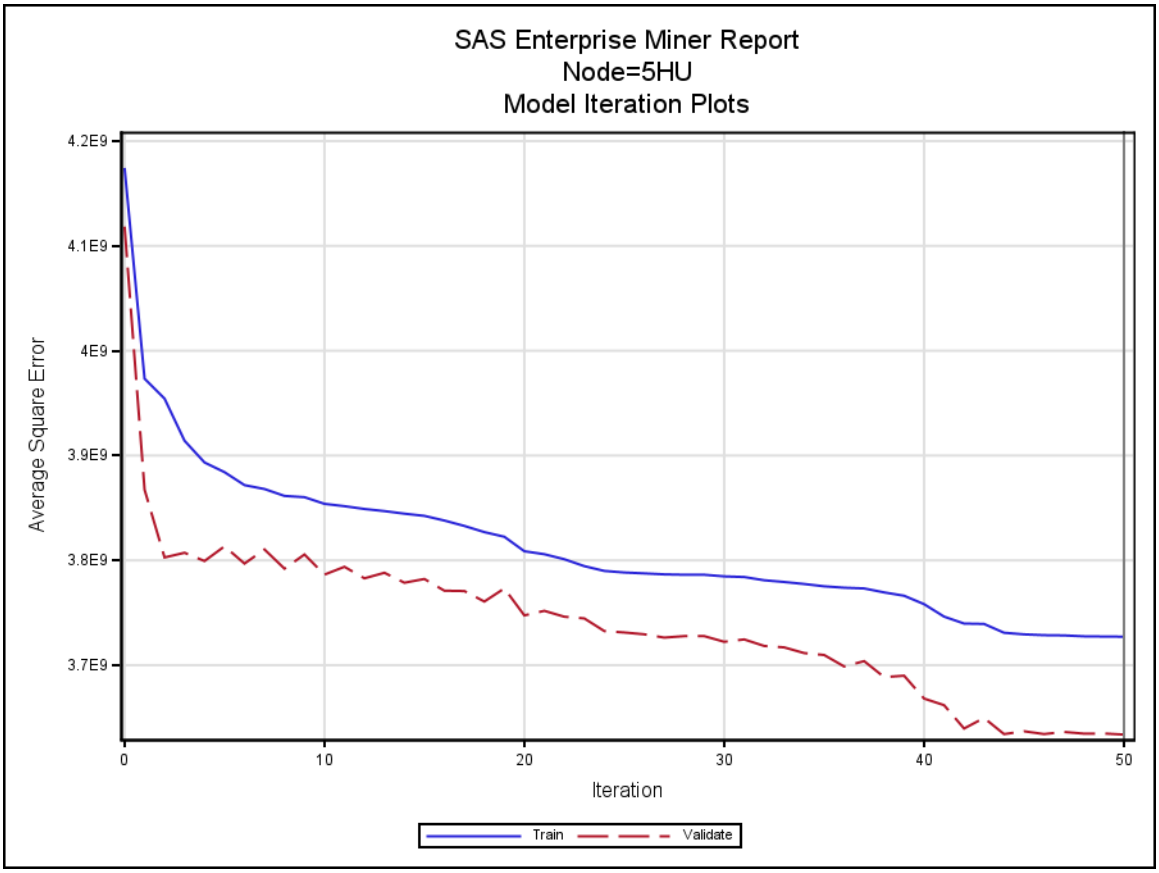
Node=5HU
Model Fit Statistics

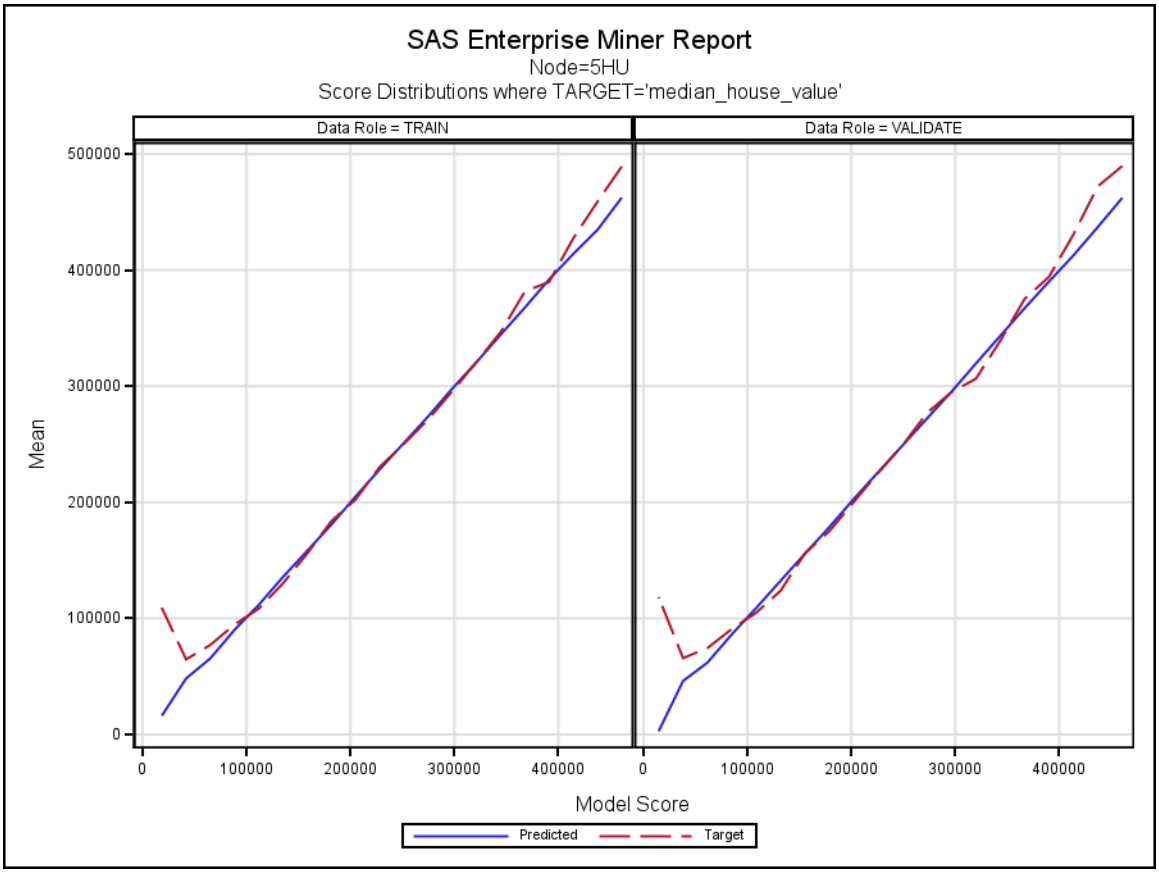
Target=median_house_value Target Label=' '

Label of Statistic	Train	Validation	Test
Total Degrees of Freedom	6605.00	.	.
Degrees of Freedom for Error	6534.00	.	.
Model Degrees of Freedom	71.00	.	.
Number of Estimated Weights	71.00	.	.
Akaike's Information Criterion	145709.03	.	.

Target=median_house_value Target Label=' '

Label of Statistic	Train	Validation	Test
Schwarz's Bayesian Criterion	146191.52	.	.
Average Squared Error	3727167485.11	3633790314.39	3532613610.65
Maximum Absolute Error	383215.05	390075.99	423145.96
Divisor for ASE	6605.00	4954.00	4953.00
Sum of Frequencies	6605.00	4954.00	4953.00
Root Average Squared Error	61050.53	60280.93	59435.79
Sum of Squared Errors	2.46179412E13	1.80017972E13	1.74970352E13
Sum of Case Weights Times Freq	6605.00	4954.00	4953.00
Final Prediction Error	3808168064.06	.	.
Mean Squared Error	3767667774.58	3633790314.39	3532613610.65
Root Final Prediction Error	61710.36	.	.
Root Mean Squared Error	61381.33	60280.93	59435.79
Average Error Function	3727167485.11	3633790314.39	3532613610.65
Error Function	2.46179412E13	1.80017972E13	1.74970352E13





Node=5HU
Score Distributions

Target Variable=median_house_value Data Role=TRAIN

Range for Predicted	Mean Predicted	Max Predicted	Min Predicted	Mean Target	Max Target	Min Target
449402.777 - 472692.706	462360.75	472692.71	449460.79	489194.65	500001	350000
426112.849 - 449402.777	434678.55	447995.93	426129.03	458963.71	500001	225000
402822.920 - 426112.849	413956.94	425817.33	403172.74	427458.19	500001	112500
379532.991 - 402822.920	391961.36	402806.38	379558.94	389786.22	500001	67500
356243.063 - 379532.991	367741.10	379307.42	356324.70	381539.82	500001	196400
332953.134 - 356243.063	344454.66	356087.00	332960.54	346687.93	500001	146300
309663.206 - 332953.134	320694.14	332807.96	309673.14	320673.92	500001	140600
286373.277 - 309663.206	297947.18	309537.49	286527.07	295710.17	500001	134400
263083.349 - 286373.277	273904.25	286286.16	263159.93	271302.05	500001	99100
239793.420 - 263083.349	251110.33	263035.91	239798.64	250276.19	500001	55000
216503.492 - 239793.420	228314.23	239792.84	216506.99	230459.61	500001	47500
193213.563 - 216503.492	204913.53	216426.68	193219.08	202337.84	500001	45800
169923.634 - 193213.563	180658.90	193176.54	169946.60	183683.54	500001	26900
146633.706 - 169923.634	158036.35	169876.37	146683.93	155709.94	500001	22500
123343.777 - 146633.706	135575.94	146600.47	123544.29	130067.73	500001	41300
100053.849 - 123343.777	111578.71	123295.28	100099.11	108301.51	500001	14999
76763.920 - 100053.849	90054.72	99966.65	76826.57	94589.23	450000	32500
53473.992 - 76763.920	65490.91	76667.63	53478.68	77045.51	275000	32900
30184.063 - 53473.992	48205.57	53227.25	30724.42	64675.95	143300	34200
6894.135 - 30184.063	16197.78	25501.42	6894.13	108950.00	170000	47900

Target Variable=median_house_value Data Role=VALIDATE

Range for Predicted	Mean Predicted	Max Predicted	Min Predicted	Mean Target	Max Target	Min Target
448865.993 - 472341.351	462101.77	472341.35	448947.49	489559.57	500001	131300
425390.636 - 448865.993	437066.09	448783.59	425550.32	472096.94	500001	331800
401915.278 - 425390.636	412615.14	425335.46	402071.45	430621.23	500001	225000
378439.920 - 401915.278	390096.08	401892.30	378538.45	394200.20	500001	206800
354964.562 - 378439.920	367143.53	377649.05	355027.13	375105.93	500001	112500
331489.205 - 354964.562	343285.99	354821.67	331596.33	339015.30	500001	158000
308013.847 - 331489.205	319433.54	331431.52	308127.64	306334.71	500001	112500
284538.489 - 308013.847	295153.21	307799.87	284587.44	294833.72	500001	105400
261063.132 - 284538.489	272187.70	284436.48	261077.30	276960.62	500001	51300
237587.774 - 261063.132	249034.32	261047.37	237618.15	249097.98	500001	90800
214112.416 - 237587.774	225969.57	237484.89	214137.36	225389.07	500001	73800
190637.058 - 214112.416	202898.68	214103.62	190647.20	200350.99	500001	45000
167161.701 - 190637.058	178688.63	190605.81	167192.65	175134.85	500000	17500
143686.343 - 167161.701	155731.70	167153.22	143795.80	156175.54	500001	70000
120210.985 - 143686.343	132479.25	143632.34	120281.26	123819.91	500001	37500
96735.628 - 120210.985	109006.96	120134.69	96738.35	104906.65	325000	47500
73260.270 - 96735.628	86263.77	96683.80	73298.16	90903.53	225000	32500
49784.912 - 73260.270	61906.77	73201.16	50248.48	74419.24	450000	14999
26309.554 - 49784.912	46041.02	49736.42	35851.91	65737.84	275000	27500
2834.197 - 26309.554	2834.20	2834.20	2834.20	118300.00	118300	118300

SAS Enterprise Miner Report

Node=3HU
Summary

Node id = Neural
Node label = 3HU
Meta path = lds2 => Part => Neural
Notes =

Node=3HU
Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	NeuralNetwork		Hidden	3		Prelim	Y	
AbsConvValue	-1.34078E154	-7.237006E75	HiddenActivation	DEFAULT		PrelimMaxTime	1 HOUR	
AbsFTime	1		HiddenBias	Y		PrelimMaxiter	10	
AbsFValue	0		HiddenCombFunction	DEFAULT		PrelimOutest		
AbsGTime	1		HiddenUnits	N		PreliminaryRuns	5	
AbsGValue	0.00001		InitialDs			RandDist	NORMAL	
AbsXTime	1		InitialSeed	12345		RandLoc	0	
AbsXValue	1E-8		InputStandardization	STD		RandScale	0.1	
Accelerate	1.2		Learn	0.1		Residuals	Y	
AddHidden	Y		MaxLearn	50		Standardizations	N	
CodefileNoRes			MaxMomentum	1.75		SuppressOutput	N	
CodefileRes			Maxiter	50		TargetActivation	DEFAULT	
ConvDefaults	Y		Maxtime	4 HOURS		TargetBias	Y	
Decelerate	0.5		MinLearn	0.00001		TargetCombFunction	DEFAULT	
DirectConnection	N		ModelSelectionCriterion	PROFIT/LOSS		TargetError	DEFAULT	
FConvTime	1		Momentum	0		Tilt	0	
FConvValue	0		NetworkArchitecture	MLP		TrainCode		
GConvTime	1		Outest			TrainingTechnique	DEFAULT	
GConvValue	1E-6		Outfit			UseEstimates	N	

Node=3HU
Variable Summary

Role	Level	Frequency Count	Name
TARGET	INTERVAL	1	median_house_value
INPUT	INTERVAL	8	households housing_median_age latitude longitude median_income population total_bedrooms total_rooms
INPUT	NOMINAL	1	ocean_proximity

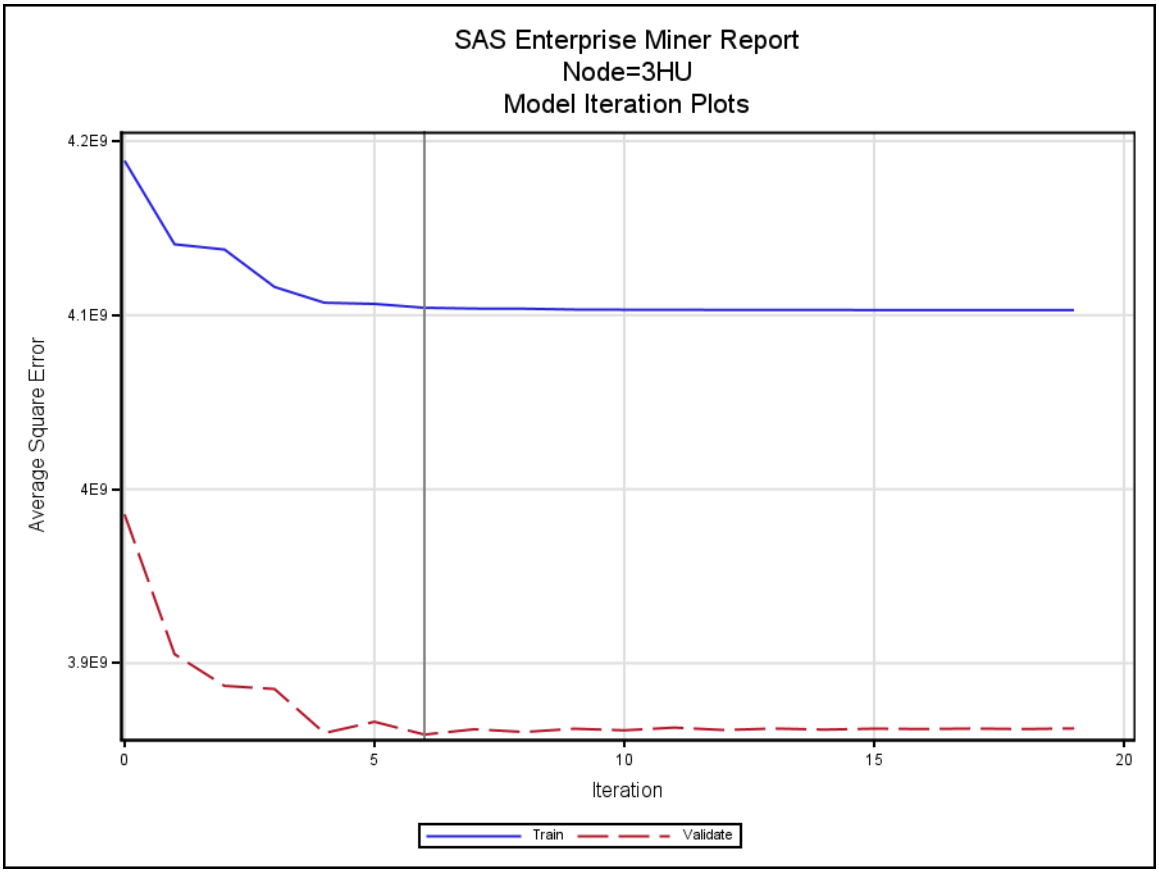
Node=3HU
Model Fit Statistics

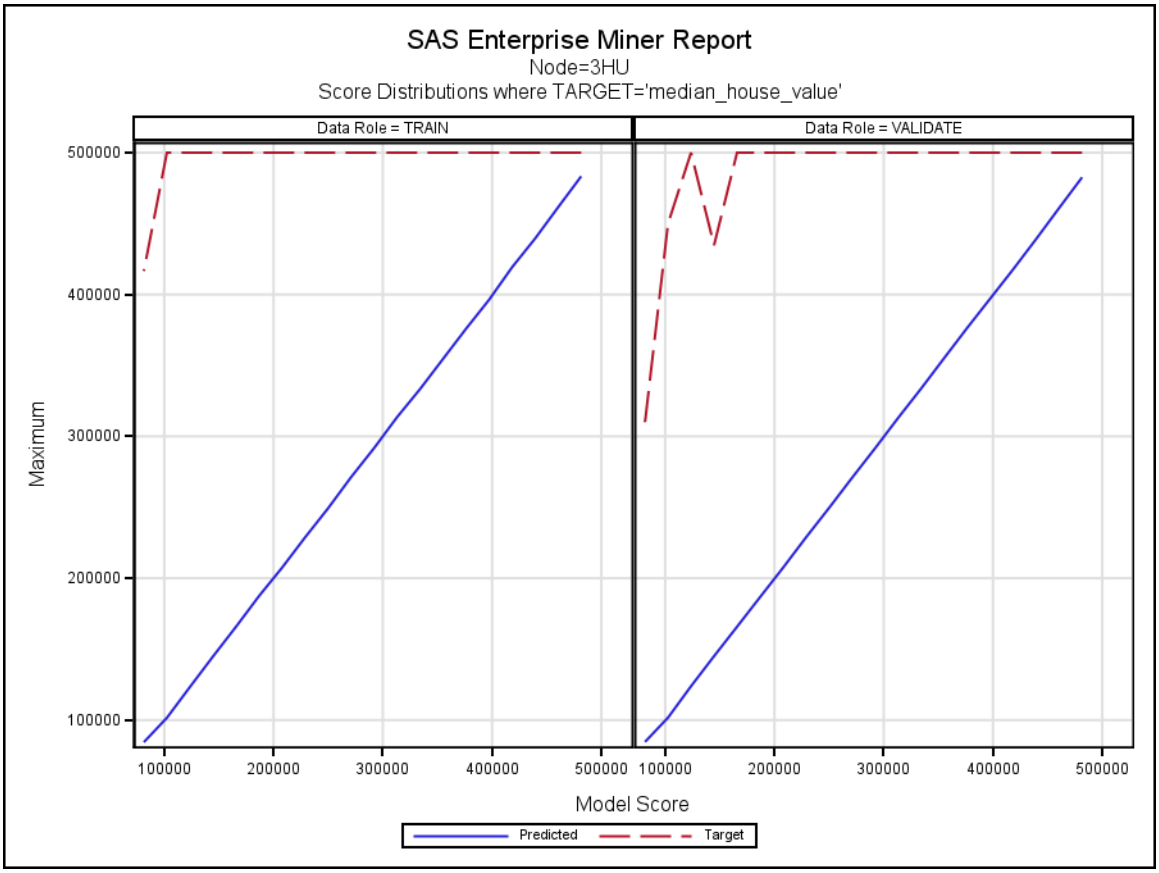
Target=median_house_value Target Label=' '

Label of Statistic	Train	Validation	Test
Total Degrees of Freedom	6605.00	.	.
Degrees of Freedom for Error	6562.00	.	.
Model Degrees of Freedom	43.00	.	.
Number of Estimated Weights	43.00	.	.
Akaike's Information Criterion	146289.54	.	.

Target=median_house_value Target Label=' '

Label of Statistic	Train	Validation	Test
Schwarz's Bayesian Criterion	146581.75	.	.
Average Squared Error	4104222196.14	3858989033.00	3813450391.11
Maximum Absolute Error	407701.72	382345.65	380226.25
Divisor for ASE	6605.00	4954.00	4953.00
Sum of Frequencies	6605.00	4954.00	4953.00
Root Average Squared Error	64064.20	62120.76	61753.14
Sum of Squared Errors	2.71083876E13	1.91174317E13	1.88880198E13
Sum of Case Weights Times Freq	6605.00	4954.00	4953.00
Final Prediction Error	4158011149.03	.	.
Mean Squared Error	4131116672.59	3858989033.00	3813450391.11
Root Final Prediction Error	64482.64	.	.
Root Mean Squared Error	64273.76	62120.76	61753.14
Average Error Function	4104222196.14	3858989033.00	3813450391.11
Error Function	2.71083876E13	1.91174317E13	1.88880198E13





Node=3HU
Score Distributions

Target Variable=median_house_value Data Role=TRAIN

Range for Predicted	Mean Predicted	Max Predicted	Min Predicted	Mean Target	Max Target	Min Target
471018.596 - 492082.407	483295.29	492082.41	471479.47	480867.99	500001	112500
449954.785 - 471018.596	461530.54	470900.95	450287.55	482590.31	500001	259600
428890.974 - 449954.785	439479.93	449679.35	429365.47	462784.74	500001	275000
407827.162 - 428890.974	419257.20	428720.49	408156.17	421208.09	500001	67500
386763.351 - 407827.162	396319.27	407722.60	386774.24	392297.89	500001	194500
365699.540 - 386763.351	375900.53	386592.56	365712.70	373326.51	500001	184900
344635.729 - 365699.540	354638.88	365685.92	344643.70	360161.03	500001	191000
323571.918 - 344635.729	333330.90	344539.75	323742.75	335310.41	500001	61300
302508.107 - 323571.918	313527.07	323509.34	302574.17	316874.41	500001	111700
281444.296 - 302508.107	291670.03	302453.02	281465.35	291668.38	500001	140600
260380.484 - 281444.296	271185.17	281364.07	260459.21	266054.10	500001	99100
239316.673 - 260380.484	249209.27	260365.03	239389.94	245027.98	500001	47500
218252.862 - 239316.673	228703.69	239215.32	218310.67	227638.49	500001	50800
197189.051 - 218252.862	207339.97	218252.60	197218.23	207564.07	500001	26900
176125.240 - 197189.051	187379.71	197149.04	176228.84	186031.30	500001	28300
155061.429 - 176125.240	165725.01	176121.52	155113.69	160170.89	500001	60900
133997.617 - 155061.429	144789.14	155047.81	134158.61	145846.67	500001	22500
112933.806 - 133997.617	123309.63	133958.69	112993.72	126170.88	500001	40400
91869.995 - 112933.806	101595.68	112829.61	91870.74	107974.80	500001	37500
70806.184 - 91869.995	84445.67	91838.68	70806.18	81168.58	416700	14999

Target Variable=median_house_value Data Role=VALIDATE

Range for Predicted	Mean Predicted	Max Predicted	Min Predicted	Mean Target	Max Target	Min Target
470994.903 - 492045.698	482577.47	492045.70	471049.69	494819.03	500001	131300
449944.109 - 470994.903	460897.71	470346.27	450133.69	482314.13	500001	112500
428893.314 - 449944.109	439005.96	449860.73	429646.45	458123.30	500001	260600
407842.519 - 428893.314	417629.68	428724.49	407864.82	426518.43	500001	186500
386791.725 - 407842.519	396980.55	407746.87	386813.65	403224.55	500001	225000
365740.930 - 386791.725	376339.14	386082.38	366018.88	388007.54	500001	206800
344690.135 - 365740.930	355056.71	365689.95	344781.96	365651.40	500001	156000
323639.341 - 344690.135	333615.43	344467.82	323662.20	333839.42	500001	137500
302588.546 - 323639.341	312922.32	323583.75	302742.62	306312.04	500001	158900
281537.751 - 302588.546	291556.62	302440.46	281596.05	288416.62	500001	105400
260486.957 - 281537.751	270773.51	281296.56	260685.22	269151.97	500001	112500
239436.162 - 260486.957	249519.03	260484.87	239526.23	252216.87	500001	87500
218385.368 - 239436.162	228815.92	239384.23	218441.06	222180.17	500001	67500
197334.573 - 218385.368	207383.77	218353.16	197399.33	205672.76	500001	30000
176283.778 - 197334.573	186647.02	197242.97	176285.05	181250.33	500001	37500
155232.984 - 176283.778	165784.28	176278.35	155266.28	159778.40	500001	17500
134182.189 - 155232.984	145045.48	155227.12	134182.41	145198.54	434700	48500
113131.394 - 134182.189	123874.36	134136.79	113144.74	129390.61	500001	47500
92080.600 - 113131.394	101723.58	113128.22	92121.36	104821.58	450000	39400
71029.805 - 92080.600	84642.81	92074.57	71029.81	80946.57	310000	14999

SAS Enterprise Miner Report

Node=Exhaustive Regression Summary

Node id = Reg
Node label = Exhaustive Regression
Meta path = Ids2 => Part => Reg
Notes =

Node=Exhaustive Regression Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	Regression		Force	0		PolynomialDegree	2	
AbsConvValue	-1.34078E154	-7.237006E75	GConvTimes	1		PrintDesignMatrix	N	
AbsFTime	1		GConvValue	1E-6		Rule	NONE	
AbsFValue	0		Hierarchy	CLASS		SASSPDS	N	
AbsGTime	1		InputCoding	DEVIATION		SelectionCriterion	DEFAULT	
AbsGValue	0.00001		Interactions			SelectionDefault	Y	
AbsXTime	1		LinkFunction	LOGIT		Sequential	N	
AbsXValue	1E-8		MainEffect	Y		Simple	N	
CIParm	N		MaxCPUTime	1 HOUR		SIEntry	0.05	
ConvDefaults	Y		MaxFunctionCalls	.		SIStay	0.05	
CorB	N		MaxIterations	.		Start	0	
CovB	N		MaxStep	.		StepOutput	N	
Covout	N		MinResourceUse	N		Stop	0	
Details	N		ModelDefaults	Y		SuppressIntercept	N	
Error	LOGISTIC		ModelSelection	NONE		SuppressOutput	N	
ExcludedVariable	REJECT		OptimizationTechnique	DEFAULT		Terms	N	
FConvTimes	1		Performance	N		TwoFactor	N	
FConvValue	0		Polynomial	N				

Node=Exhaustive Regression Variable Summary

Role	Level	Frequency Count	Name
TARGET	INTERVAL	1	median_house_value
INPUT	INTERVAL	8	households housing_median_age latitude longitude median_income population total_bedrooms total_rooms
INPUT	NOMINAL	1	ocean_proximity

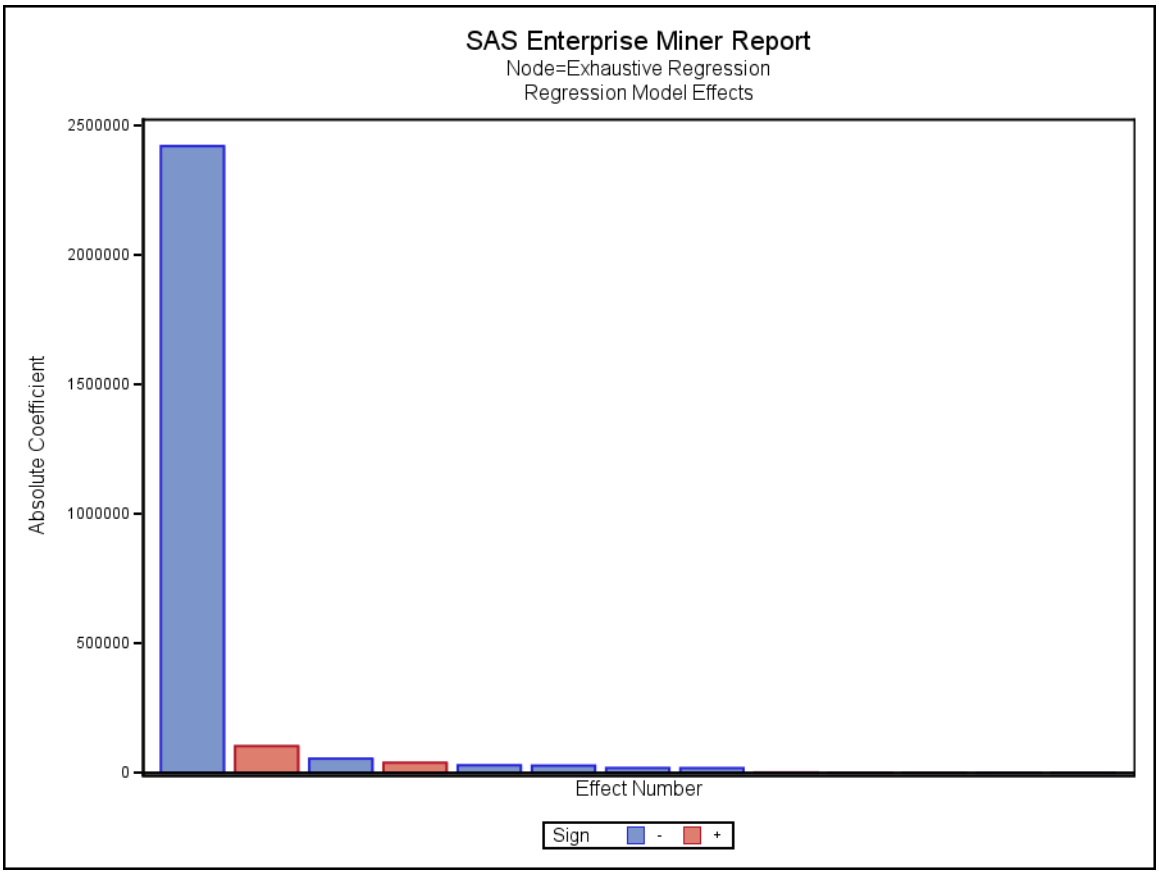
Node=Exhaustive Regression Model Fit Statistics

Target=median_house_value Target Label=' '

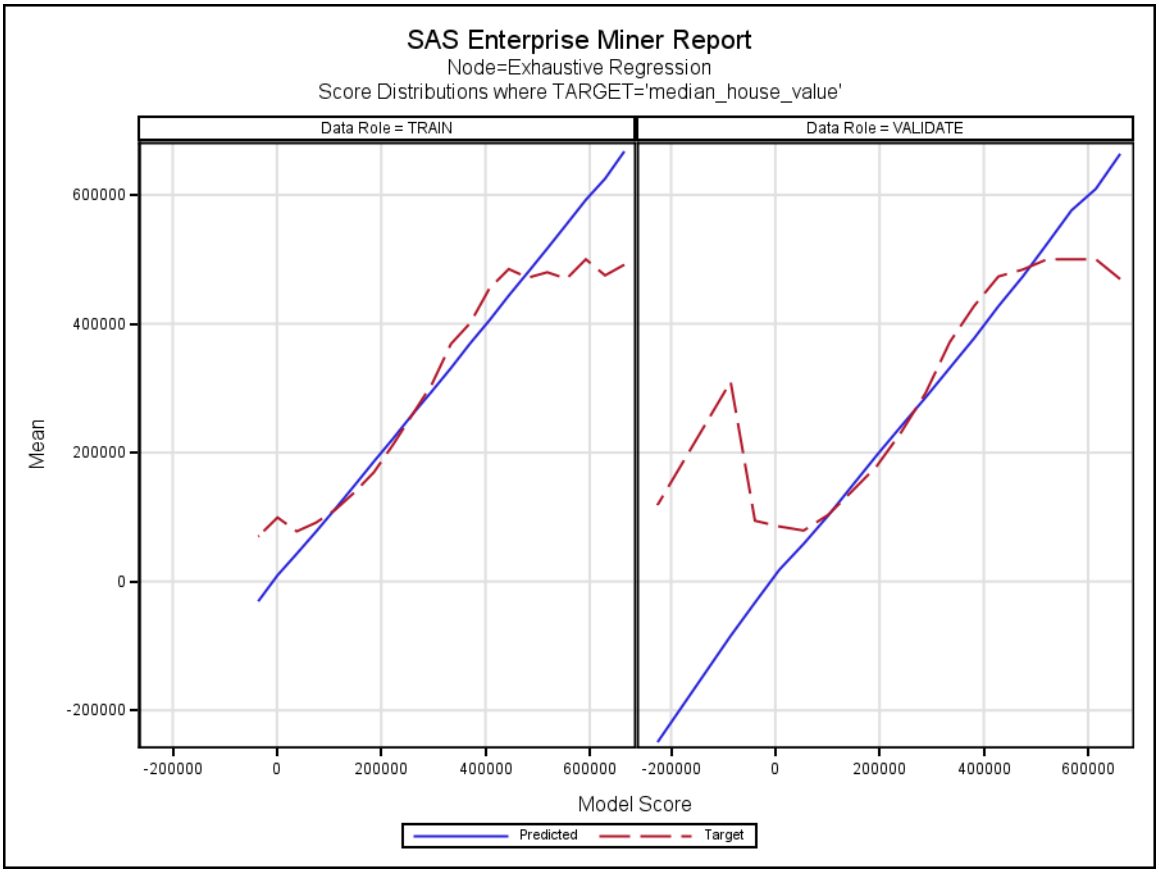
Label of Statistic	Train	Validation	Test
Akaike's Information Criterion	147398.73	.	.
Average Squared Error	4899002782.96	4798889268.60	4828197295.33
Average Error Function	4899002782.96	4798889268.60	4828197295.33
Degrees of Freedom for Error	6592.00	.	.
Model Degrees of Freedom	13.00	.	.
Total Degrees of Freedom	6605.00	.	.

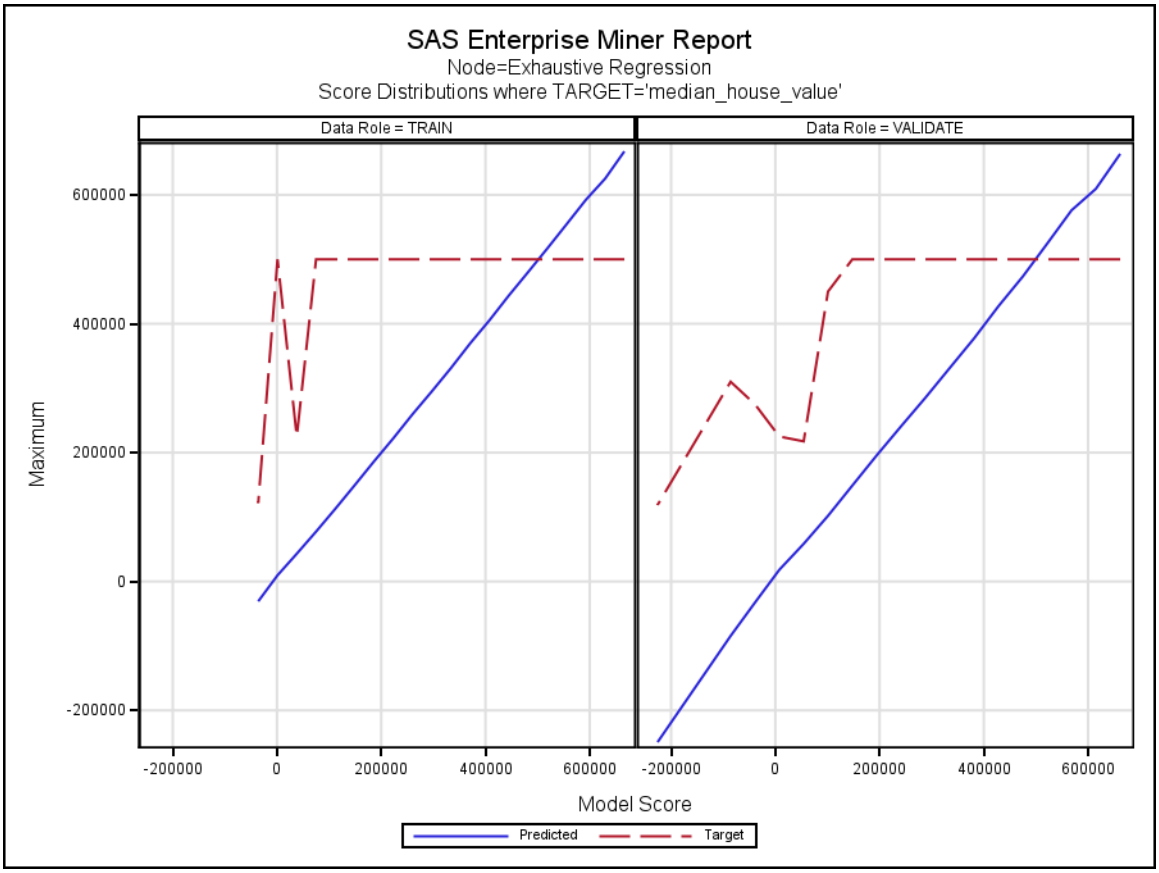
Target=median_house_value Target Label=' '

Label of Statistic	Train	Validation	Test
Divisor for ASE	6605.00	4954.00	4953.00
Error Function	3.23579134E13	2.37736974E13	2.39140612E13
Final Prediction Error	4918325306.07	.	.
Maximum Absolute Error	497860.93	553715.28	922428.05
Mean Square Error	4908664044.52	4798889268.60	4828197295.33
Sum of Frequencies	6605.00	4954.00	4953.00
Number of Estimate Weights	13.00	.	.
Root Average Sum of Squares	69992.88	69274.02	69485.23
Root Final Prediction Error	70130.77	.	.
Root Mean Squared Error	70061.86	69274.02	69485.23
Schwarz's Bayesian Criterion	147487.07	.	.
Sum of Squared Errors	3.23579134E13	2.37736974E13	2.39140612E13
Sum of Case Weights Times Freq	6605.00	4954.00	4953.00



Effect Number	Variable	Level	Coefficient	T-value	P Value	Effect Number	Variable	Level	Coefficient	T-value	P Value
1	Intercept		-2419684.95	-15.3958	0.000000	8	ocean_proximity	_1H_OCEA	-17930.29	-1.8050	0.071119
2	ocean_proximity	ISLAND	102841.59	2.6139	0.008972	9	housing_median_age		1135.56	14.4869	0.000000
3	ocean_proximity	INLAND	-54326.48	-5.3327	0.000000	10	total_bedrooms		105.29	7.6676	0.000000
4	median_income		38746.85	62.9974	0.000000	11	households		52.36	3.3834	0.000720
5	longitude		-28788.13	-15.8698	0.000000	12	population		-43.96	-20.0150	0.000000
6	latitude		-27406.01	-15.3558	0.000000	13	total_rooms		-4.81	-3.3507	0.000811
7	ocean_proximity	NEAR_BAY	-18337.29	-1.8073	0.070759	.			.	.	.





Node=Exhaustive Regression
Score Distributions

Target Variable=median_house_value Data Role=TRAIN

Range for Predicted	Mean Predicted	Max Predicted	Min Predicted	Mean Target	Max Target	Min Target
648018.233 - 684990.121	667541.18	684990.12	658551.43	491667.58	500001	400000
611046.346 - 648018.233	625102.84	647677.07	612863.41	475000.83	500001	350000
574074.458 - 611046.346	592257.97	601528.08	576945.80	500001.00	500001	500001
537102.571 - 574074.458	554354.28	571324.92	538197.08	469770.80	500001	281000
500130.683 - 537102.571	516781.13	535458.37	500612.64	479874.58	500001	225000
463158.796 - 500130.683	479957.27	499896.53	463529.99	471463.66	500001	112500
426186.908 - 463158.796	443642.38	462179.87	426981.90	484832.39	500001	351700
389215.021 - 426186.908	405397.53	426120.48	390544.01	454862.09	500001	225000
352243.133 - 389215.021	369614.30	389070.98	352333.46	401036.51	500001	67500
315271.246 - 352243.133	331191.90	352134.38	315309.69	368635.06	500001	187500
278299.358 - 315271.246	294722.54	315182.38	278338.90	305487.53	500001	99100
241327.471 - 278299.358	259296.91	278282.89	241334.60	259104.44	500001	61300
204355.583 - 241327.471	221863.37	241325.52	204370.09	212569.13	500001	45800
167383.696 - 204355.583	186128.50	204336.95	167385.80	169463.66	500001	22500
130411.808 - 167383.696	149215.71	167327.47	130467.76	138424.84	500001	26900
93439.921 - 130411.808	112841.68	130408.47	93476.28	111398.08	500001	37500
56468.033 - 93439.921	77405.40	93402.71	56538.84	91352.84	500001	28300
19496.146 - 56468.033	43100.08	56431.23	19603.40	77778.23	225000	37500
-17475.742 - 19496.146	9622.13	19433.43	-7236.21	99341.03	500001	14999
-54447.629 - -17475.742	-30762.16	-18832.00	-54447.63	69871.43	121900	32900

Target Variable=median_house_value Data Role=VALIDATE

Range for Predicted	Mean Predicted	Max Predicted	Min Predicted	Mean Target	Max Target	Min Target
638299.716 - 685015.279	663601.10	685015.28	642157.15	469275.92	500001	131300
591584.152 - 638299.716	609310.64	633779.09	591975.18	500001.00	500001	500001
544868.589 - 591584.152	575958.35	590329.22	549436.37	500001.00	500001	500001
498153.025 - 544868.589	524230.01	544179.02	498314.90	500001.00	500001	500001
451437.462 - 498153.025	473287.90	495366.62	452172.41	484057.57	500001	112500
404721.898 - 451437.462	427213.56	451421.08	405171.06	473397.02	500001	228300
358006.335 - 404721.898	377361.75	403171.41	358215.57	427531.67	500001	225000
311290.771 - 358006.335	331373.33	357779.72	311312.41	371445.59	500001	156000
264575.208 - 311290.771	285224.47	311221.26	264581.46	291960.96	500001	105400
217859.644 - 264575.208	240840.02	264562.38	217905.69	231316.14	500001	81800
171144.081 - 217859.644	195762.60	217840.97	171171.12	179020.93	500001	17500
124428.517 - 171144.081	148948.94	171032.23	124449.18	140644.01	500001	30000
77712.954 - 124428.517	101959.91	124204.00	77774.29	103301.00	450000	39200
30997.390 - 77712.954	58446.71	77549.40	31014.25	79072.51	217500	32500
-15718.173 - 30997.390	18168.97	30945.50	-10356.75	85389.78	225000	14999
-62433.737 - -15718.173	-32691.11	-25016.95	-45246.20	94275.00	275000	27500
-109149.300 - -62433.737	-84442.87	-84442.87	-84442.87	310000.00	310000	310000
-249295.991 - -202580.427	-249295.99	-249295.99	-249295.99	118800.00	118800	118800

SAS Enterprise Miner Report

Node=Forward Regression
Summary

Node id = Reg2
Node label = Forward Regression
Meta path = Ids2 => Part => Reg2
Notes =

Node=Forward Regression
Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	Regression		Force	0		PolynomialDegree	2	
AbsConvValue	-1.34078E154	-7.237006E75	GConvTimes	1		PrintDesignMatrix	N	
AbsFTime	1		GConvValue	1E-6		Rule	NONE	
AbsFValue	0		Hierarchy	CLASS		SASSPDS	N	
AbsGTime	1		InputCoding	DEVIATION		SelectionCriterion	DEFAULT	
AbsGValue	0.00001		Interactions			SelectionDefault	Y	
AbsXTime	1		LinkFunction	LOGIT		Sequential	N	
AbsXValue	1E-8		MainEffect	Y		Simple	N	
CIParm	N		MaxCPUTime	1 HOUR		SIEntry	0.05	
ConvDefaults	Y		MaxFunctionCalls	.		SIStay	0.05	
CorB	N		MaxIterations	.		Start	0	
CovB	N		MaxStep	.		StepOutput	N	
Covout	N		MinResourceUse	N		Stop	0	
Details	N		ModelDefaults	Y		SuppressIntercept	N	
Error	LOGISTIC		ModelSelection	FORWARD	NONE	SuppressOutput	N	
ExcludedVariable	REJECT		OptimizationTechnique	DEFAULT		Terms	N	
FConvTimes	1		Performance	N		TwoFactor	N	
FConvValue	0		Polynomial	N				

Node=Forward Regression
Variable Summary

Role	Level	Frequency Count	Name
TARGET	INTERVAL	1	median_house_value
INPUT	INTERVAL	8	households housing_median_age latitude longitude median_income population total_bedrooms total_rooms
INPUT	NOMINAL	1	ocean_proximity

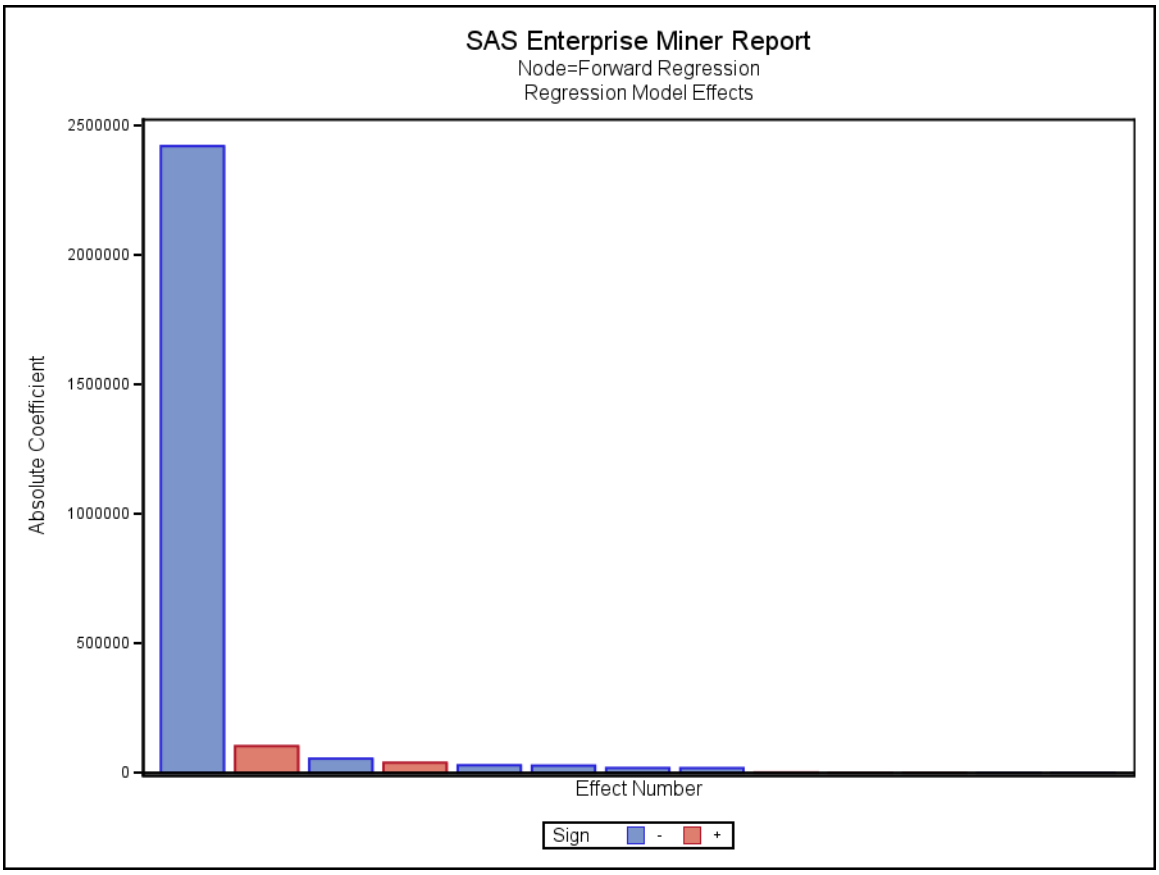
Node=Forward Regression
Model Fit Statistics

Target=median_house_value Target Label=' '

Label of Statistic	Train	Validation	Test
Akaike's Information Criterion	147398.73	.	.
Average Squared Error	4899002782.96	4798889268.60	4828197295.33
Average Error Function	4899002782.96	4798889268.60	4828197295.33
Degrees of Freedom for Error	6592.00	.	.
Model Degrees of Freedom	13.00	.	.
Total Degrees of Freedom	6605.00	.	.

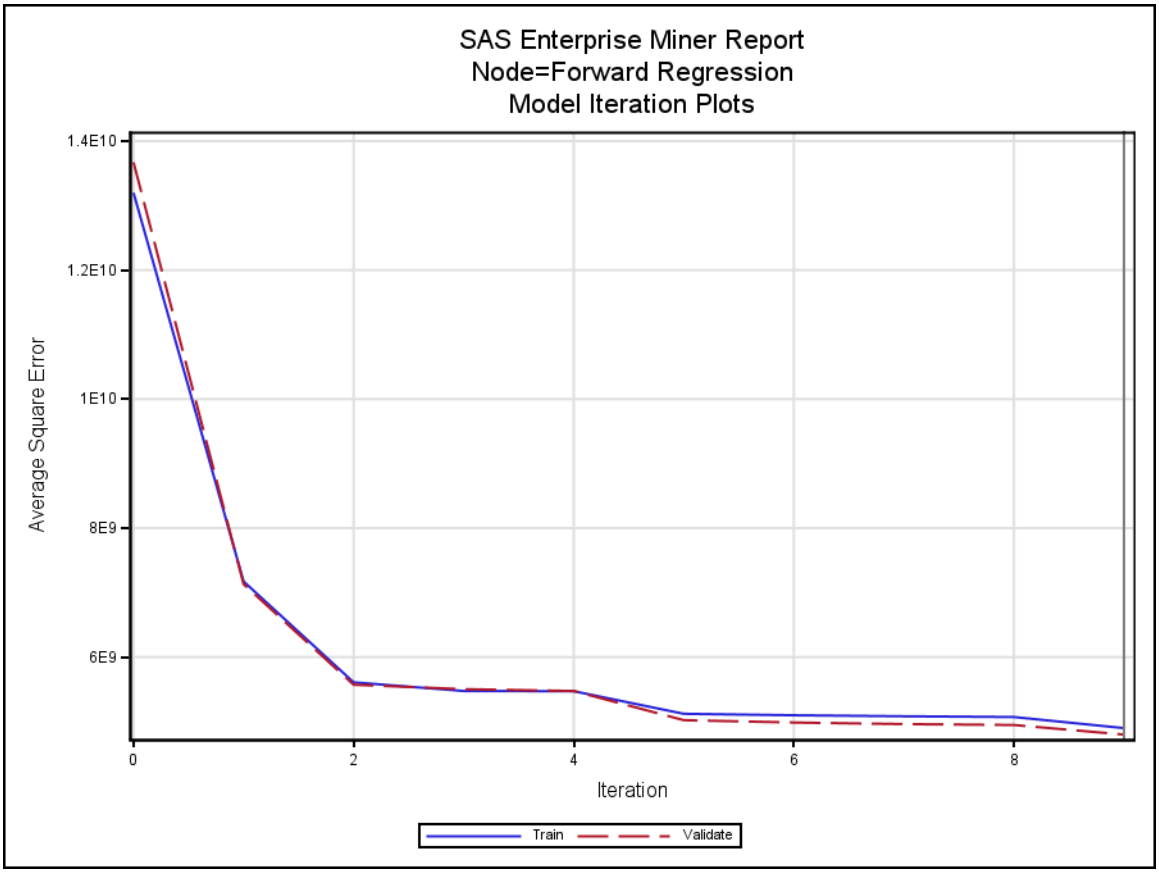
Target=median_house_value Target Label=' '

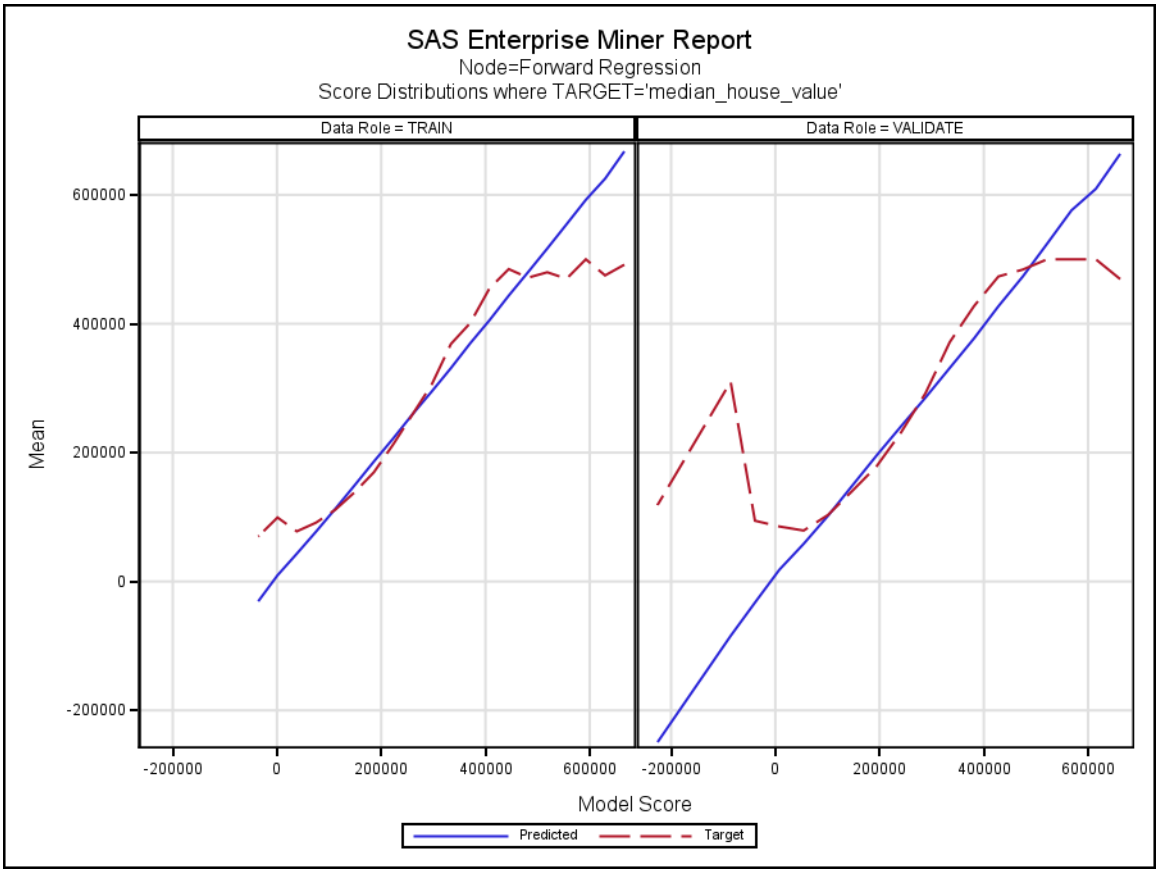
Label of Statistic	Train	Validation	Test
Divisor for ASE	6605.00	4954.00	4953.00
Error Function	3.23579134E13	2.37736974E13	2.39140612E13
Final Prediction Error	4918325306.07	.	.
Maximum Absolute Error	497860.93	553715.28	922428.05
Mean Square Error	4908664044.52	4798889268.60	4828197295.33
Sum of Frequencies	6605.00	4954.00	4953.00
Number of Estimate Weights	13.00	.	.
Root Average Sum of Squares	69992.88	69274.02	69485.23
Root Final Prediction Error	70130.77	.	.
Root Mean Squared Error	70061.86	69274.02	69485.23
Schwarz's Bayesian Criterion	147487.07	.	.
Sum of Squared Errors	3.23579134E13	2.37736974E13	2.39140612E13
Sum of Case Weights Times Freq	6605.00	4954.00	4953.00

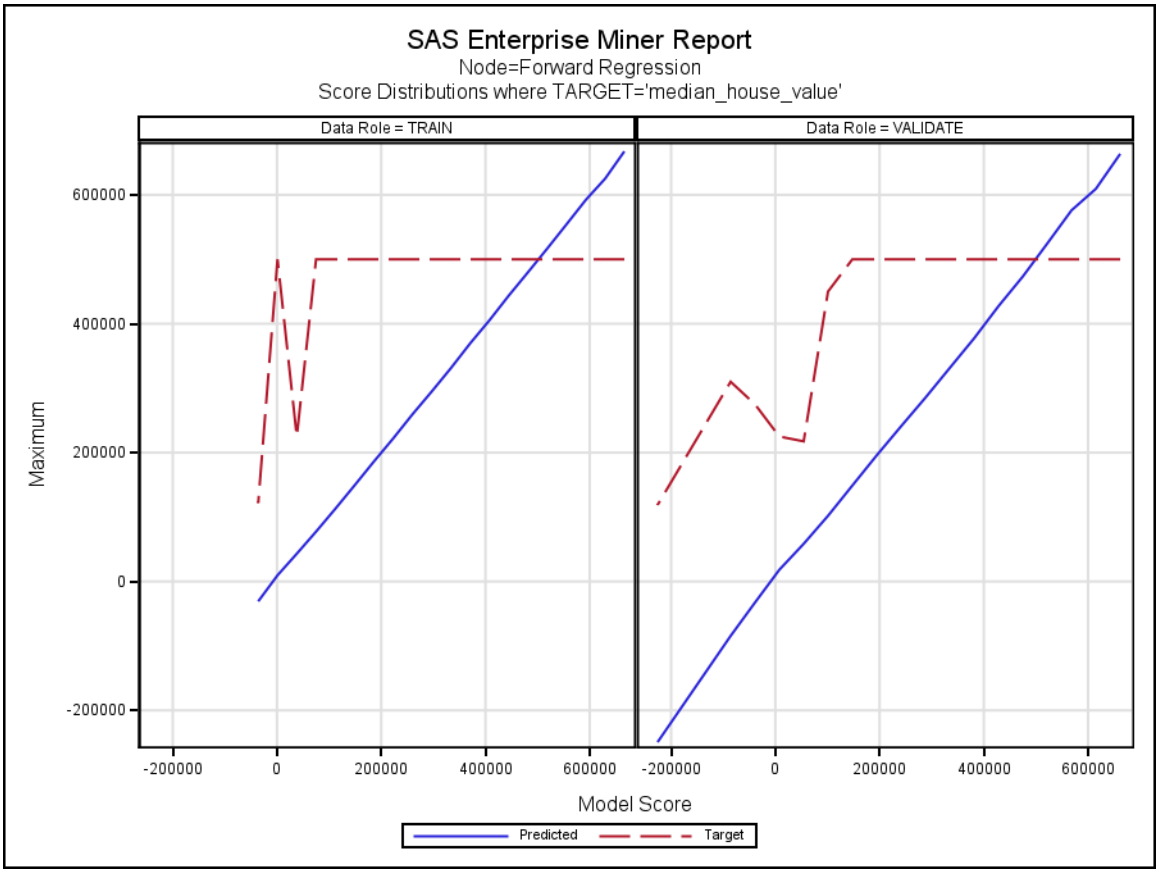


Effect Number	Variable	Level	Coefficient	T-value	P Value	Effect Number	Variable	Level	Coefficient	T-value	P Value
1	Intercept		-2419684.95	-15.3958	0.000000	8	ocean_proximity	_1H_OCEA	-17930.29	-1.8050	0.071119
2	ocean_proximity	ISLAND	102841.59	2.6139	0.008972	9	housing_median_age		1135.56	14.4869	0.000000
3	ocean_proximity	INLAND	-54326.48	-5.3327	0.000000	10	total_bedrooms		105.29	7.6676	0.000000
4	median_income		38746.85	62.9974	0.000000	11	households		52.36	3.3834	0.000720
5	longitude		-28788.13	-15.8698	0.000000	12	population		-43.96	-20.0150	0.000000
6	latitude		-27406.01	-15.3558	0.000000	13	total_rooms		-4.81	-3.3507	0.000811
7	ocean_proximity	NEAR_BAY	-18337.29	-1.8073	0.070759	.			.	.	.

SAS Enterprise Miner Report
Node=Forward Regression
Model Iteration Plots







Node=Forward Regression
Score Distributions

Target Variable=median_house_value Data Role=TRAIN

Range for Predicted	Mean Predicted	Max Predicted	Min Predicted	Mean Target	Max Target	Min Target
648018.233 - 684990.121	667541.18	684990.12	658551.43	491667.58	500001	400000
611046.346 - 648018.233	625102.84	647677.07	612863.41	475000.83	500001	350000
574074.458 - 611046.346	592257.97	601528.08	576945.80	500001.00	500001	500001
537102.571 - 574074.458	554354.28	571324.92	538197.08	469770.80	500001	281000
500130.683 - 537102.571	516781.13	535458.37	500612.64	479874.58	500001	225000
463158.796 - 500130.683	479957.27	499896.53	463529.99	471463.66	500001	112500
426186.908 - 463158.796	443642.38	462179.87	426981.90	484832.39	500001	351700
389215.021 - 426186.908	405397.53	426120.48	390544.01	454862.09	500001	225000
352243.133 - 389215.021	369614.30	389070.98	352333.46	401036.51	500001	67500
315271.246 - 352243.133	331191.90	352134.38	315309.69	368635.06	500001	187500
278299.358 - 315271.246	294722.54	315182.38	278338.90	305487.53	500001	99100
241327.471 - 278299.358	259296.91	278282.89	241334.60	259104.44	500001	61300
204355.583 - 241327.471	221863.37	241325.52	204370.09	212569.13	500001	45800
167383.696 - 204355.583	186128.50	204336.95	167385.80	169463.66	500001	22500
130411.808 - 167383.696	149215.71	167327.47	130467.76	138424.84	500001	26900
93439.921 - 130411.808	112841.68	130408.47	93476.28	111398.08	500001	37500
56468.033 - 93439.921	77405.40	93402.71	56538.84	91352.84	500001	28300
19496.146 - 56468.033	43100.08	56431.23	19603.40	77778.23	225000	37500
-17475.742 - 19496.146	9622.13	19433.43	-7236.21	99341.03	500001	14999
-54447.629 - -17475.742	-30762.16	-18832.00	-54447.63	69871.43	121900	32900

Target Variable=median_house_value Data Role=VALIDATE

Range for Predicted	Mean Predicted	Max Predicted	Min Predicted	Mean Target	Max Target	Min Target
638299.716 - 685015.279	663601.10	685015.28	642157.15	469275.92	500001	131300
591584.152 - 638299.716	609310.64	633779.09	591975.18	500001.00	500001	500001
544868.589 - 591584.152	575958.35	590329.22	549436.37	500001.00	500001	500001
498153.025 - 544868.589	524230.01	544179.02	498314.90	500001.00	500001	500001
451437.462 - 498153.025	473287.90	495366.62	452172.41	484057.57	500001	112500
404721.898 - 451437.462	427213.56	451421.08	405171.06	473397.02	500001	228300
358006.335 - 404721.898	377361.75	403171.41	358215.57	427531.67	500001	225000
311290.771 - 358006.335	331373.33	357779.72	311312.41	371445.59	500001	156000
264575.208 - 311290.771	285224.47	311221.26	264581.46	291960.96	500001	105400
217859.644 - 264575.208	240840.02	264562.38	217905.69	231316.14	500001	81800
171144.081 - 217859.644	195762.60	217840.97	171171.12	179020.93	500001	17500
124428.517 - 171144.081	148948.94	171032.23	124449.18	140644.01	500001	30000
77712.954 - 124428.517	101959.91	124204.00	77774.29	103301.00	450000	39200
30997.390 - 77712.954	58446.71	77549.40	31014.25	79072.51	217500	32500
-15718.173 - 30997.390	18168.97	30945.50	-10356.75	85389.78	225000	14999
-62433.737 - -15718.173	-32691.11	-25016.95	-45246.20	94275.00	275000	27500
-109149.300 - -62433.737	-84442.87	-84442.87	-84442.87	310000.00	310000	310000
-249295.991 - -202580.427	-249295.99	-249295.99	-249295.99	118800.00	118800	118800

SAS Enterprise Miner Report

Node=Backward Regression
Summary

Node id = Reg3
Node label = Backward Regression
Meta path = lds2 => Part => Reg3
Notes =

Node=Backward Regression
Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	Regression		Force	0		PolynomialDegree	2	
AbsConvValue	-1.34078E154	-7.237006E75	GConvTimes	1		PrintDesignMatrix	N	
AbsFTime	1		GConvValue	1E-6		Rule	NONE	
AbsFValue	0		Hierarchy	CLASS		SASSPDS	N	
AbsGTime	1		InputCoding	DEVIATION		SelectionCriterion	DEFAULT	
AbsGValue	0.00001		Interactions			SelectionDefault	Y	
AbsXTime	1		LinkFunction	LOGIT		Sequential	N	
AbsXValue	1E-8		MainEffect	Y		Simple	N	
CIParm	N		MaxCPUTime	1 HOUR		SIEntry	0.05	
ConvDefaults	Y		MaxFunctionCalls	.		SIStay	0.05	
CorB	N		MaxIterations	.		Start	0	
CovB	N		MaxStep	.		StepOutput	N	
Covout	N		MinResourceUse	N		Stop	0	
Details	N		ModelDefaults	Y		SuppressIntercept	N	
Error	LOGISTIC		ModelSelection	BACKWARD	NONE	SuppressOutput	N	
ExcludedVariable	REJECT		OptimizationTechnique	DEFAULT		Terms	N	
FConvTimes	1		Performance	N		TwoFactor	N	
FConvValue	0		Polynomial	N				

Node=Backward Regression
Variable Summary

Role	Level	Frequency Count	Name
TARGET	INTERVAL	1	median_house_value
INPUT	INTERVAL	8	households housing_median_age latitude longitude median_income population total_bedrooms total_rooms
INPUT	NOMINAL	1	ocean_proximity

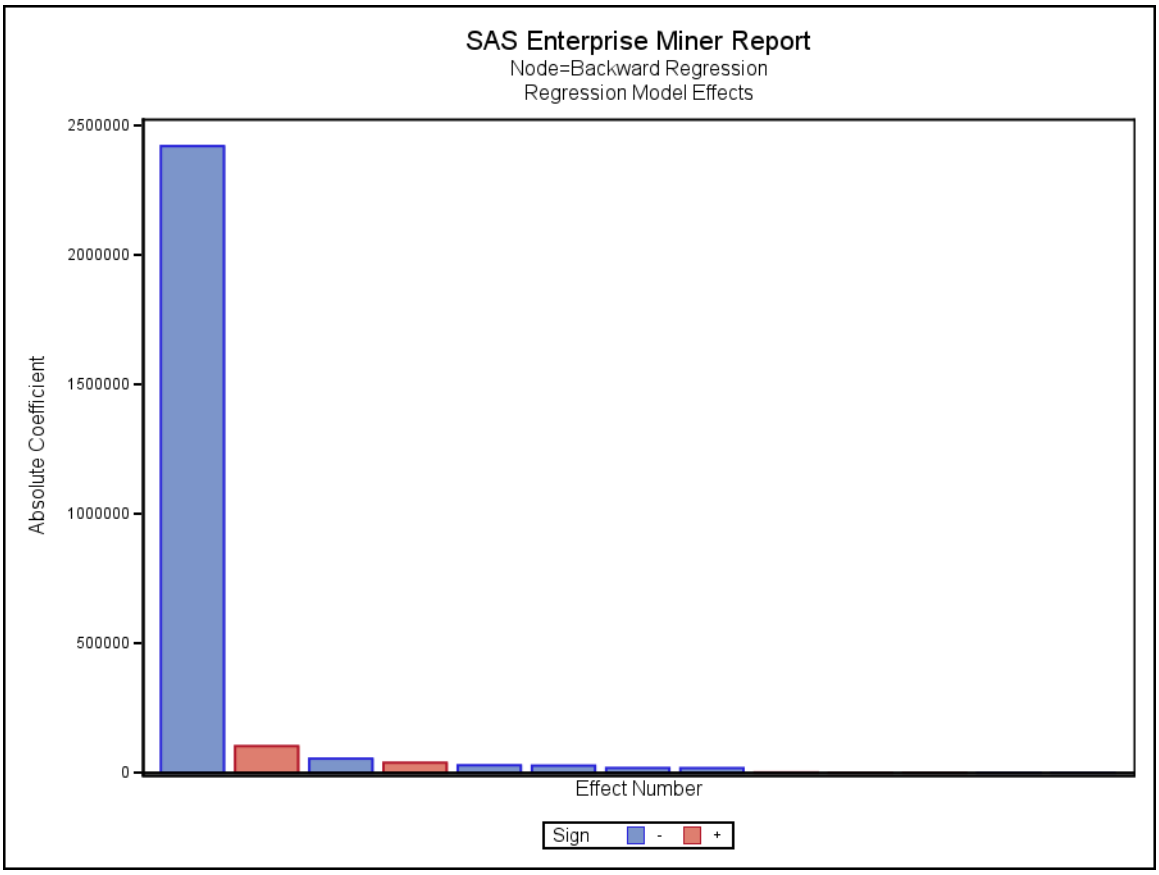
Node=Backward Regression
Model Fit Statistics

Target=median_house_value Target Label=' '

Label of Statistic	Train	Validation	Test
Akaike's Information Criterion	147398.73	.	.
Average Squared Error	4899002782.96	4798889268.60	4828197295.33
Average Error Function	4899002782.96	4798889268.60	4828197295.33
Degrees of Freedom for Error	6592.00	.	.
Model Degrees of Freedom	13.00	.	.
Total Degrees of Freedom	6605.00	.	.

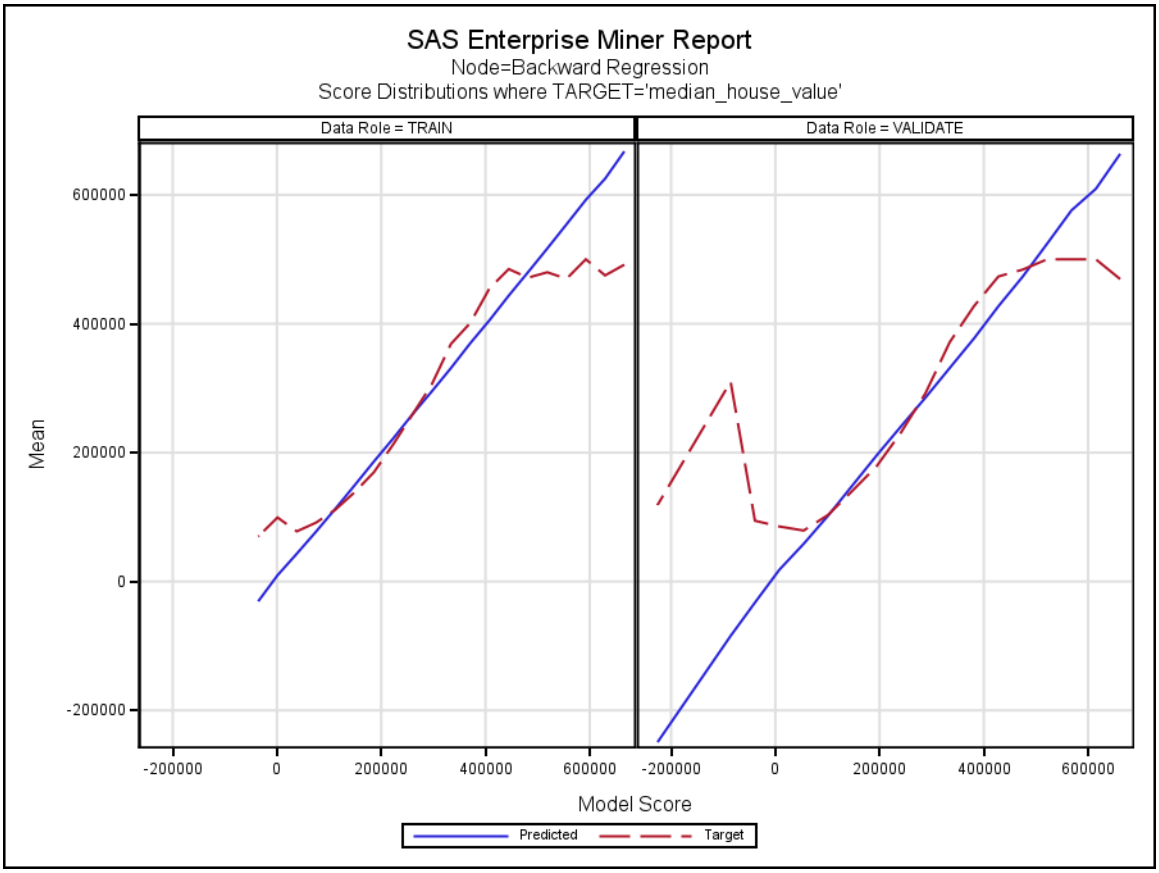
Target=median_house_value Target Label=' '

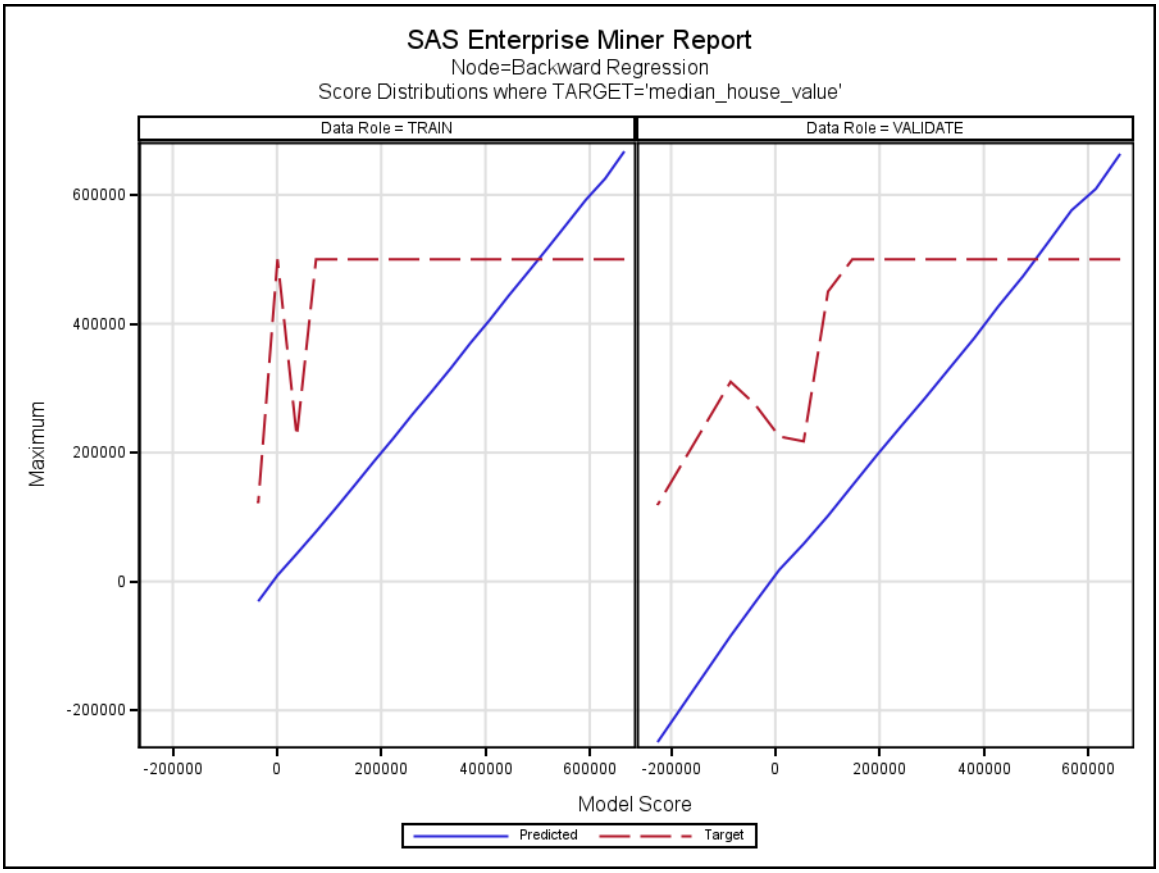
Label of Statistic	Train	Validation	Test
Divisor for ASE	6605.00	4954.00	4953.00
Error Function	3.23579134E13	2.37736974E13	2.39140612E13
Final Prediction Error	4918325306.07	.	.
Maximum Absolute Error	497860.93	553715.28	922428.05
Mean Square Error	4908664044.52	4798889268.60	4828197295.33
Sum of Frequencies	6605.00	4954.00	4953.00
Number of Estimate Weights	13.00	.	.
Root Average Sum of Squares	69992.88	69274.02	69485.23
Root Final Prediction Error	70130.77	.	.
Root Mean Squared Error	70061.86	69274.02	69485.23
Schwarz's Bayesian Criterion	147487.07	.	.
Sum of Squared Errors	3.23579134E13	2.37736974E13	2.39140612E13
Sum of Case Weights Times Freq	6605.00	4954.00	4953.00



Effect Number	Variable	Level	Coefficient	T-value	P Value	Effect Number	Variable	Level	Coefficient	T-value	P Value
1	Intercept		-2419684.95	-15.3958	0.000000	8	ocean_proximity	_1H_OCEA	-17930.29	-1.8050	0.071119
2	ocean_proximity	ISLAND	102841.59	2.6139	0.008972	9	housing_median_age		1135.56	14.4869	0.000000
3	ocean_proximity	INLAND	-54326.48	-5.3327	0.000000	10	total_bedrooms		105.29	7.6676	0.000000
4	median_income		38746.85	62.9974	0.000000	11	households		52.36	3.3834	0.000720
5	longitude		-28788.13	-15.8698	0.000000	12	population		-43.96	-20.0150	0.000000
6	latitude		-27406.01	-15.3558	0.000000	13	total_rooms		-4.81	-3.3507	0.000811
7	ocean_proximity	NEAR_BAY	-18337.29	-1.8073	0.070759	.			.	.	.

SAS Enterprise Miner Report
Node=Backward Regression
Model Iteration Plots





Node=Backward Regression
Score Distributions

Target Variable=median_house_value Data Role=TRAIN

Range for Predicted	Mean Predicted	Max Predicted	Min Predicted	Mean Target	Max Target	Min Target
648018.233 - 684990.121	667541.18	684990.12	658551.43	491667.58	500001	400000
611046.346 - 648018.233	625102.84	647677.07	612863.41	475000.83	500001	350000
574074.458 - 611046.346	592257.97	601528.08	576945.80	500001.00	500001	500001
537102.571 - 574074.458	554354.28	571324.92	538197.08	469770.80	500001	281000
500130.683 - 537102.571	516781.13	535458.37	500612.64	479874.58	500001	225000
463158.796 - 500130.683	479957.27	499896.53	463529.99	471463.66	500001	112500
426186.908 - 463158.796	443642.38	462179.87	426981.90	484832.39	500001	351700
389215.021 - 426186.908	405397.53	426120.48	390544.01	454862.09	500001	225000
352243.133 - 389215.021	369614.30	389070.98	352333.46	401036.51	500001	67500
315271.246 - 352243.133	331191.90	352134.38	315309.69	368635.06	500001	187500
278299.358 - 315271.246	294722.54	315182.38	278338.90	305487.53	500001	99100
241327.471 - 278299.358	259296.91	278282.89	241334.60	259104.44	500001	61300
204355.583 - 241327.471	221863.37	241325.52	204370.09	212569.13	500001	45800
167383.696 - 204355.583	186128.50	204336.95	167385.80	169463.66	500001	22500
130411.808 - 167383.696	149215.71	167327.47	130467.76	138424.84	500001	26900
93439.921 - 130411.808	112841.68	130408.47	93476.28	111398.08	500001	37500
56468.033 - 93439.921	77405.40	93402.71	56538.84	91352.84	500001	28300
19496.146 - 56468.033	43100.08	56431.23	19603.40	77778.23	225000	37500
-17475.742 - 19496.146	9622.13	19433.43	-7236.21	99341.03	500001	14999
-54447.629 - -17475.742	-30762.16	-18832.00	-54447.63	69871.43	121900	32900

Target Variable=median_house_value Data Role=VALIDATE

Range for Predicted	Mean Predicted	Max Predicted	Min Predicted	Mean Target	Max Target	Min Target
638299.716 - 685015.279	663601.10	685015.28	642157.15	469275.92	500001	131300
591584.152 - 638299.716	609310.64	633779.09	591975.18	500001.00	500001	500001
544868.589 - 591584.152	575958.35	590329.22	549436.37	500001.00	500001	500001
498153.025 - 544868.589	524230.01	544179.02	498314.90	500001.00	500001	500001
451437.462 - 498153.025	473287.90	495366.62	452172.41	484057.57	500001	112500
404721.898 - 451437.462	427213.56	451421.08	405171.06	473397.02	500001	228300
358006.335 - 404721.898	377361.75	403171.41	358215.57	427531.67	500001	225000
311290.771 - 358006.335	331373.33	357779.72	311312.41	371445.59	500001	156000
264575.208 - 311290.771	285224.47	311221.26	264581.46	291960.96	500001	105400
217859.644 - 264575.208	240840.02	264562.38	217905.69	231316.14	500001	81800
171144.081 - 217859.644	195762.60	217840.97	171171.12	179020.93	500001	17500
124428.517 - 171144.081	148948.94	171032.23	124449.18	140644.01	500001	30000
77712.954 - 124428.517	101959.91	124204.00	77774.29	103301.00	450000	39200
30997.390 - 77712.954	58446.71	77549.40	31014.25	79072.51	217500	32500
-15718.173 - 30997.390	18168.97	30945.50	-10356.75	85389.78	225000	14999
-62433.737 - -15718.173	-32691.11	-25016.95	-45246.20	94275.00	275000	27500
-109149.300 - -62433.737	-84442.87	-84442.87	-84442.87	310000.00	310000	310000
-249295.991 - -202580.427	-249295.99	-249295.99	-249295.99	118800.00	118800	118800

SAS Enterprise Miner Report

Node=Stepwise Regression
Summary

Node id = Reg4
Node label = Stepwise Regression
Meta path = Ids2 => Part => Reg4
Notes =

Node=Stepwise Regression
Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	Regression		Force	0		PolynomialDegree	2	
AbsConvValue	-1.34078E154	-7.237006E75	GConvTimes	1		PrintDesignMatrix	N	
AbsFTime	1		GConvValue	1E-6		Rule	NONE	
AbsFValue	0		Hierarchy	CLASS		SASSPDS	N	
AbsGTime	1		InputCoding	DEVIATION		SelectionCriterion	DEFAULT	
AbsGValue	0.00001		Interactions			SelectionDefault	Y	
AbsXTime	1		LinkFunction	LOGIT		Sequential	N	
AbsXValue	1E-8		MainEffect	Y		Simple	N	
ClParm	N		MaxCPUTime	1 HOUR		SlEntry	0.05	
ConvDefaults	Y		MaxFunctionCalls	.		SlStay	0.05	
CorB	N		MaxIterations	.		Start	0	
CovB	N		MaxStep	.		StepOutput	N	
Covout	N		MinResourceUse	N		Stop	0	
Details	N		ModelDefaults	Y		SuppressIntercept	N	
Error	LOGISTIC		ModelSelection	STEPWISE	NONE	SuppressOutput	N	
ExcludedVariable	REJECT		OptimizationTechnique	DEFAULT		Terms	N	
FConvTimes	1		Performance	N		TwoFactor	N	
FConvValue	0		Polynomial	N				

Node=Stepwise Regression
Variable Summary

Role	Level	Frequency Count	Name
TARGET	INTERVAL	1	median_house_value
INPUT	INTERVAL	8	households housing_median_age latitude longitude median_income population total_bedrooms total_rooms
INPUT	NOMINAL	1	ocean_proximity

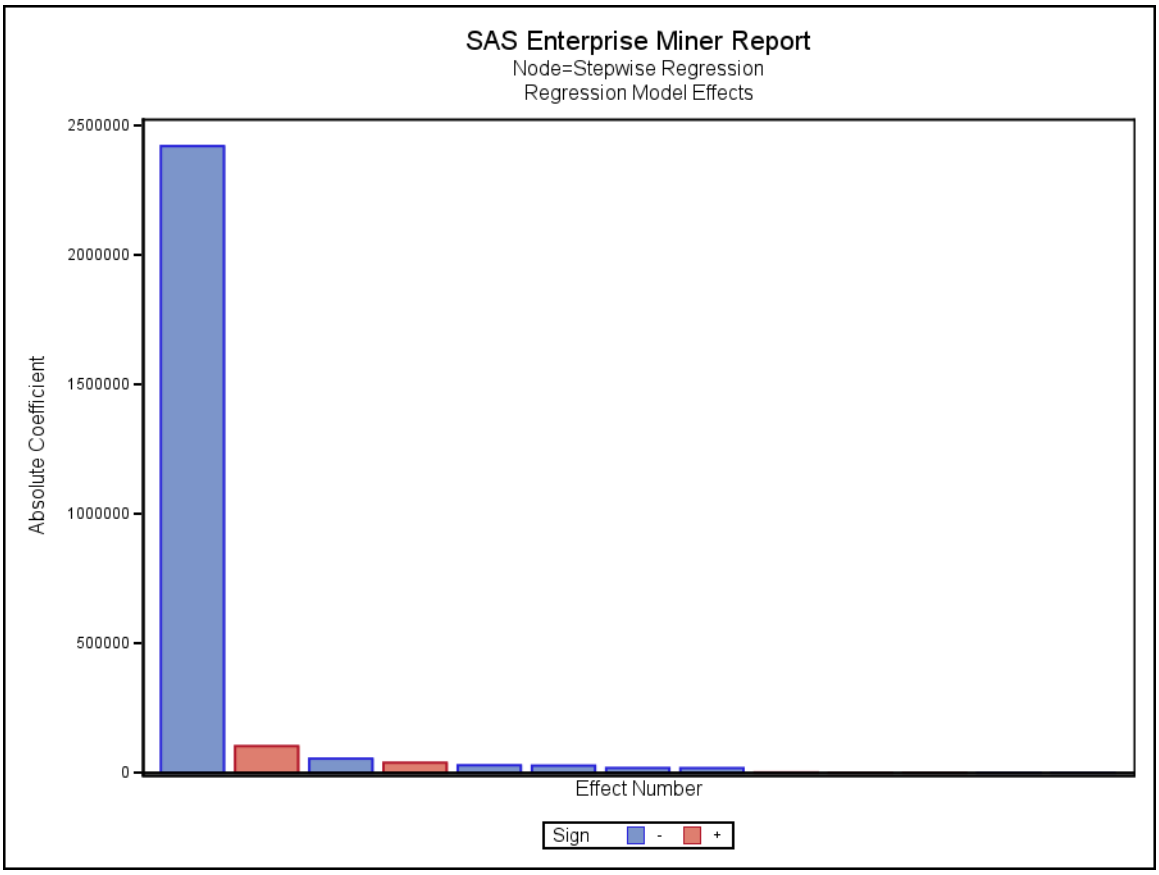
Node=Stepwise Regression
Model Fit Statistics

Target=median_house_value Target Label=' '

Label of Statistic	Train	Validation	Test
Akaike's Information Criterion	147398.73	.	.
Average Squared Error	4899002782.96	4798889268.60	4828197295.33
Average Error Function	4899002782.96	4798889268.60	4828197295.33
Degrees of Freedom for Error	6592.00	.	.
Model Degrees of Freedom	13.00	.	.
Total Degrees of Freedom	6605.00	.	.

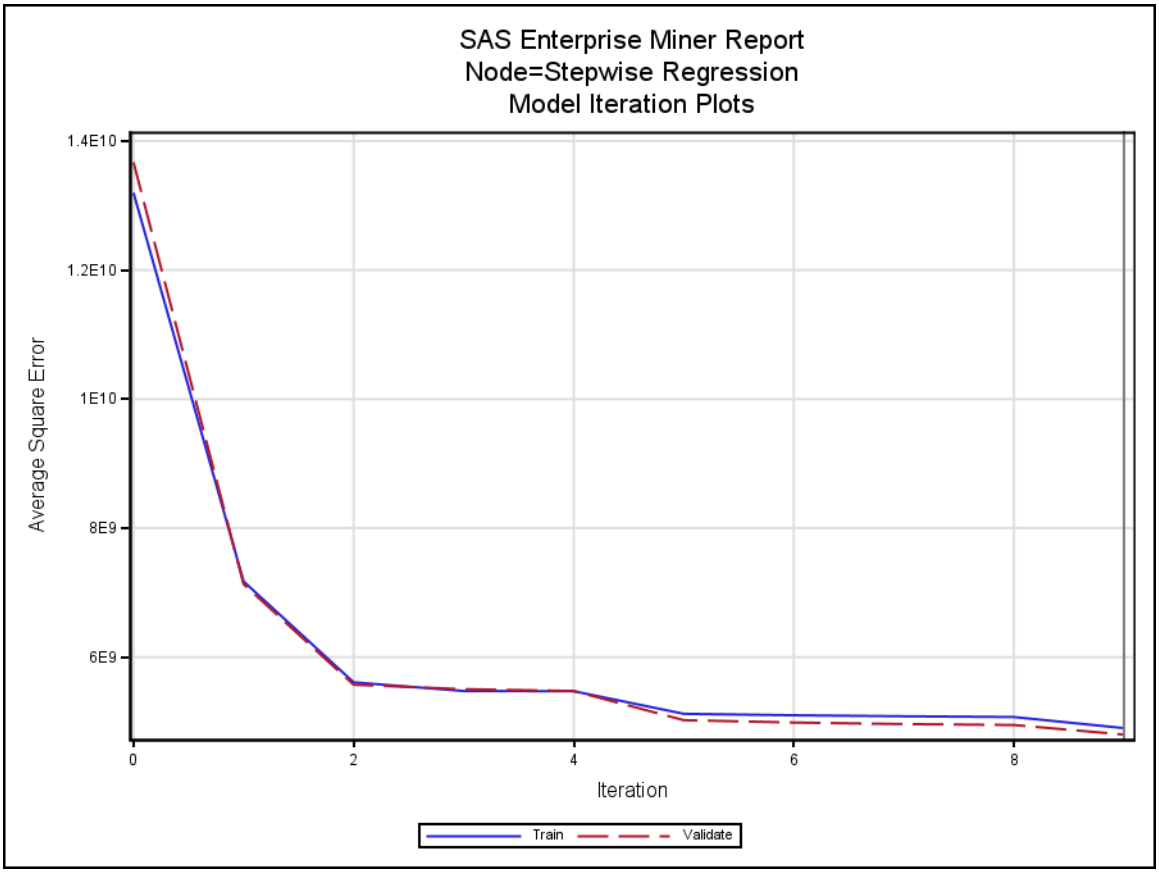
Target=median_house_value Target Label=' '

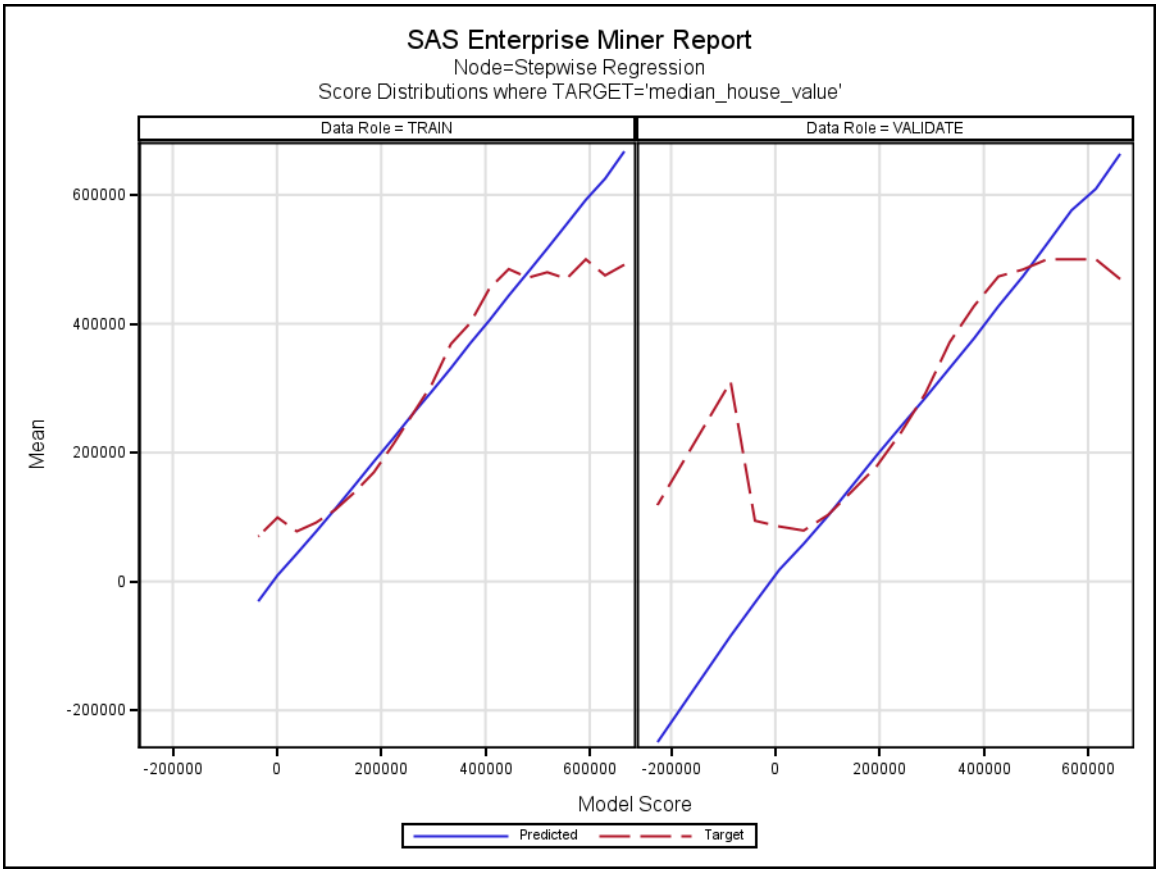
Label of Statistic	Train	Validation	Test
Divisor for ASE	6605.00	4954.00	4953.00
Error Function	3.23579134E13	2.37736974E13	2.39140612E13
Final Prediction Error	4918325306.07	.	.
Maximum Absolute Error	497860.93	553715.28	922428.05
Mean Square Error	4908664044.52	4798889268.60	4828197295.33
Sum of Frequencies	6605.00	4954.00	4953.00
Number of Estimate Weights	13.00	.	.
Root Average Sum of Squares	69992.88	69274.02	69485.23
Root Final Prediction Error	70130.77	.	.
Root Mean Squared Error	70061.86	69274.02	69485.23
Schwarz's Bayesian Criterion	147487.07	.	.
Sum of Squared Errors	3.23579134E13	2.37736974E13	2.39140612E13
Sum of Case Weights Times Freq	6605.00	4954.00	4953.00

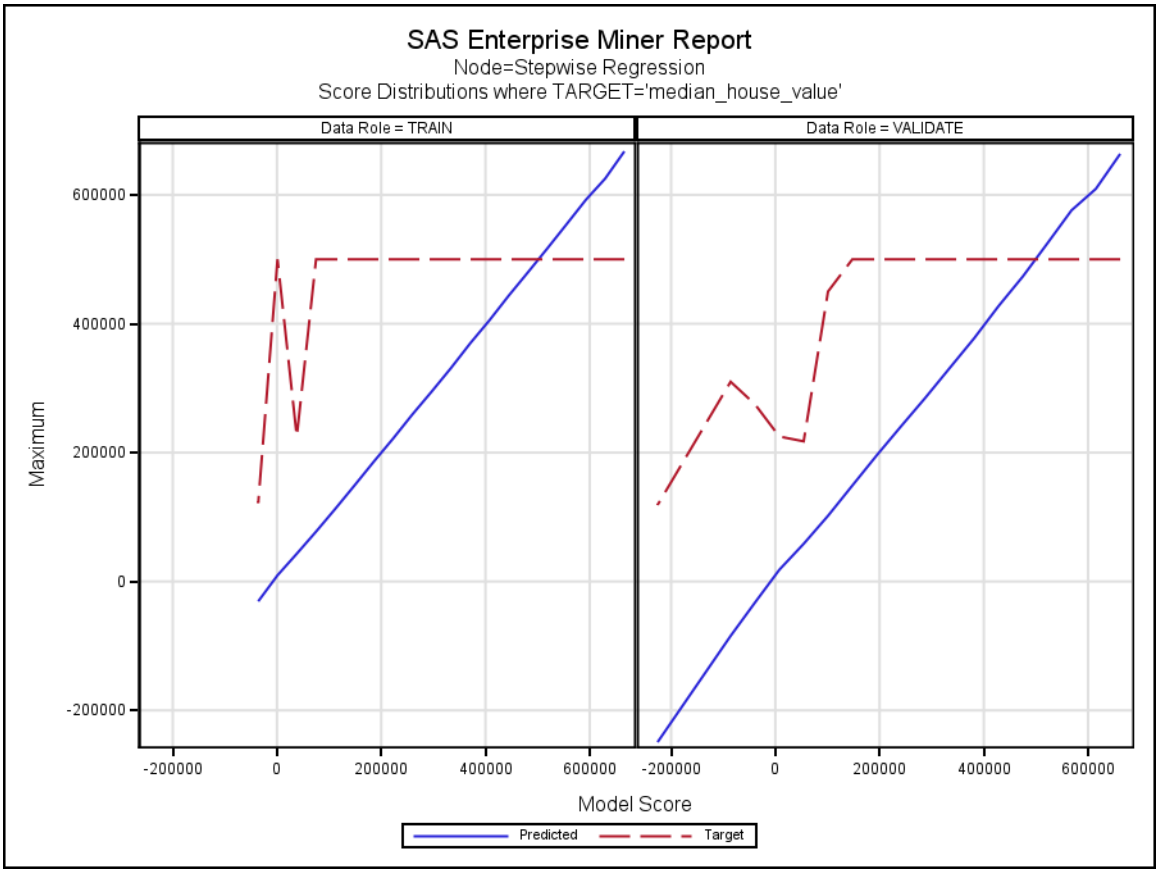


Effect Number	Variable	Level	Coefficient	T-value	P Value	Effect Number	Variable	Level	Coefficient	T-value	P Value
1	Intercept		-2419684.95	-15.3958	0.000000	8	ocean_proximity	_1H_OCEA	-17930.29	-1.8050	0.071119
2	ocean_proximity	ISLAND	102841.59	2.6139	0.008972	9	housing_median_age		1135.56	14.4869	0.000000
3	ocean_proximity	INLAND	-54326.48	-5.3327	0.000000	10	total_bedrooms		105.29	7.6676	0.000000
4	median_income		38746.85	62.9974	0.000000	11	households		52.36	3.3834	0.000720
5	longitude		-28788.13	-15.8698	0.000000	12	population		-43.96	-20.0150	0.000000
6	latitude		-27406.01	-15.3558	0.000000	13	total_rooms		-4.81	-3.3507	0.000811
7	ocean_proximity	NEAR_BAY	-18337.29	-1.8073	0.070759	.			.	.	.

SAS Enterprise Miner Report
Node=Stepwise Regression
Model Iteration Plots







Node=Stepwise Regression
Score Distributions

Target Variable=median_house_value Data Role=TRAIN

Range for Predicted	Mean Predicted	Max Predicted	Min Predicted	Mean Target	Max Target	Min Target
648018.233 - 684990.121	667541.18	684990.12	658551.43	491667.58	500001	400000
611046.346 - 648018.233	625102.84	647677.07	612863.41	475000.83	500001	350000
574074.458 - 611046.346	592257.97	601528.08	576945.80	500001.00	500001	500001
537102.571 - 574074.458	554354.28	571324.92	538197.08	469770.80	500001	281000
500130.683 - 537102.571	516781.13	535458.37	500612.64	479874.58	500001	225000
463158.796 - 500130.683	479957.27	499896.53	463529.99	471463.66	500001	112500
426186.908 - 463158.796	443642.38	462179.87	426981.90	484832.39	500001	351700
389215.021 - 426186.908	405397.53	426120.48	390544.01	454862.09	500001	225000
352243.133 - 389215.021	369614.30	389070.98	352333.46	401036.51	500001	67500
315271.246 - 352243.133	331191.90	352134.38	315309.69	368635.06	500001	187500
278299.358 - 315271.246	294722.54	315182.38	278338.90	305487.53	500001	99100
241327.471 - 278299.358	259296.91	278282.89	241334.60	259104.44	500001	61300
204355.583 - 241327.471	221863.37	241325.52	204370.09	212569.13	500001	45800
167383.696 - 204355.583	186128.50	204336.95	167385.80	169463.66	500001	22500
130411.808 - 167383.696	149215.71	167327.47	130467.76	138424.84	500001	26900
93439.921 - 130411.808	112841.68	130408.47	93476.28	111398.08	500001	37500
56468.033 - 93439.921	77405.40	93402.71	56538.84	91352.84	500001	28300
19496.146 - 56468.033	43100.08	56431.23	19603.40	77778.23	225000	37500
-17475.742 - 19496.146	9622.13	19433.43	-7236.21	99341.03	500001	14999
-54447.629 - -17475.742	-30762.16	-18832.00	-54447.63	69871.43	121900	32900

Target Variable=median_house_value Data Role=VALIDATE

Range for Predicted	Mean Predicted	Max Predicted	Min Predicted	Mean Target	Max Target	Min Target
638299.716 - 685015.279	663601.10	685015.28	642157.15	469275.92	500001	131300
591584.152 - 638299.716	609310.64	633779.09	591975.18	500001.00	500001	500001
544868.589 - 591584.152	575958.35	590329.22	549436.37	500001.00	500001	500001
498153.025 - 544868.589	524230.01	544179.02	498314.90	500001.00	500001	500001
451437.462 - 498153.025	473287.90	495366.62	452172.41	484057.57	500001	112500
404721.898 - 451437.462	427213.56	451421.08	405171.06	473397.02	500001	228300
358006.335 - 404721.898	377361.75	403171.41	358215.57	427531.67	500001	225000
311290.771 - 358006.335	331373.33	357779.72	311312.41	371445.59	500001	156000
264575.208 - 311290.771	285224.47	311221.26	264581.46	291960.96	500001	105400
217859.644 - 264575.208	240840.02	264562.38	217905.69	231316.14	500001	81800
171144.081 - 217859.644	195762.60	217840.97	171171.12	179020.93	500001	17500
124428.517 - 171144.081	148948.94	171032.23	124449.18	140644.01	500001	30000
77712.954 - 124428.517	101959.91	124204.00	77774.29	103301.00	450000	39200
30997.390 - 77712.954	58446.71	77549.40	31014.25	79072.51	217500	32500
-15718.173 - 30997.390	18168.97	30945.50	-10356.75	85389.78	225000	14999
-62433.737 - -15718.173	-32691.11	-25016.95	-45246.20	94275.00	275000	27500
-109149.300 - -62433.737	-84442.87	-84442.87	-84442.87	310000.00	310000	310000
-249295.991 - -202580.427	-249295.99	-249295.99	-249295.99	118800.00	118800	118800

SAS Enterprise Miner Report

Node=RegTree B2D6
Summary

Node id = Tree
Node label = RegTree B2D6
Meta path = Ids2 => Part => Tree
Notes =

Node=RegTree B2D6
Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	DecisionTree		Kass	Y		Pred	N	
AVG	Y		KassApply	BEFORE		Predict	Y	
AssessMeasure	PROFIT/LOSS		LeafSize	5		ProfitLoss	NONE	
AssessPercentage	0.25		Leafid	Y		RASE	N	
CV	N		Maxbranch	2		SampleMethod	RANDOM	
CVNIter	10		Maxdepth	6		SampleSeed	12345	
CVRepeat	1		MinCatSize	5		SampleSize	10000	
CVSeed	12345		MissingValue	USEINSEARCH		ShowNodeId	Y	
ClassColorBy	PERCENTCORRECT		NSubtree	1		ShowValid	Y	
Count	Y		NodeRole	SEGMENT		SigLevel	0.2	
CreateSample	DEFAULT		NodeSample	20000		SplitPrecision	4	
Criterion	DEFAULT		NominalCriterion	PROBCHISQ		Splitsize	.	
Depth	Y		Nrules	5		Subtree	ASSESSMENT	
Dummy	N		Nsurrs	0		Target	ALL	
Exhaustive	5000		NumInputs	1		ToolType	MODEL	
Freeze	N		NumSingleImp	5		TrainMode	BATCH	
ImportModel	N		ObsImportance	N		UseDecision	N	
ImportedTreeData			OrdinalCriterion	ENTROPY		UseMultipleTarget	N	
Inputs	N		PercentCorrect	N		UsePriors	N	
IntColorBy	AVG		Performance	DISK		UseVarOnce	N	
IntervalCriterion	PROBF		Precision	4		VarSelection	Y	

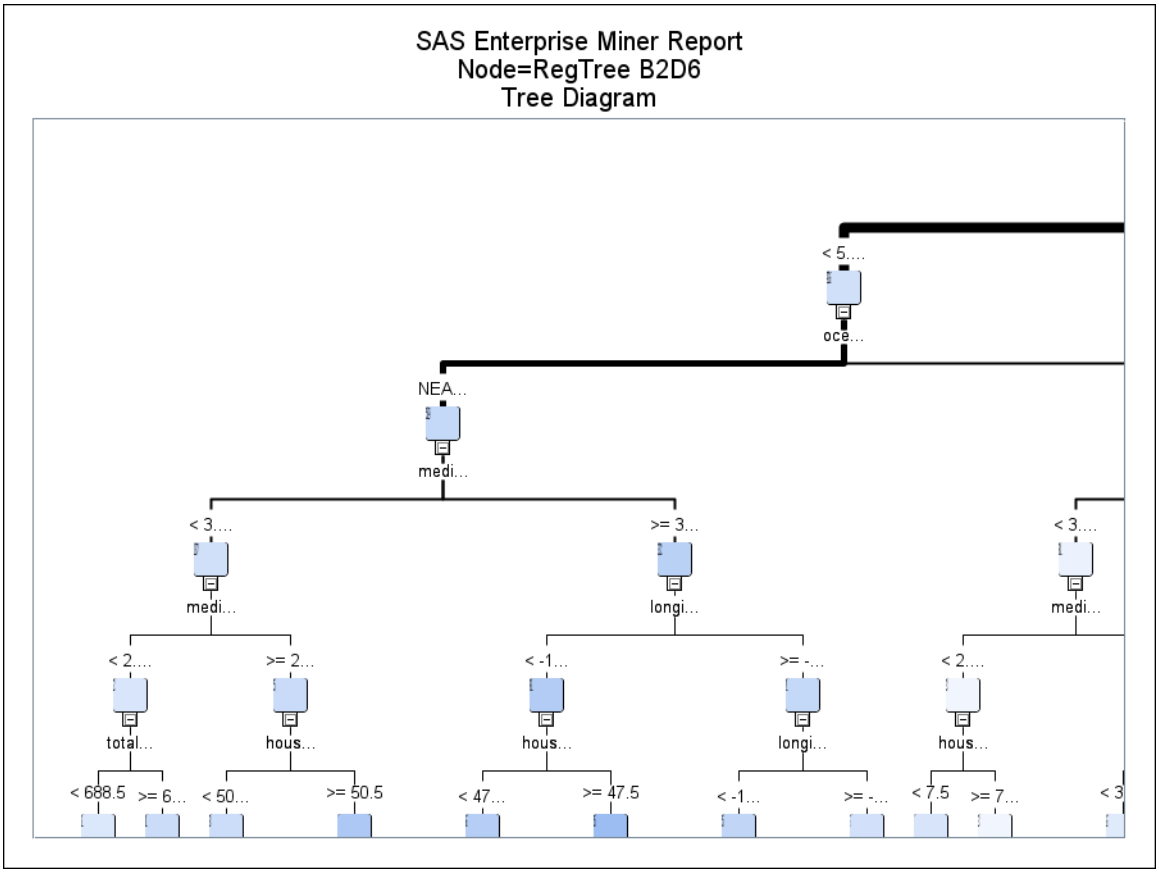
Node=RegTree B2D6
Variable Summary

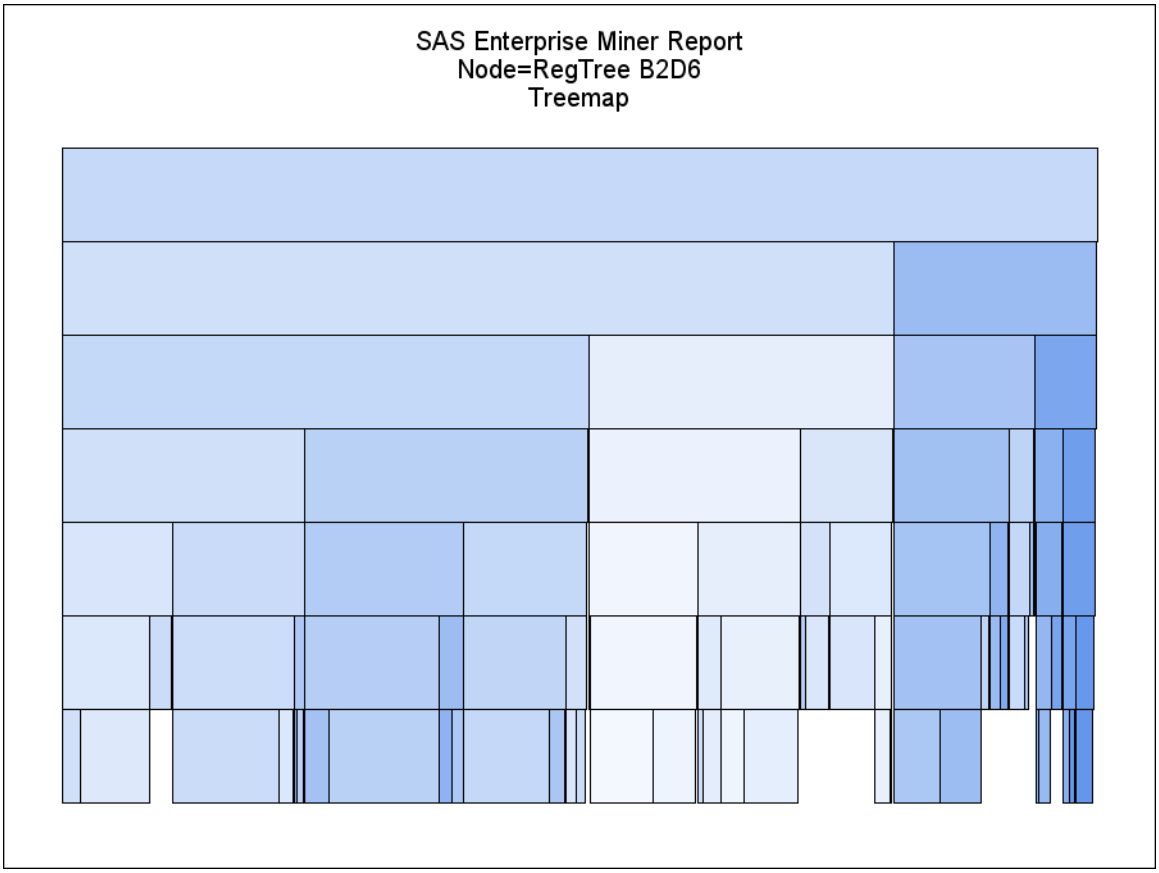
Role	Level	Frequency Count	Name
TARGET	INTERVAL	1	median_house_value
INPUT	INTERVAL	8	households housing_median_age latitude longitude median_income population total_bedrooms total_rooms
INPUT	NOMINAL	1	ocean_proximity
ID	INTERVAL	1	_dataobs_

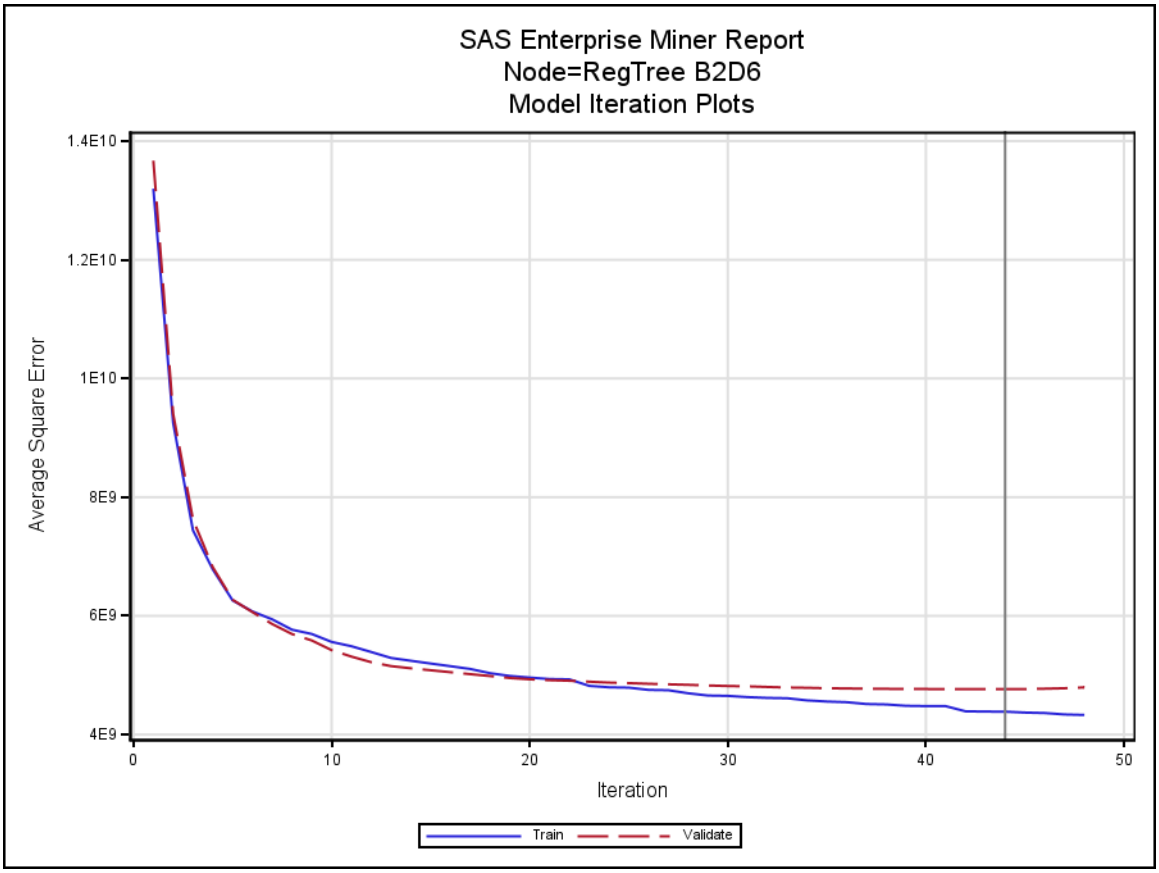
Node=RegTree B2D6
Model Fit Statistics

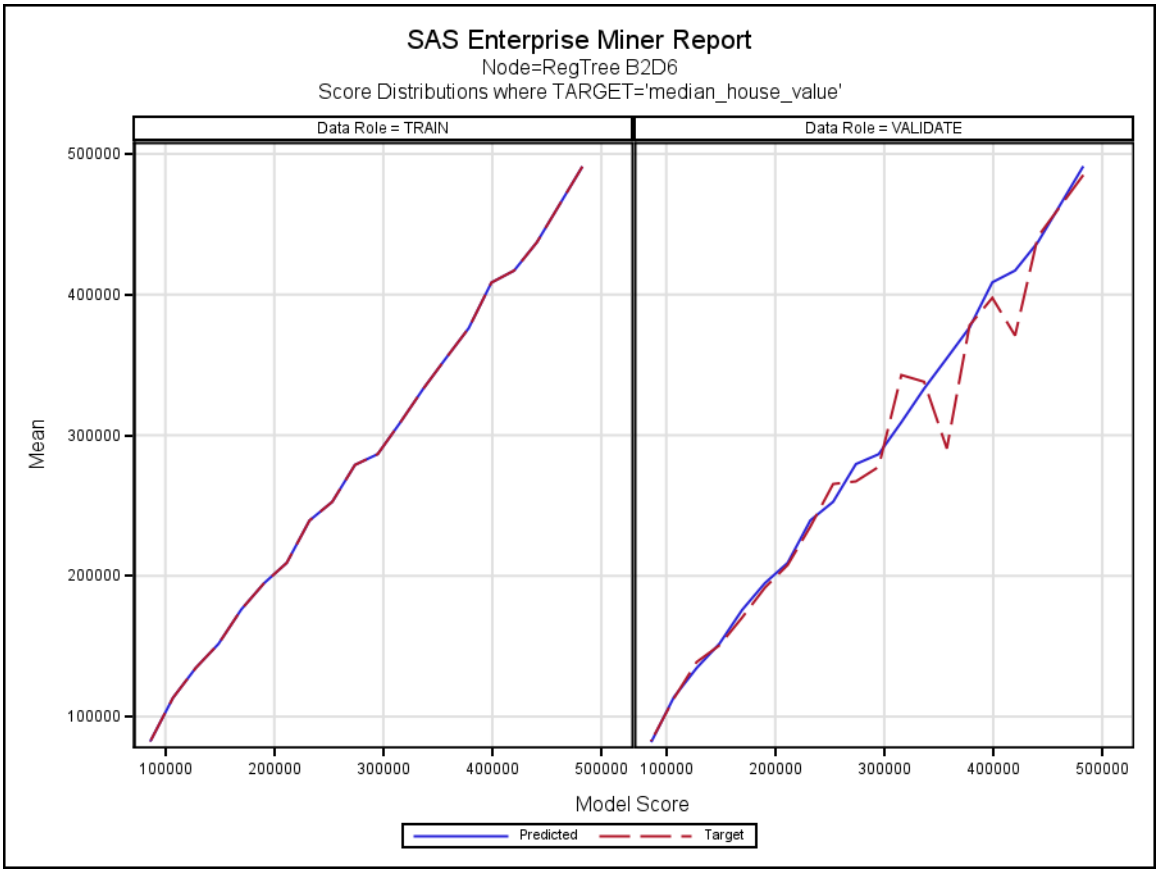
Target=median_house_value Target Label=' '

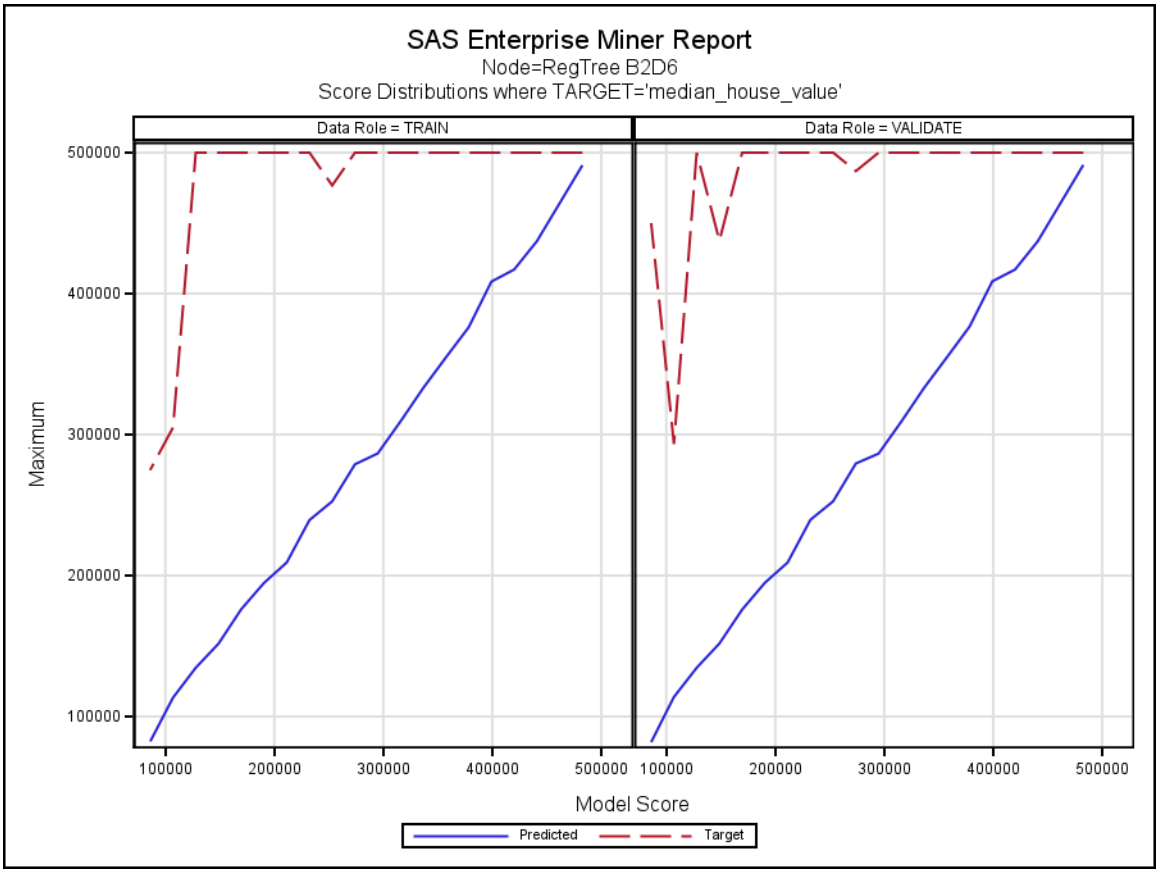
Label of Statistic	Train	Validation	Test
Sum of Frequencies	6605.00	4954.00	4953.00
Maximum Absolute Error	378861.50	374259.56	378861.50
Sum of Squared Errors	2.89692878E13	2.36050533E13	2.33639201E13
Average Squared Error	4385963336.45	4764847246.40	4717124991.06
Root Average Squared Error	66226.61	69027.87	68681.33
Divisor for ASE	6605.00	4954.00	4953.00
Total Degrees of Freedom	6605.00	.	.




SAS Enterprise Miner Report
Node=RegTree B2D6
Treemap


SAS Enterprise Miner Report
Node=RegTree B2D6
Model Iteration Plots
Average Square Error
1.4E10
1.2E10
1E10
8E9
6E9
4E9
0
10
20
30
40
50
Iteration
Train
Validate





Node=RegTree B2D6
Score Distributions

Target Variable=median_house_value Data Role=TRAIN

Range for Predicted	Mean Predicted	Max Predicted	Min Predicted	Mean Target	Max Target	Min Target
472177.719 - 493042.839	491025.34	493042.84	485231.51	491025.34	500001	383900
430447.480 - 451312.599	437200.35	437200.35	437200.35	437200.35	500001	171700
409582.360 - 430447.480	417086.14	417086.14	417086.14	417086.14	500001	242000
388717.240 - 409582.360	408509.82	409004.14	407363.91	408509.82	500001	241700
367852.120 - 388717.240	376100.94	387566.09	369685.92	376100.94	500001	187300
346987.000 - 367852.120	354770.92	354770.92	354770.92	354770.92	500001	200800
326121.880 - 346987.000	332980.79	339837.69	329685.46	332980.79	500001	166300
305256.760 - 326121.880	309265.69	309265.69	309265.69	309265.69	500001	108900
284391.640 - 305256.760	286602.19	302333.41	285408.85	286602.19	500001	87500
263526.520 - 284391.640	278919.47	281789.51	271807.41	278919.47	500001	112500
242661.400 - 263526.520	252661.11	253500.00	252338.46	252661.11	476700	67500
221796.281 - 242661.400	239363.57	240554.56	226600.00	239363.57	500001	72000
200931.161 - 221796.281	209282.71	209282.71	209282.71	209282.71	500001	93100
180066.041 - 200931.161	194766.47	199209.43	190364.06	194766.47	500001	28300
159200.921 - 180066.041	176120.01	179145.45	175266.68	176120.01	500001	38800
138335.801 - 159200.921	151691.34	157423.08	145190.53	151691.34	500001	22500
117470.681 - 138335.801	134612.61	138201.13	121139.50	134612.61	500001	46900
96605.561 - 117470.681	113392.94	116165.04	104090.38	113392.94	305000	41300
75740.441 - 96605.561	82363.72	89907.61	75740.44	82363.72	275000	14999

Target Variable=median_house_value Data Role=VALIDATE

Range for Predicted	Mean Predicted	Max Predicted	Min Predicted	Mean Target	Max Target	Min Target
472177.719 - 493042.839	491250.08	493042.84	485231.51	485009.85	500001	131300
430447.480 - 451312.599	437200.35	437200.35	437200.35	441092.89	500001	254400
409582.360 - 430447.480	417086.14	417086.14	417086.14	370850.50	500001	137500
388717.240 - 409582.360	408708.36	409004.14	407363.91	397692.02	500001	156000
367852.120 - 388717.240	376629.67	387566.09	369685.92	378178.82	500001	113300
346987.000 - 367852.120	354770.92	354770.92	354770.92	290528.62	500001	94600
326121.880 - 346987.000	333137.94	339837.69	329685.46	338023.33	500001	142000
305256.760 - 326121.880	309265.69	309265.69	309265.69	342818.56	500001	104300
284391.640 - 305256.760	286516.88	302333.41	285408.85	277506.61	500001	67500
263526.520 - 284391.640	279498.93	281789.51	271807.41	267201.82	486900	93800
242661.400 - 263526.520	252683.78	253500.00	252338.46	265343.38	500001	126800
221796.281 - 242661.400	239539.24	240554.56	226600.00	235126.80	500001	67500
200931.161 - 221796.281	209282.71	209282.71	209282.71	207726.71	500001	87500
180066.041 - 200931.161	194898.45	199209.43	190364.06	191666.36	500001	45000
159200.921 - 180066.041	175860.73	179145.45	175266.68	170437.86	500001	17500
138335.801 - 159200.921	151812.80	157423.08	145190.53	150646.40	437500	61300
117470.681 - 138335.801	134586.34	138201.13	121139.50	138802.35	500001	37500
96605.561 - 117470.681	113886.05	116165.04	104090.38	113018.21	293300	54900
75740.441 - 96605.561	81846.76	89907.61	75740.44	82611.31	450000	14999

SAS Enterprise Miner Report

Node=RegTree B2D4
Summary

Node id = Tree2
Node label = RegTree B2D4
Meta path = Ids2 => Part => Tree2
Notes =

Node=RegTree B2D4
Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	DecisionTree		Kass	Y		Pred	N	
AVG	Y		KassApply	BEFORE		Predict	Y	
AssessMeasure	PROFIT/LOSS		LeafSize	5		ProfitLoss	NONE	
AssessPercentage	0.25		Leafid	Y		RASE	N	
CV	N		Maxbranch	2		SampleMethod	RANDOM	
CVNIter	10		Maxdepth	4	6	SampleSeed	12345	
CVRepeat	1		MinCatSize	5		SampleSize	10000	
CVSeed	12345		MissingValue	USEINSEARCH		ShowNodeId	Y	
ClassColorBy	PERCENTCORRECT		NSubtree	1		ShowValid	Y	
Count	Y		NodeRole	SEGMENT		SigLevel	0.2	
CreateSample	DEFAULT		NodeSample	20000		SplitPrecision	4	
Criterion	DEFAULT		NominalCriterion	PROBCHISQ		Splitsize	.	
Depth	Y		Nrules	5		Subtree	ASSESSMENT	
Dummy	N		Nsurrs	0		Target	ALL	
Exhaustive	5000		NumInputs	1		ToolType	MODEL	
Freeze	N		NumSingleImp	5		TrainMode	BATCH	
ImportModel	N		ObsImportance	N		UseDecision	N	
ImportedTreeData			OrdinalCriterion	ENTROPY		UseMultipleTarget	N	
Inputs	N		PercentCorrect	N		UsePriors	N	
IntColorBy	AVG		Performance	DISK		UseVarOnce	N	
IntervalCriterion	PROBF		Precision	4		VarSelection	Y	

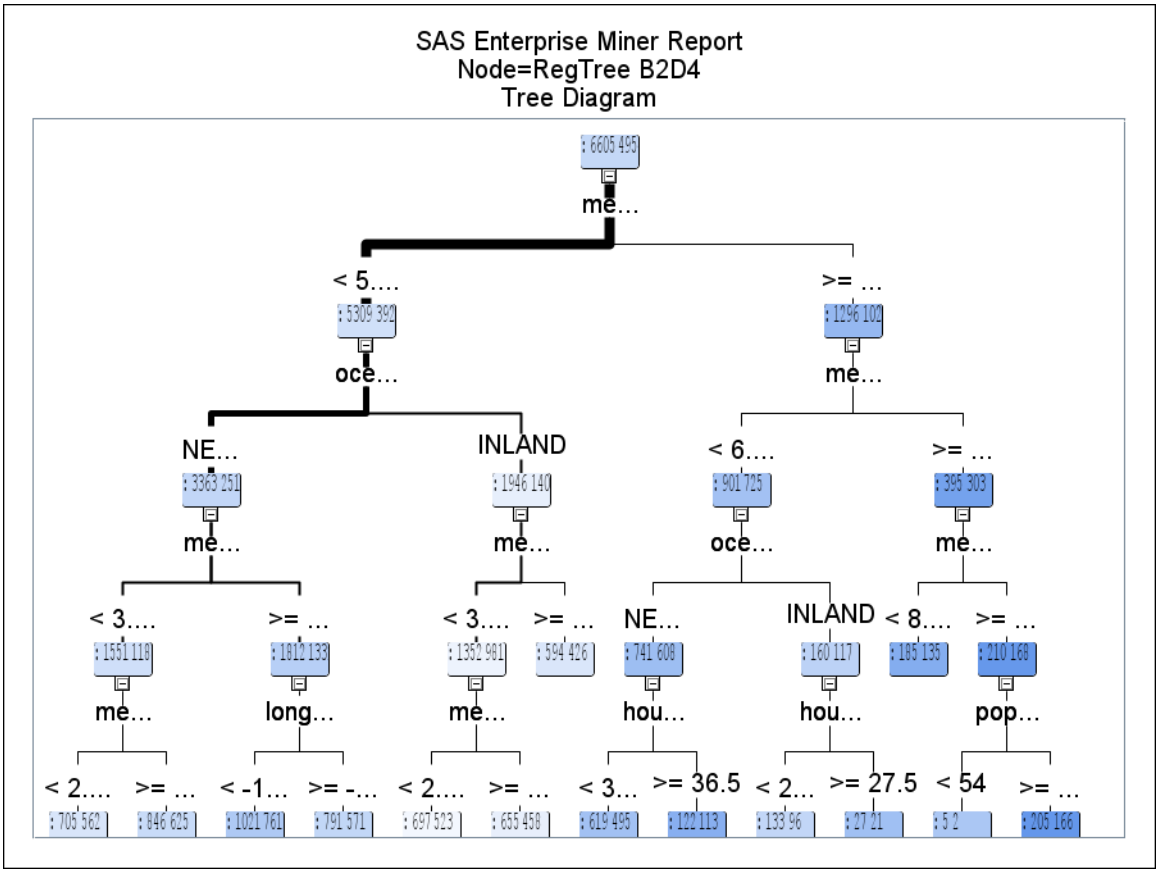
Node=RegTree B2D4
Variable Summary

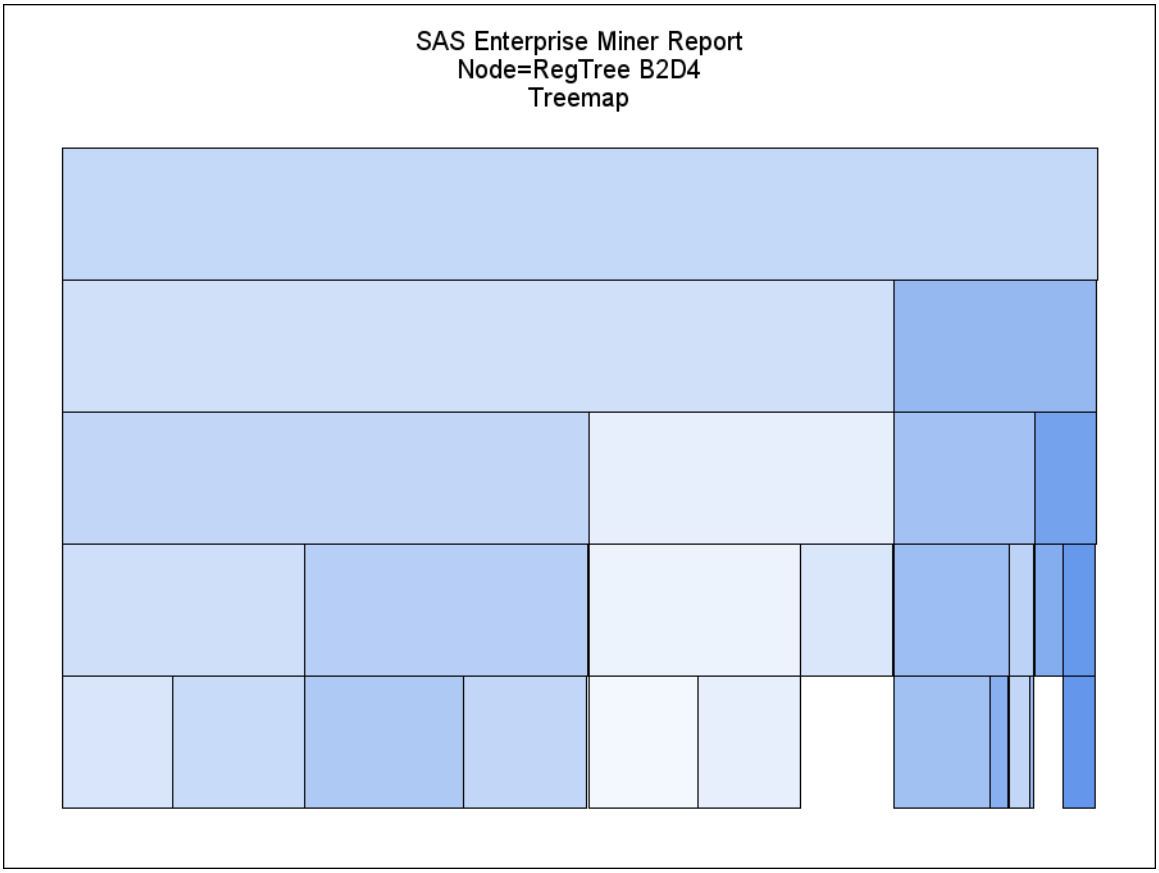
Role	Level	Frequency Count	Name
TARGET	INTERVAL	1	median_house_value
INPUT	INTERVAL	8	households housing_median_age latitude longitude median_income population total_bedrooms total_rooms
INPUT	NOMINAL	1	ocean_proximity
ID	INTERVAL	1	_dataobs_

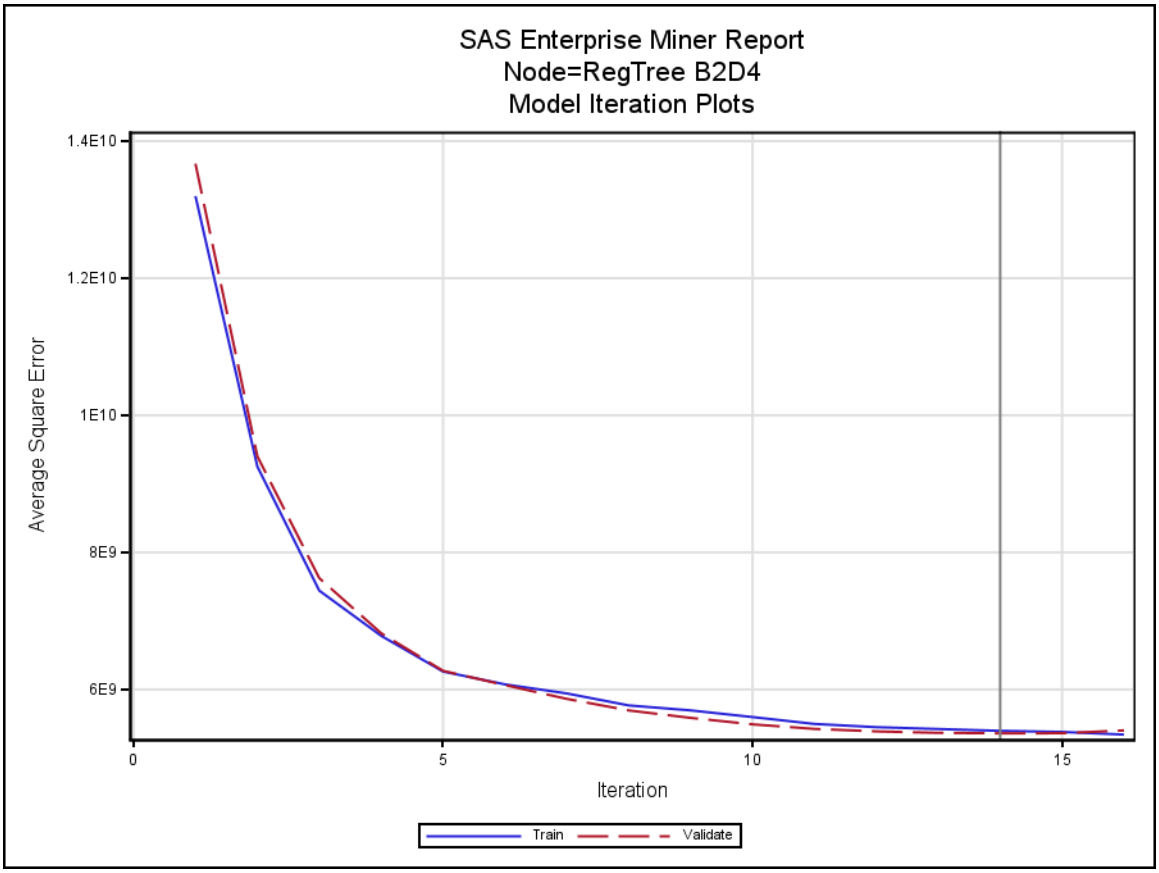
Node=RegTree B2D4
Model Fit Statistics

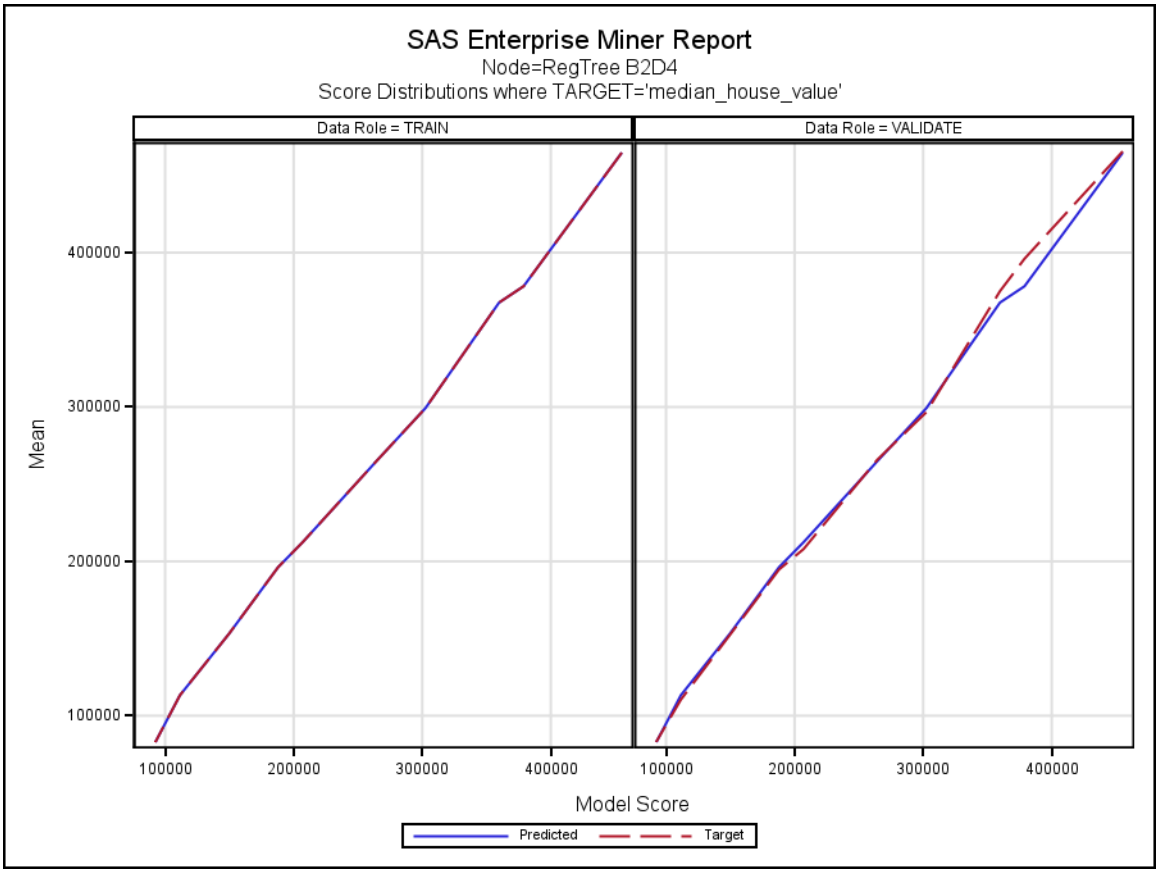
Target=median_house_value Target Label=' '

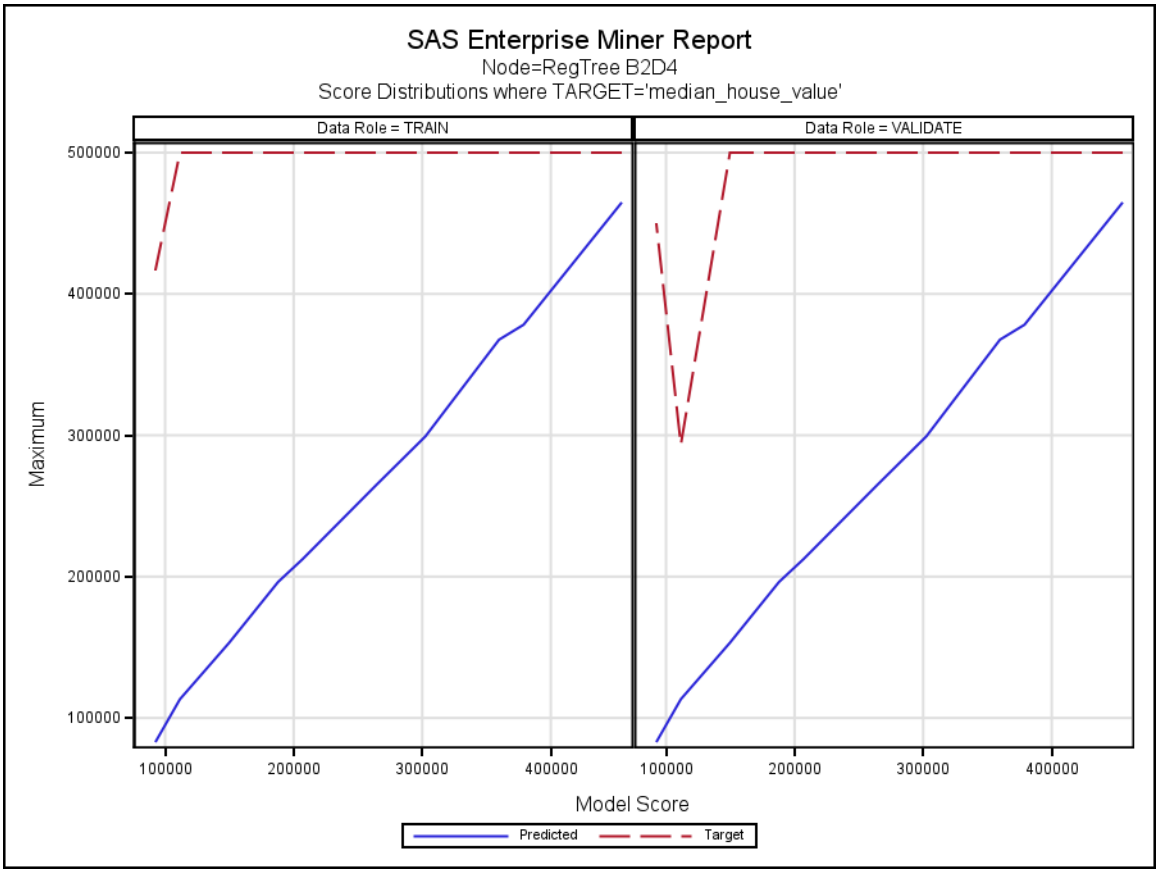
Label of Statistic	Train	Validation	Test
Sum of Frequencies	6605.00	4954.00	4953.00
Maximum Absolute Error	386759.47	367126.11	386759.47
Sum of Squared Errors	3.56431642E13	2.65528364E13	2.67221605E13
Average Squared Error	5396391252.18	5359878164.12	5395146469.85
Root Average Squared Error	73460.13	73211.19	73451.66
Divisor for ASE	6605.00	4954.00	4953.00
Total Degrees of Freedom	6605.00	.	.




SAS Enterprise Miner Report
Node=RegTree B2D4
Treemap


SAS Enterprise Miner Report
Node=RegTree B2D4
Model Iteration Plots
Average Square Error
1.4E10
1.2E10
1E10
8E9
6E9
0
5
10
15
Iteration
Train
Validate





Node=RegTree B2D4
Score Distributions

Target Variable=median_house_value Data Role=TRAIN

Range for Predicted	Mean Predicted	Max Predicted	Min Predicted	Mean Target	Max Target	Min Target
445685.325 - 464780.663	464780.66	464780.66	464780.66	464780.66	500001	187300
369303.969 - 388399.308	378233.69	378233.69	378233.69	378233.69	500001	67500
350208.630 - 369303.969	367645.22	367645.22	367645.22	367645.22	500001	174200
292922.614 - 312017.953	299227.29	302333.41	299091.81	299227.29	500001	136000
254731.936 - 273827.275	264856.96	273700.00	264813.65	264856.96	500001	73600
197445.920 - 216541.259	212418.30	213947.37	212161.20	212418.30	500001	71800
178350.581 - 197445.920	196227.08	196227.08	196227.08	196227.08	500001	22500
140159.903 - 159255.242	153083.22	155294.62	150458.59	153083.22	500001	38800
101969.225 - 121064.564	113241.53	113241.53	113241.53	113241.53	500001	40400
82873.887 - 101969.225	82873.89	82873.89	82873.89	82873.89	416700	14999

Target Variable=median_house_value Data Role=VALIDATE

Range for Predicted	Mean Predicted	Max Predicted	Min Predicted	Mean Target	Max Target	Min Target
445685.325 - 464780.663	464780.66	464780.66	464780.66	465633.21	500001	131300
369303.969 - 388399.308	378233.69	378233.69	378233.69	396061.75	500001	155600
350208.630 - 369303.969	367645.22	367645.22	367645.22	375040.91	500001	142000
292922.614 - 312017.953	299223.73	302333.41	299091.81	296628.15	500001	105400
254731.936 - 273827.275	264836.95	273700.00	264813.65	265781.44	500001	67500
197445.920 - 216541.259	212418.28	213947.37	212161.20	207963.28	500001	67500
178350.581 - 197445.920	196227.08	196227.08	196227.08	194701.45	500001	45000
140159.903 - 159255.242	153209.45	155294.62	150458.59	152371.46	500001	17500
101969.225 - 121064.564	113241.53	113241.53	113241.53	110364.63	293300	30000
82873.887 - 101969.225	82873.89	82873.89	82873.89	83119.50	450000	14999

SAS Enterprise Miner Report

Node=RegTree B2D2
Summary

Node id = Tree3
Node label = RegTree B2D2
Meta path = Ids2 => Part => Tree3
Notes =

Node=RegTree B2D2
Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	DecisionTree		Kass	Y		Pred	N	
AVG	Y		KassApply	BEFORE		Predict	Y	
AssessMeasure	PROFIT/LOSS		LeafSize	5		ProfitLoss	NONE	
AssessPercentage	0.25		Leafid	Y		RASE	N	
CV	N		Maxbranch	2		SampleMethod	RANDOM	
CVNIter	10		Maxdepth	2	6	SampleSeed	12345	
CVRepeat	1		MinCatSize	5		SampleSize	10000	
CVSeed	12345		MissingValue	USEINSEARCH		ShowNodeId	Y	
ClassColorBy	PERCENTCORRECT		NSubtree	1		ShowValid	Y	
Count	Y		NodeRole	SEGMENT		SigLevel	0.2	
CreateSample	DEFAULT		NodeSample	20000		SplitPrecision	4	
Criterion	DEFAULT		NominalCriterion	PROBCHISQ		Splitsize	.	
Depth	Y		Nrules	5		Subtree	ASSESSMENT	
Dummy	N		Nsurrs	0		Target	ALL	
Exhaustive	5000		NumInputs	1		ToolType	MODEL	
Freeze	N		NumSingleImp	5		TrainMode	BATCH	
ImportModel	N		ObsImportance	N		UseDecision	N	
ImportedTreeData			OrdinalCriterion	ENTROPY		UseMultipleTarget	N	
Inputs	N		PercentCorrect	N		UsePriors	N	
IntColorBy	AVG		Performance	DISK		UseVarOnce	N	
IntervalCriterion	PROBF		Precision	4		VarSelection	Y	

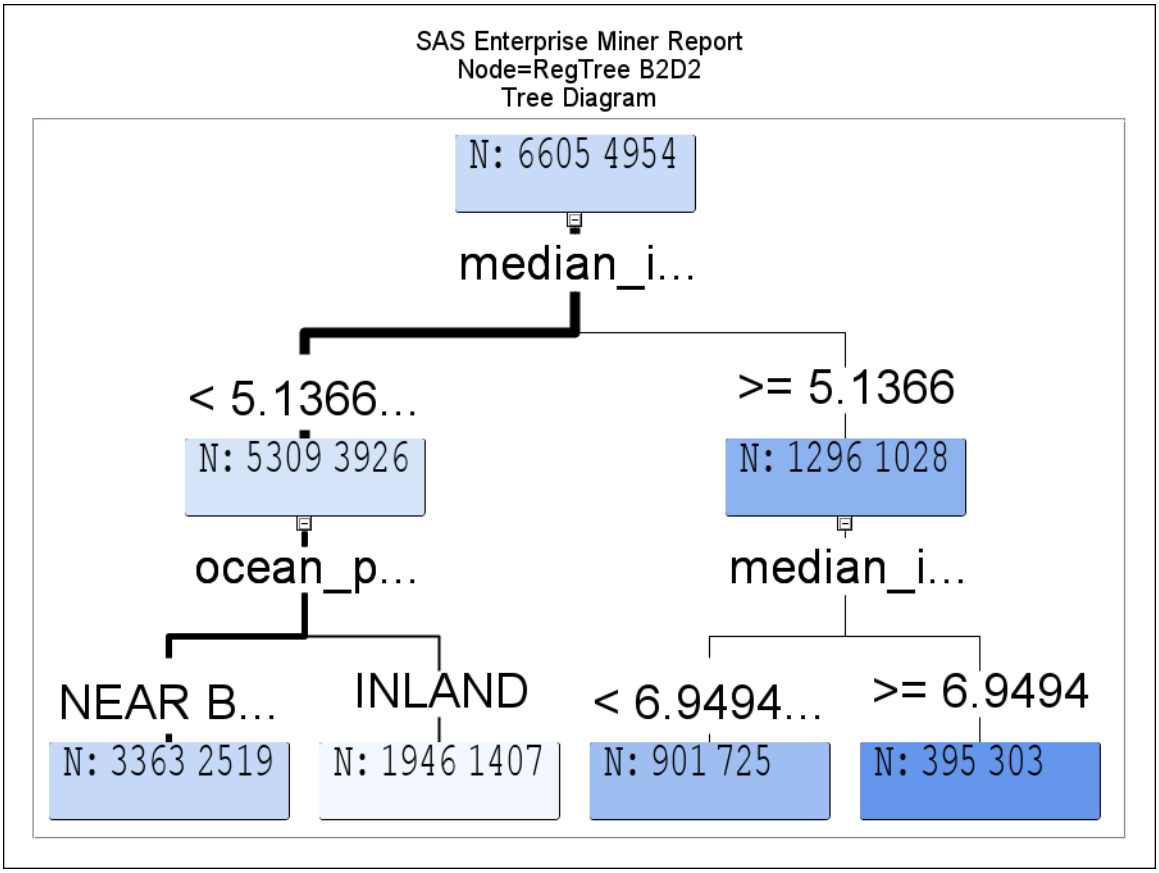
Node=RegTree B2D2
Variable Summary

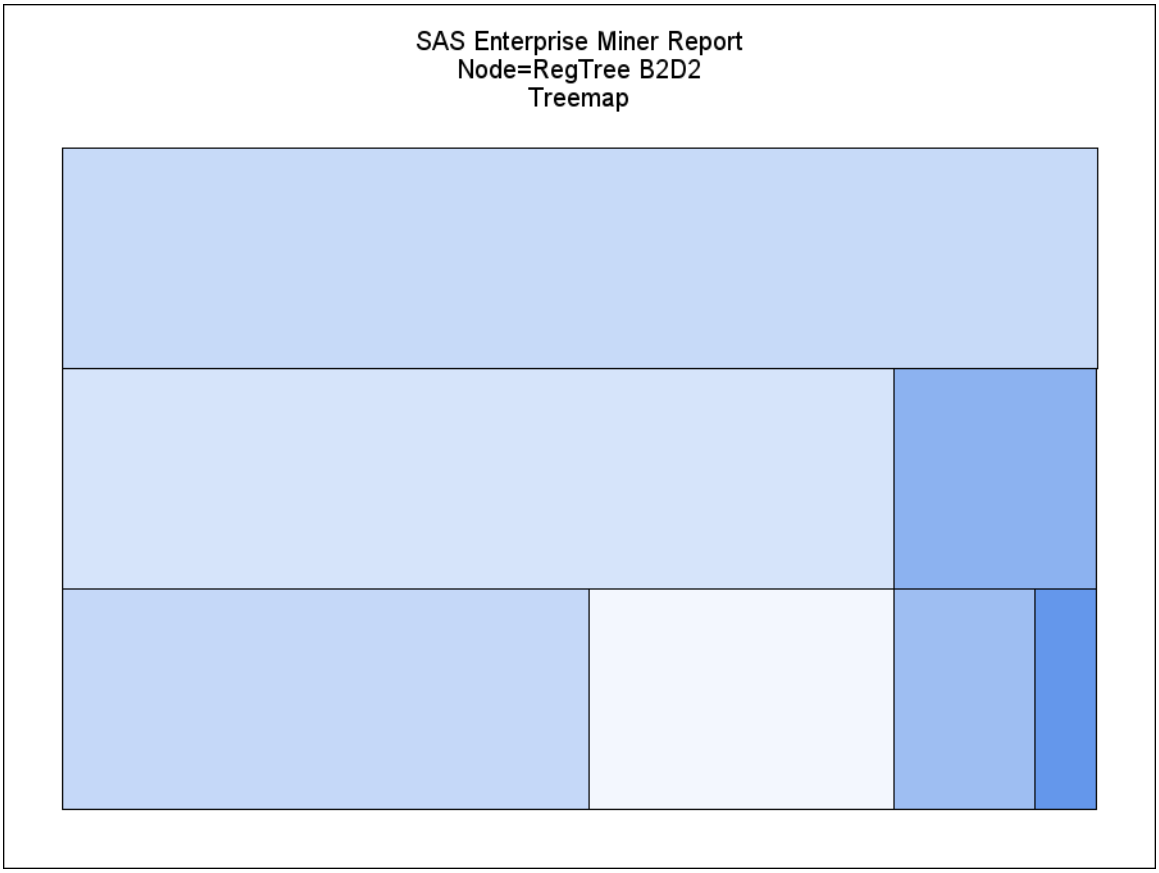
Role	Level	Frequency Count	Name
TARGET	INTERVAL	1	median_house_value
INPUT	INTERVAL	8	households housing_median_age latitude longitude median_income population total_bedrooms total_rooms
INPUT	NOMINAL	1	ocean_proximity
ID	INTERVAL	1	_dataobs_

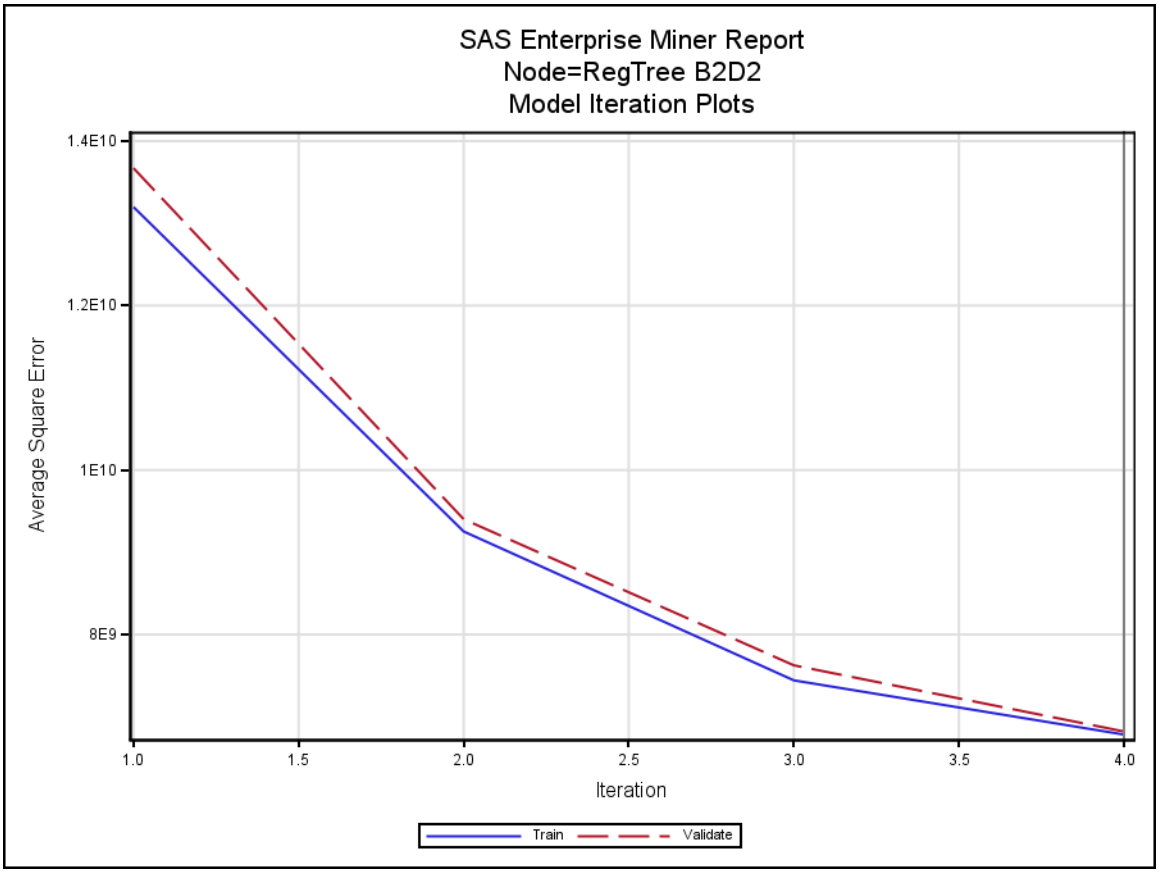
Node=RegTree B2D2
Model Fit Statistics

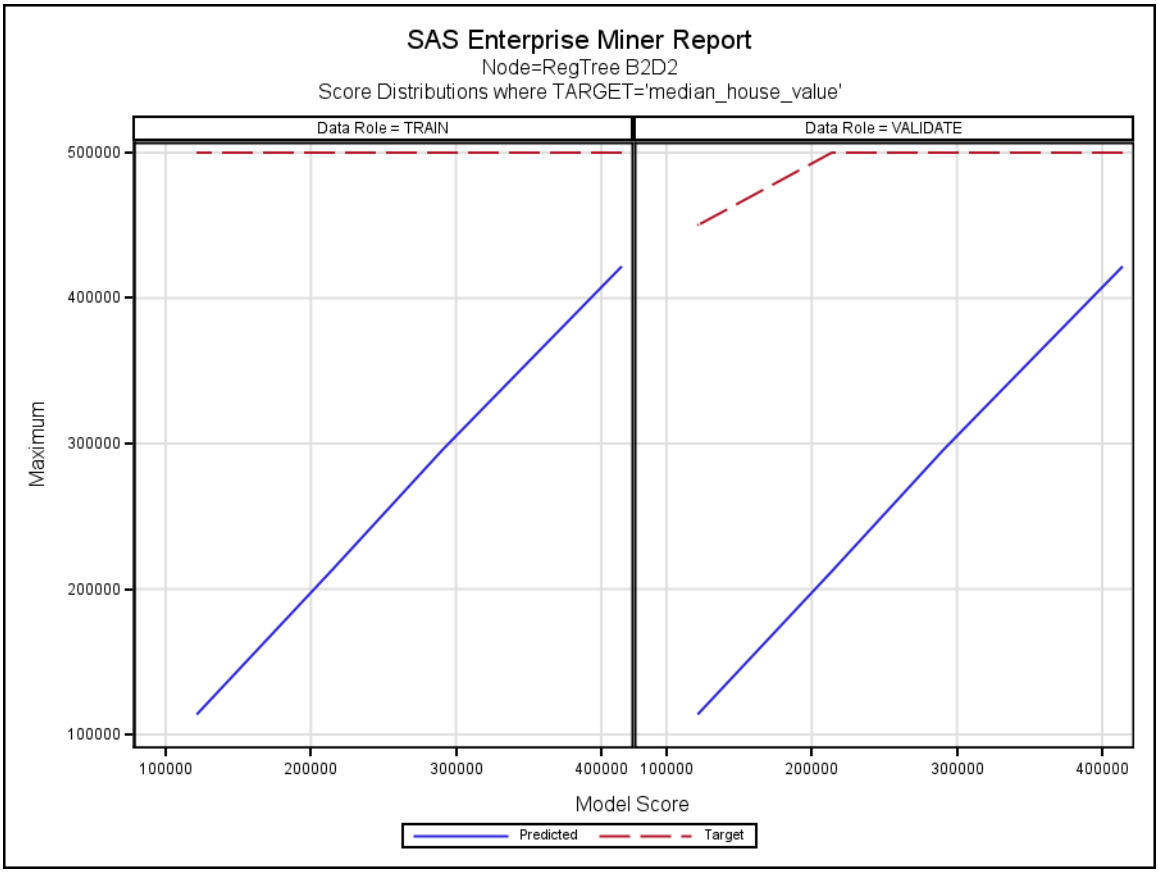
Target=median_house_value Target Label=' '

Label of Statistic	Train	Validation	Test
Sum of Frequencies	6605.00	4954.00	4953.00
Maximum Absolute Error	386276.08	336275.08	386276.08
Sum of Squared Errors	4.48015218E13	3.37893677E13	3.32942578E13
Average Squared Error	6782970748.61	6820623269.20	6722038733.52
Root Average Squared Error	82358.79	82587.06	81988.04
Divisor for ASE	6605.00	4954.00	4953.00
Total Degrees of Freedom	6605.00	.	.




SAS Enterprise Miner Report
Node=RegTree B2D2
Treemap


SAS Enterprise Miner Report
Node=RegTree B2D2
Model Iteration Plots
Average Square Error
1.4E10
1.2E10
1E10
8E9
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Iteration
Train
Validate



Node=RegTree B2D2
Score Distributions

Target Variable=median_house_value Data Role=TRAIN

Range for Predicted	Mean Predicted	Max Predicted	Min Predicted	Mean Target	Max Target	Min Target
406422.146 - 421827.263	421827.26	421827.26	421827.26	421827.26	500001	67500
283181.211 - 298586.328	295902.94	295902.94	295902.94	295902.94	500001	71800
206155.626 - 221560.743	212216.79	212216.79	212216.79	212216.79	500001	22500
113724.924 - 129130.041	113724.92	113724.92	113724.92	113724.92	500001	14999

Target Variable=median_house_value Data Role=VALIDATE

Range for Predicted	Mean Predicted	Max Predicted	Min Predicted	Mean Target	Max Target	Min Target
406422.146 - 421827.263	421827.26	421827.26	421827.26	433088.94	500001	112500
283181.211 - 298586.328	295902.94	295902.94	295902.94	295963.66	500001	97200
206155.626 - 221560.743	212216.79	212216.79	212216.79	210687.12	500001	17500
113724.924 - 129130.041	113724.92	113724.92	113724.92	111894.38	450000	14999

SAS Enterprise Miner Report

Node=RegTree B3D6
Summary

Node id = Tree4
Node label = RegTree B3D6
Meta path = Ids2 => Part => Tree4
Notes =

Node=RegTree B3D6
Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	DecisionTree		Kass	Y		Pred	N	
AVG	Y		KassApply	BEFORE		Predict	Y	
AssessMeasure	PROFIT/LOSS		LeafSize	5		ProfitLoss	NONE	
AssessPercentage	0.25		Leafid	Y		RASE	N	
CV	N		Maxbranch	3	2	SampleMethod	RANDOM	
CVNIter	10		Maxdepth	6		SampleSeed	12345	
CVRepeat	1		MinCatSize	5		SampleSize	10000	
CVSeed	12345		MissingValue	USEINSEARCH		ShowNodeId	Y	
ClassColorBy	PERCENTCORRECT		NSubtree	1		ShowValid	Y	
Count	Y		NodeRole	SEGMENT		SigLevel	0.2	
CreateSample	DEFAULT		NodeSample	20000		SplitPrecision	4	
Criterion	DEFAULT		NominalCriterion	PROBCHISQ		Splitsize	.	
Depth	Y		Nrules	5		Subtree	ASSESSMENT	
Dummy	N		Nsurrs	0		Target	ALL	
Exhaustive	5000		NumInputs	1		ToolType	MODEL	
Freeze	N		NumSingleImp	5		TrainMode	BATCH	
ImportModel	N		ObsImportance	N		UseDecision	N	
ImportedTreeData			OrdinalCriterion	ENTROPY		UseMultipleTarget	N	
Inputs	N		PercentCorrect	N		UsePriors	N	
IntColorBy	AVG		Performance	DISK		UseVarOnce	N	
IntervalCriterion	PROBF		Precision	4		VarSelection	Y	

Node=RegTree B3D6
Variable Summary

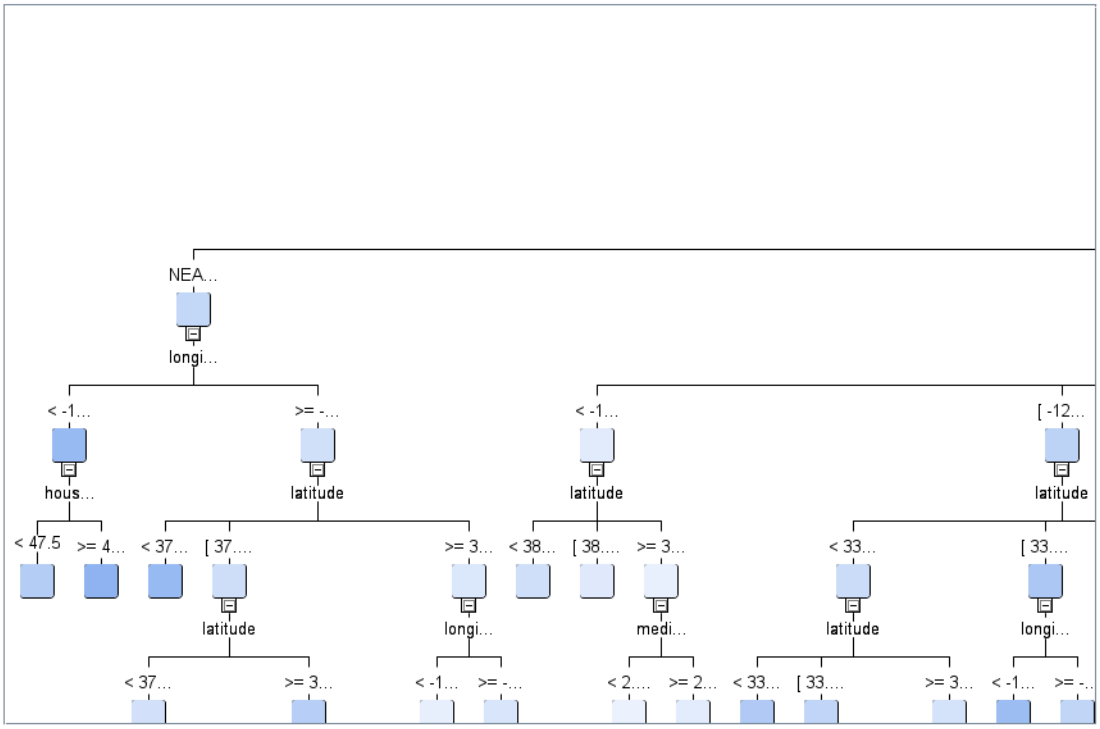
Role	Level	Frequency Count	Name
TARGET	INTERVAL	1	median_house_value
INPUT	INTERVAL	8	households housing_median_age latitude longitude median_income population total_bedrooms total_rooms
INPUT	NOMINAL	1	ocean_proximity
ID	INTERVAL	1	_dataobs_

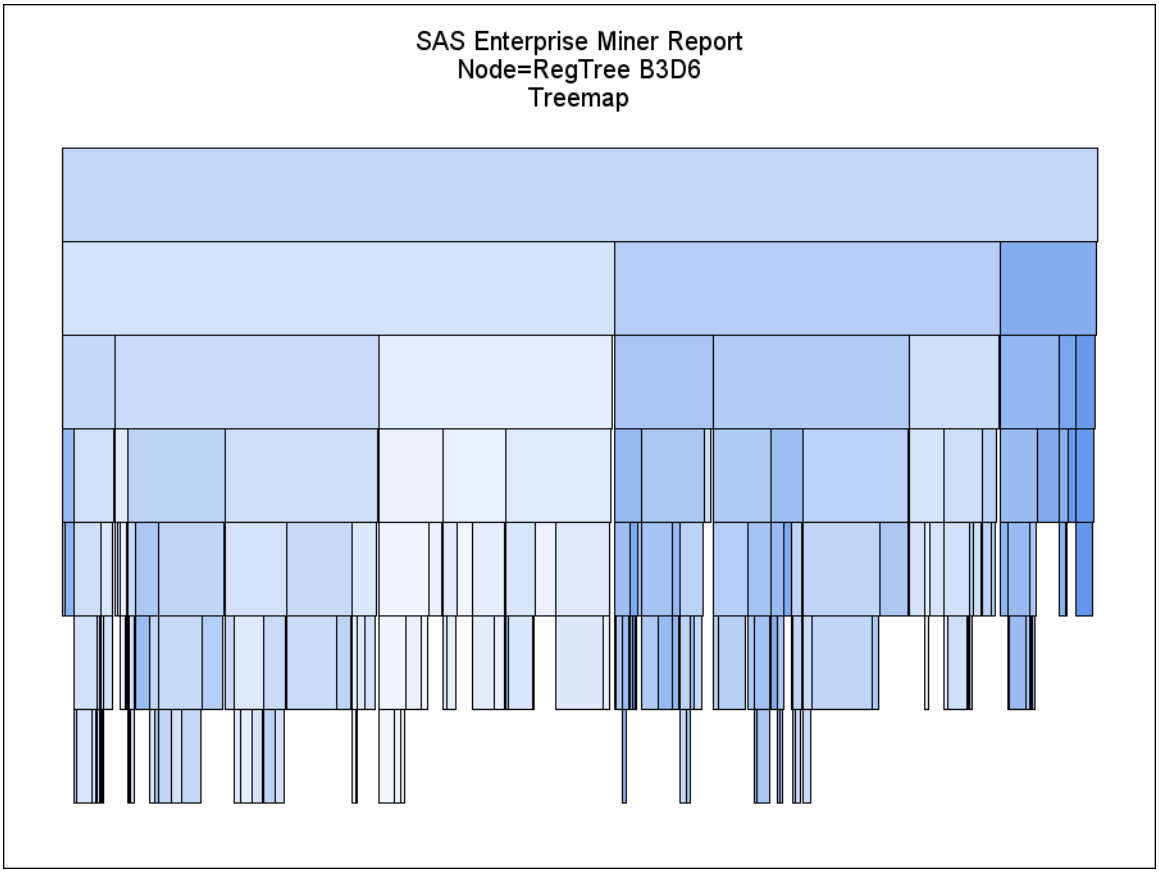
Node=RegTree B3D6
Model Fit Statistics

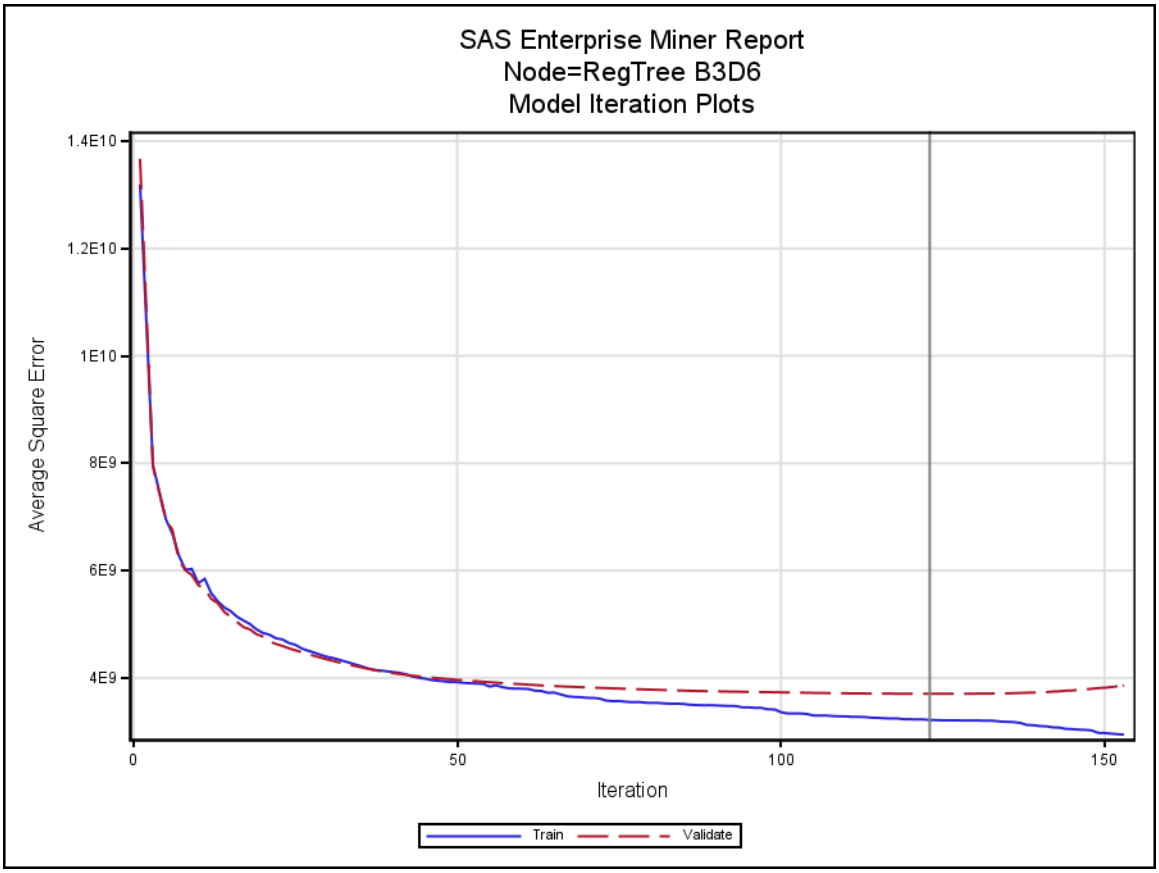
Target=median_house_value Target Label=' '

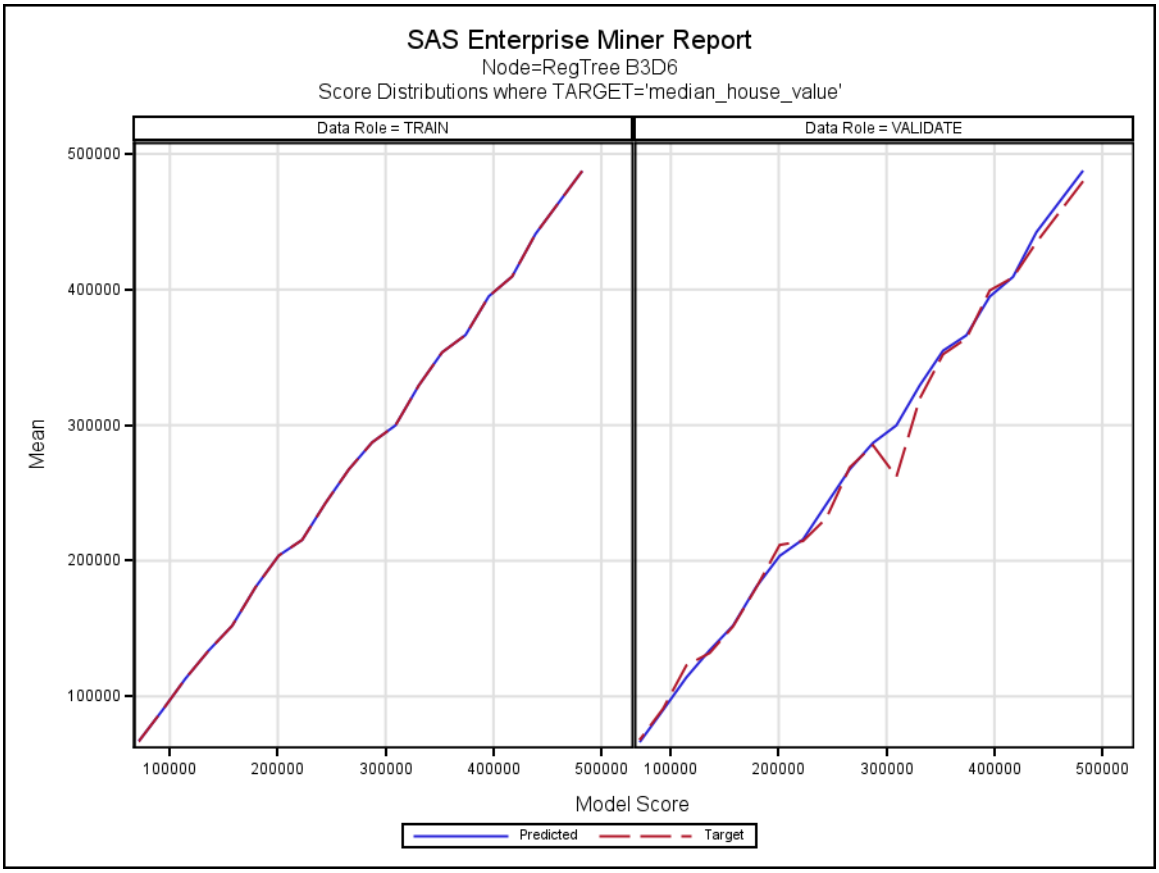
Label of Statistic	Train	Validation	Test
Sum of Frequencies	6605.00	4954.00	4953.00
Maximum Absolute Error	372213.22	361742.84	365509.22
Sum of Squared Errors	2.12321742E13	1.83457147E13	1.81088905E13
Average Squared Error	3214560819.21	3703212490.82	3656145880.26
Root Average Squared Error	56697.10	60854.03	60466.07
Divisor for ASE	6605.00	4954.00	4953.00
Total Degrees of Freedom	6605.00	.	.

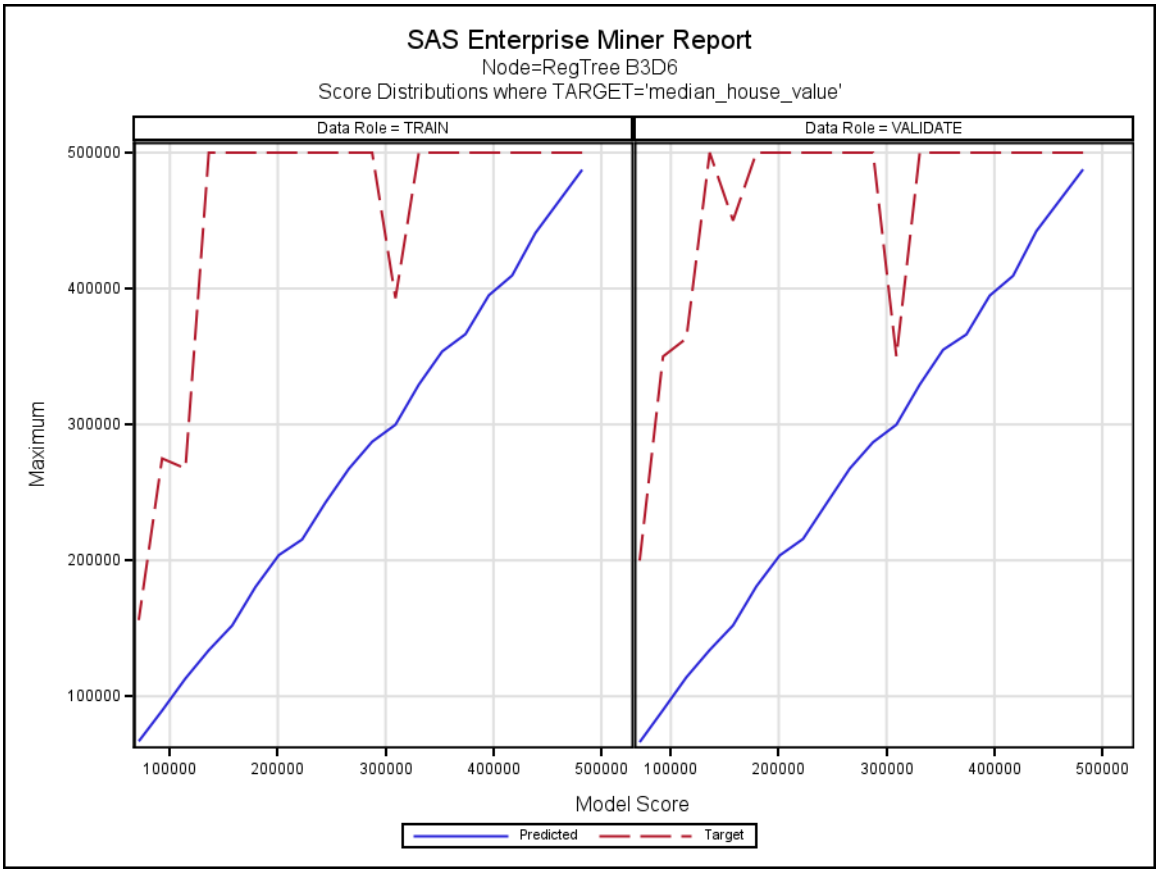
SAS Enterprise Miner Report
Node=RegTree B3D6
Tree Diagram




SAS Enterprise Miner Report
Node=RegTree B3D6
Treemap


SAS Enterprise Miner Report
Node=RegTree B3D6
Model Iteration Plots
Average Square Error
1.4E10
1.2E10
1E10
8E9
6E9
4E9
0
50
100
150
Iteration
Train
Validate





Node=RegTree B3D6
Score Distributions

Target Variable=median_house_value Data Role=TRAIN

Range for Predicted	Mean Predicted	Max Predicted	Min Predicted	Mean Target	Max Target	Min Target
471406.452 - 493042.839	487411.12	493042.84	475500.65	487411.13	500001	281900
428133.678 - 449770.065	440983.58	444641.83	437325.33	440983.58	500001	326600
406497.290 - 428133.678	409624.56	417086.14	408300.30	409624.56	500001	188400
384860.903 - 406497.290	395025.59	397126.84	388334.29	395025.59	500001	67500
363224.516 - 384860.903	366445.10	378600.17	364286.41	366445.10	500001	187300
341588.129 - 363224.516	353707.41	358674.55	346862.50	353707.41	500001	166300
319951.741 - 341588.129	329319.75	335812.50	321033.37	329319.75	500001	55000
298315.354 - 319951.741	299850.00	299850.00	299850.00	299850.00	392900	204100
276678.967 - 298315.354	287291.08	296875.00	277593.58	287291.08	500001	67500
255042.580 - 276678.967	267340.42	273700.00	258237.50	267340.42	500001	87500
233406.192 - 255042.580	242557.15	254375.61	236406.29	242557.15	500001	28300
211769.805 - 233406.192	215359.42	231587.18	211900.00	215359.42	500001	71800
190133.418 - 211769.805	203870.71	209209.30	192085.71	203870.71	500001	47500
168497.031 - 190133.418	180445.78	188133.33	168616.67	180445.78	500001	46900
146860.643 - 168497.031	152033.55	157280.02	149291.55	152033.55	500001	50000
125224.256 - 146860.643	133902.28	145803.03	125776.90	133902.28	500001	22500
103587.869 - 125224.256	113119.40	124580.00	106808.57	113119.40	267600	45000
81951.482 - 103587.869	89485.49	103044.74	82988.64	89485.49	275000	32500
60315.094 - 81951.482	66896.73	79988.89	60315.09	66896.73	156300	14999

Target Variable=median_house_value Data Role=VALIDATE

Range for Predicted	Mean Predicted	Max Predicted	Min Predicted	Mean Target	Max Target	Min Target
471406.452 - 493042.839	487679.88	493042.84	475500.65	479836.35	500001	131300
428133.678 - 449770.065	442290.10	444641.83	437325.33	434764.54	500001	262500
406497.290 - 428133.678	409356.92	417086.14	408300.30	408970.28	500001	137500
384860.903 - 406497.290	394823.99	397126.84	388334.29	399316.42	500001	143300
363224.516 - 384860.903	366253.95	378600.17	364286.41	363951.16	500001	140100
341588.129 - 363224.516	354911.49	358674.55	346862.50	352244.72	500001	159600
319951.741 - 341588.129	329322.10	335812.50	321033.37	319541.64	500001	67500
298315.354 - 319951.741	299850.00	299850.00	299850.00	261920.00	349900	159900
276678.967 - 298315.354	286924.18	296875.00	277593.58	285120.83	500001	17500
255042.580 - 276678.967	267479.98	273700.00	258237.50	268855.62	500001	81800
233406.192 - 255042.580	241736.08	254375.61	236406.29	231426.52	500001	37500
211769.805 - 233406.192	215771.39	231587.18	211900.00	214549.01	500001	67500
190133.418 - 211769.805	203637.87	209209.30	192085.71	211679.79	500001	93800
168497.031 - 190133.418	180893.87	188133.33	168616.67	180375.59	500001	45000
146860.643 - 168497.031	152162.29	157280.02	149291.55	151598.90	450000	64400
125224.256 - 146860.643	134075.16	145803.03	125776.90	132008.43	500001	55000
103587.869 - 125224.256	113985.41	124580.00	106808.57	122823.45	363300	55000
81951.482 - 103587.869	89910.48	103044.74	82988.64	90910.80	350000	30000
60315.094 - 81951.482	66275.51	79988.89	60315.09	67949.57	200000	14999

SAS Enterprise Miner Report

Node=Model Comparison (4)
Summary

Node id = MdlComp4
Node label = Model Comparison (4)
Meta path = Ids2 => Part => Neural2 => MdlComp4
Notes =

Node=Model Comparison (4)
Properties

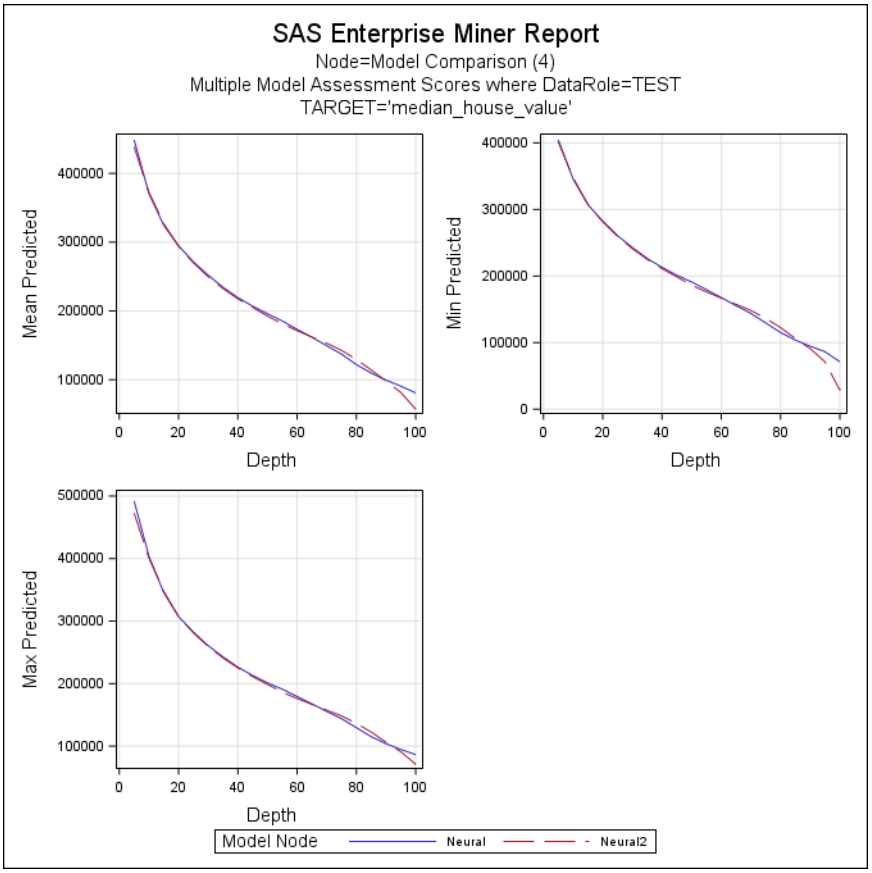
Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	ModelCompare		NumberOfReportedLevels	1E-6		SelectionData	DEFAULT	
AssessAllTargetLevels	N		NumberofBins	20		SelectionDepth	10	
DecileBin	20		ProfitEpsilon	1E-6		SelectionTable	TRAIN	TABLE
HPCriteria	DEFAULT		RecomputeAssess	N		StatisticUsed	_VASE_	
LiftEpsilon	1E-6		RocChart	Y		TargetLabel		
ModelCriteria	Valid: Average Squared Error		RocEpsilon	0.01		TargetName	median_house_value	
ModelDescription	5HU		RoiEpsilon	1E-6		classViyaCriteria	DEFAULT	
ModelId	Neural2		ScoreDistBin	20		intervalViyaCriteria	DEFAULT	
NormalizeReportingVariables	Y		SelectionCriteria	DEFAULT				

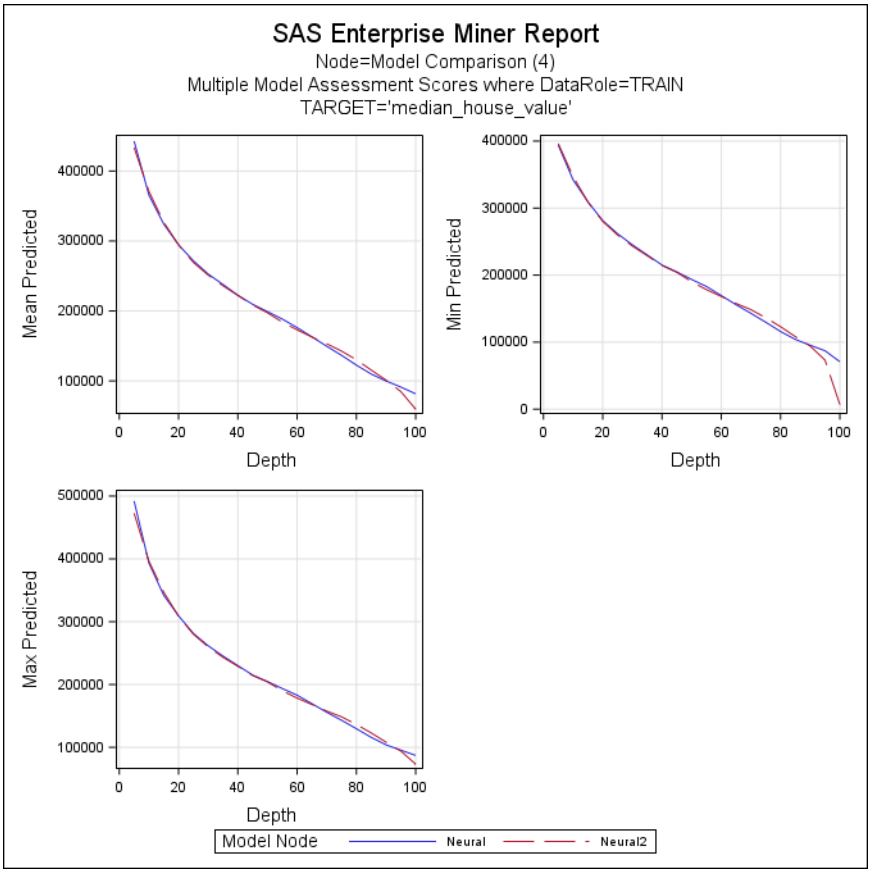
Node=Model Comparison (4)
Variable Summary

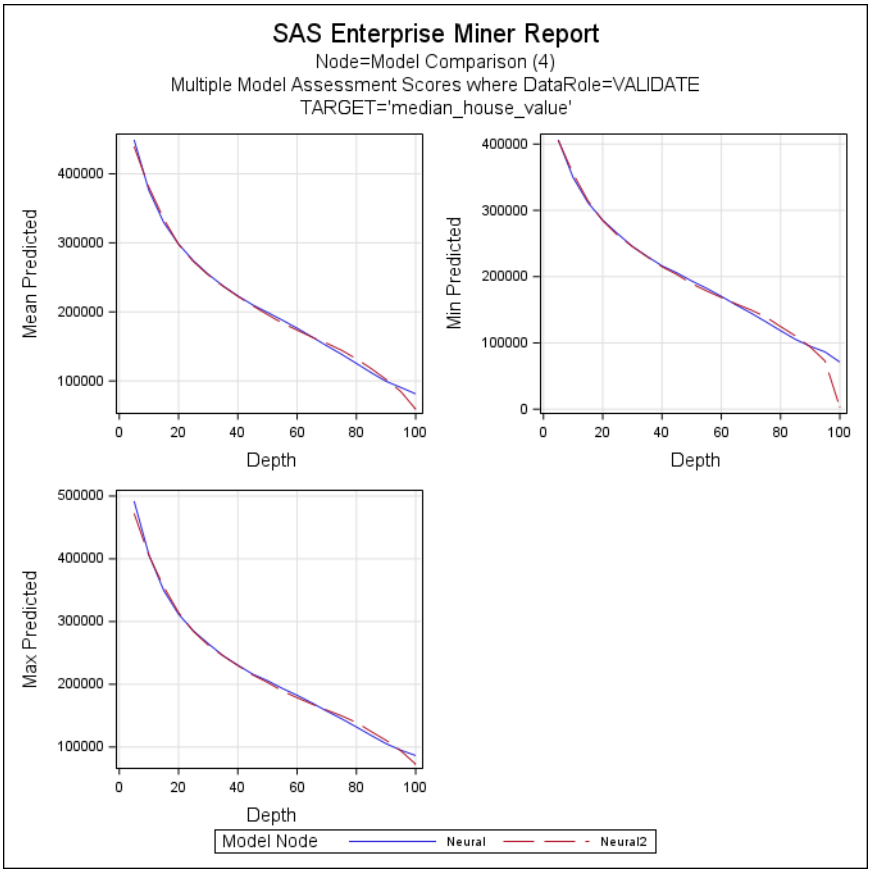
Role	Level	Frequency Count	Name
TARGET	INTERVAL	1	median_house_value

Node=Model Comparison (4)
Fit Statistics Table

Selected Model	Predecessor Node	Model Node	Model Description	Target	Target Label	Selection Criterion: Valid: Average Squared Error	Train: Average Squared Error
Y	Neural2	Neural2	5HU	median_house_value		3633790314.4	3727167485.1
	Neural	Neural	3HU	median_house_value		3858989033.0	4104222196.1


SAS Enterprise Miner Report
Node=Model Comparison (4)
Multiple Model Assessment Scores where DataRole=TEST
TARGET='median_house_value'
Mean Predicted
Min Predicted
Max Predicted
Depth
Model Node
Neural
Neural2


SAS Enterprise Miner Report
Node=Model Comparison (4)
Multiple Model Assessment Scores where DataRole=TRAIN
TARGET='median_house_value'
Mean Predicted
Min Predicted
Max Predicted
Depth
Model Node
Neural
Neural2


SAS Enterprise Miner Report
Node=Model Comparison (4)
Multiple Model Assessment Scores where DataRole=VALIDATE
TARGET='median_house_value'
Mean Predicted
Min Predicted
Max Predicted
Depth
Model Node
Neural
Neural2

SAS Enterprise Miner Report

Node=MLR ModelComparison Summary

Node id = MdlComp2
Node label = MLR ModelComparison
Meta path = Ids2 => Part => Reg => MdlComp2
Notes =

Node=MLR ModelComparison Properties

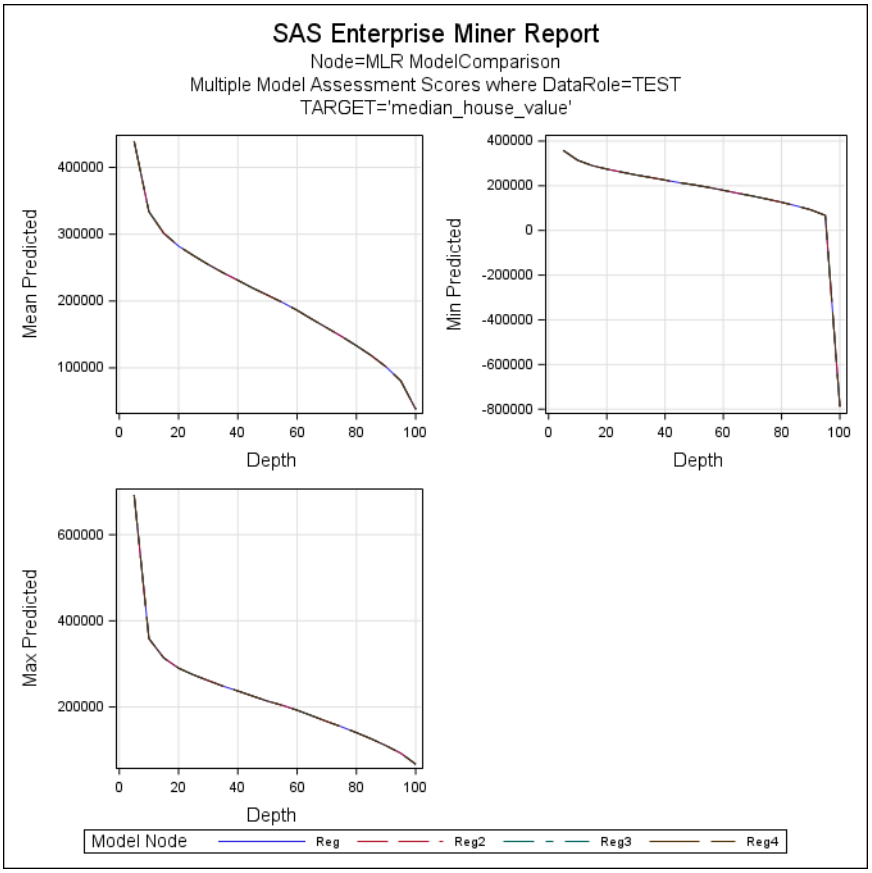
Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	ModelCompare		NumberOfReportedLevels	1E-6		SelectionData	DEFAULT	
AssessAllTargetLevels	N		NumberofBins	20		SelectionDepth	10	
DecileBin	20		ProfitEpsilon	1E-6		SelectionTable	TRAIN	TABLE
HPCriteria	DEFAULT		RecomputeAssess	N		StatisticUsed	_VASE_	
LiftEpsilon	1E-6		RocChart	Y		TargetLabel		
ModelCriteria	Valid: Average Squared Error		RocEpsilon	0.01		TargetName	median_house_value	
ModelDescription	Exhaustive Regression		RoiEpsilon	1E-6		classViyaCriteria	DEFAULT	
ModelId	Reg		ScoreDistBin	20		intervalViyaCriteria	DEFAULT	
NormalizeReportingVariables	Y		SelectionCriteria	DEFAULT				

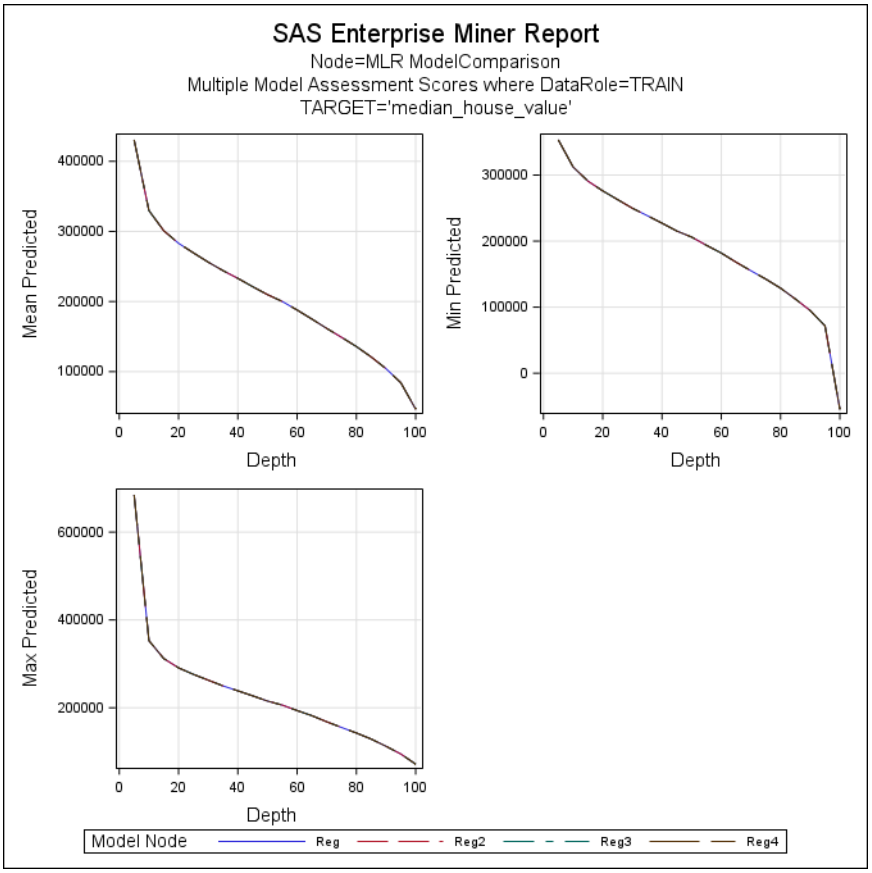
Node=MLR ModelComparison Variable Summary

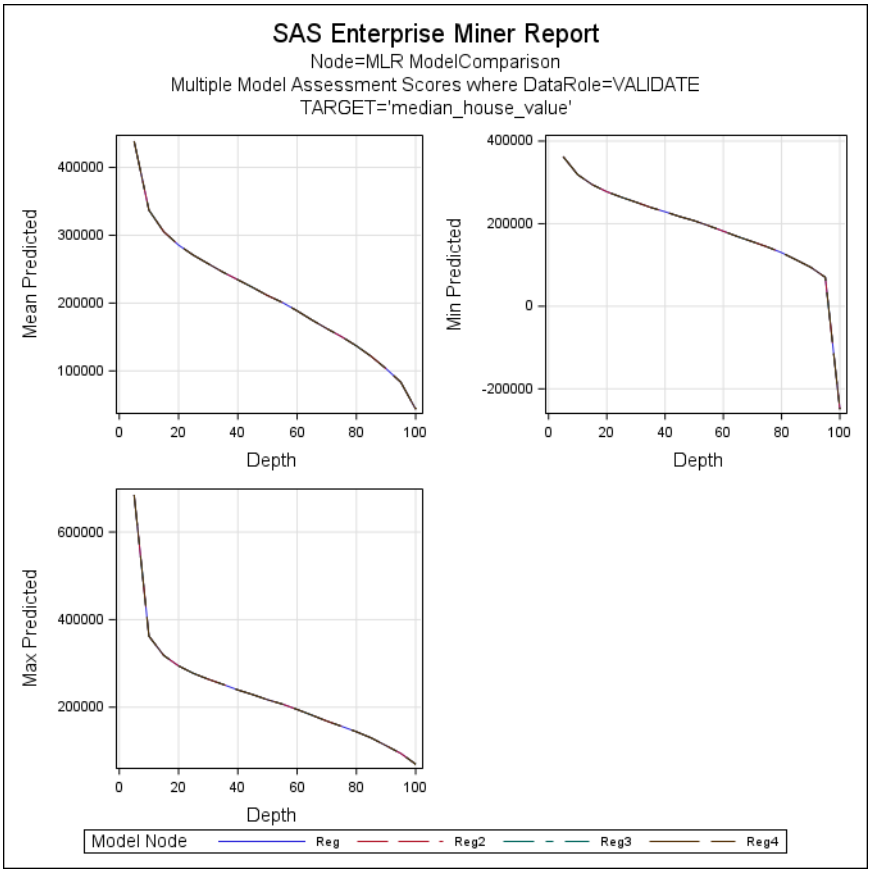
Role	Level	Frequency Count	Name
TARGET	INTERVAL	1	median_house_value

Node=MLR ModelComparison Fit Statistics Table

Selected Model	Predecessor Node	Model Node	Model Description	Target Variable	Target Label	Selection Criterion: Valid: Average Squared Error	Train: Average Squared Error
Y	Reg	Reg	Exhaustive Regression	median_house_value		4798889268.6	4899002783
	Reg2	Reg2	Forward Regression	median_house_value		4798889268.6	4899002783
	Reg3	Reg3	Backward Regression	median_house_value		4798889268.6	4899002783
	Reg4	Reg4	Stepwise Regression	median_house_value		4798889268.6	4899002783


SAS Enterprise Miner Report
Node=MLR ModelComparison
Multiple Model Assessment Scores where DataRole=TEST
TARGET='median_house_value'
Mean Predicted
Min Predicted
Max Predicted
Depth
Model Node
Reg
Reg2
Reg3
Reg4


SAS Enterprise Miner Report
Node=MLR ModelComparison
Multiple Model Assessment Scores where DataRole=TRAIN
TARGET='median_house_value'
Mean Predicted
Min Predicted
Max Predicted
Depth
Model Node
Reg
Reg2
Reg3
Reg4


SAS Enterprise Miner Report
Node=MLR ModelComparison
Multiple Model Assessment Scores where DataRole=VALIDATE
TARGET='median_house_value'
Mean Predicted
Min Predicted
Max Predicted
Depth
Model Node
Reg
Reg2
Reg3
Reg4

SAS Enterprise Miner Report

Node=RegTree Model Comparison Summary

Node id = MdlComp
Node label = RegTree Model Comparison
Meta path = Ids2 => Part => Tree4 => MdlComp
Notes =

Node=RegTree Model Comparison Properties

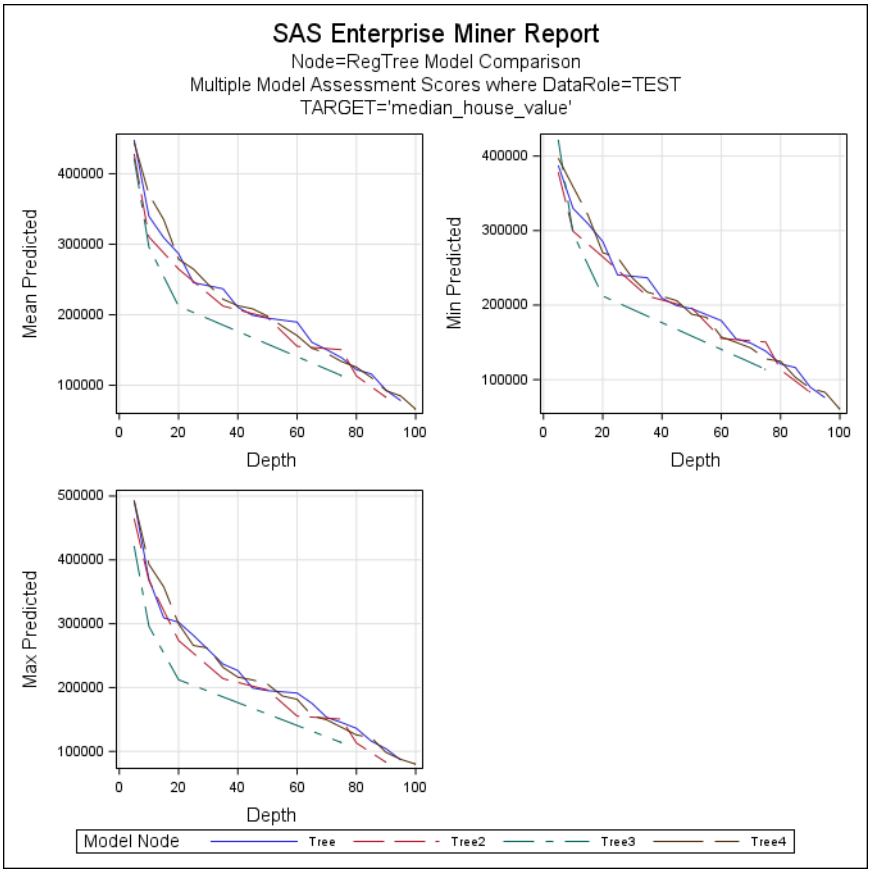
Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	ModelCompare		NumberOfReportedLevels	1E-6		SelectionData	DEFAULT	
AssessAllTargetLevels	N		NumberofBins	20		SelectionDepth	10	
DecileBin	20		ProfitEpsilon	1E-6		SelectionTable	TRAIN	TABLE
HPCriteria	DEFAULT		RecomputeAssess	N		StatisticUsed	_VASE_	
LiftEpsilon	1E-6		RocChart	Y		TargetLabel		
ModelCriteria	Valid: Average Squared Error		RocEpsilon	0.01		TargetName	median_house_value	
ModelDescription	RegTree B3D6		RoiEpsilon	1E-6		classViyaCriteria	DEFAULT	
ModelId	Tree4		ScoreDistBin	20		intervalViyaCriteria	DEFAULT	
NormalizeReportingVariables	Y		SelectionCriteria	DEFAULT				

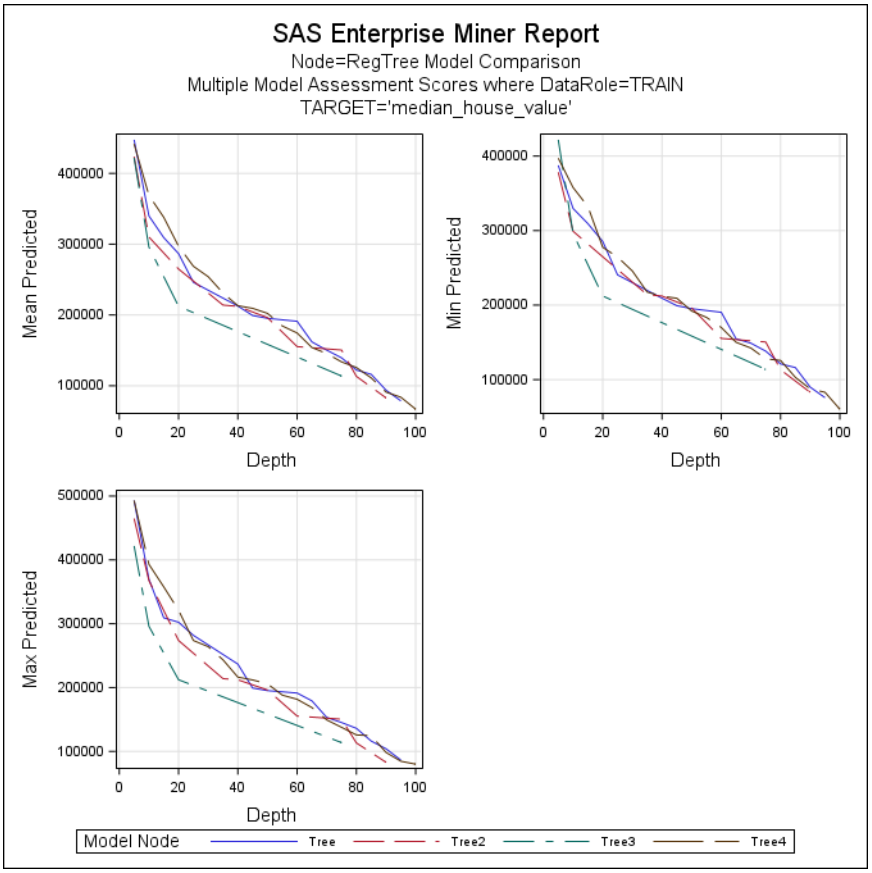
Node=RegTree Model Comparison Variable Summary

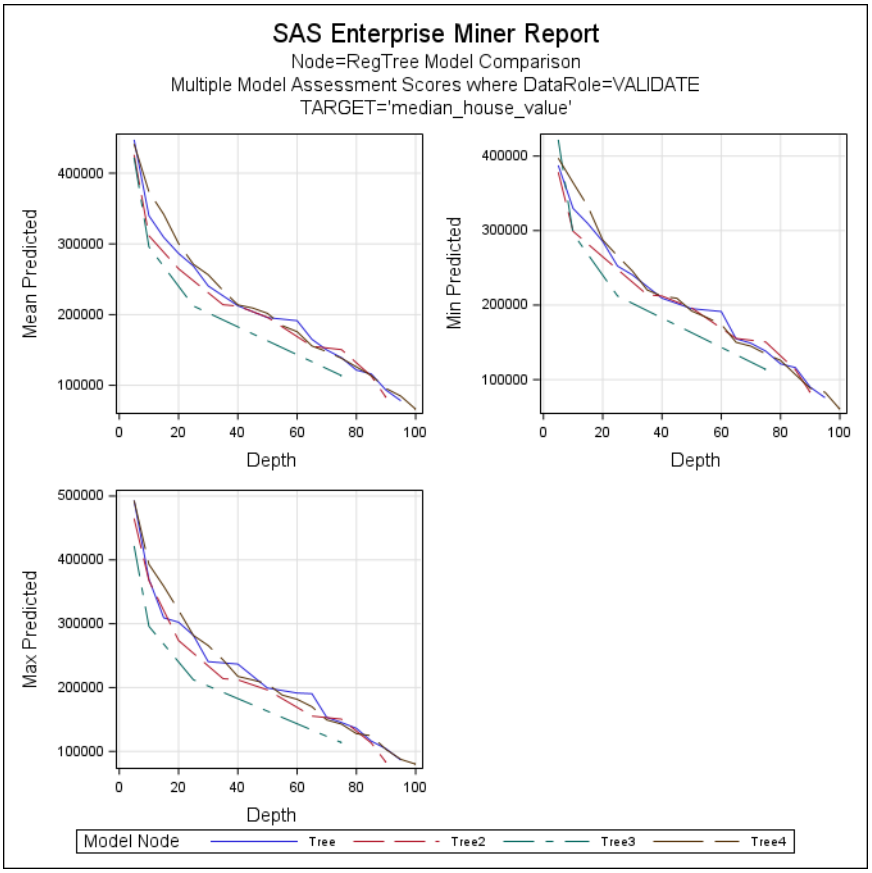
Role	Level	Frequency Count	Name
TARGET	INTERVAL	1	median_house_value

Node=RegTree Model Comparison Fit Statistics Table

Selected Model	Predecessor Node	Model Node	Model Description	Target Variable	Target Label	Selection Criterion: Valid: Average Squared Error	Train: Average Squared Error
Y	Tree4	Tree4	RegTree B3D6	median_house_value		3703212490.8	3214560819.2
	Tree	Tree	RegTree B2D6	median_house_value		4764847246.4	4385963336.4
	Tree2	Tree2	RegTree B2D4	median_house_value		5359878164.1	5396391252.2
	Tree3	Tree3	RegTree B2D2	median_house_value		6820623269.2	6782970748.6


SAS Enterprise Miner Report
Node=RegTree Model Comparison
Multiple Model Assessment Scores where DataRole=TEST
TARGET='median_house_value'
Mean Predicted
Min Predicted
Max Predicted
Depth
Model Node
Tree
Tree2
Tree3
Tree4


SAS Enterprise Miner Report
Node=RegTree Model Comparison
Multiple Model Assessment Scores where DataRole=TRAIN
TARGET='median_house_value'
Mean Predicted
Min Predicted
Max Predicted
Depth
Model Node
Tree
Tree2
Tree3
Tree4


SAS Enterprise Miner Report
Node=RegTree Model Comparison
Multiple Model Assessment Scores where DataRole=VALIDATE
TARGET='median_house_value'
Mean Predicted
Min Predicted
Max Predicted
Depth
Model Node
Tree
Tree2
Tree3
Tree4

SAS Enterprise Miner Report

Node=HOUSING_20_TRAIN
Summary

Node id = Ids
Node label = HOUSING_20_TRAIN
Meta path = Ids
Notes =

Node=HOUSING_20_TRAIN
Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	DataSource		DsCreatedBy	smupp		NBytes	394240	.
ApplyIntervalLevelLowerLimit	Y		DsId	housingtrain		NCols	11	.
ApplyMaxClassLevels	Y		DsModifiedBy	smupp		NObs	4128	.
ApplyMaxPercentMissing	Y		DsModifyDate	2049495372.7		NewTable		
CMeta	WORK.M33VX30B		DsSampleName			NewVariableRole	REJECT	
ComputeStatistics	N		DsSampleSize			OutputType	VIEW	
DBPassThrough	Y		DsSampleSizeType			Role	RAW	TRAIN
Data	HUDATA.HOUSING_20_TRAIN		DsScope	LOCAL		Sample	D	
DataSelection	DATASOURCE		IdentifyEmptyColumns	Y		SampleSizeObs	10000	
DataSource	housingtrain		IntervalLowerLimit	20		SampleSizePercent	20	
DataSourceRole	RAW		Library	HUDATA		SampleSizeType	PERCENT	
Description			MaxClassLevels	20		Scope	LOCAL	
DropMapVariables	Y		MaxPercentMissing	50		Segment		
DsCreateDate	2049495372.6		MetaAdvisor	BASIC		Table	HOUSING_20_TRAIN	

Node=HOUSING_20_TRAIN
Data Attributes

Attribute	Value	Attribute	Value	Attribute	Value
Data Name	HOUSING_20_TRAIN	Date Created	11Dec2024:00:14:49	Data Size	394240
Data Type	DATA	Date Modified	11Dec2024:00:14:49	Role	RAW
Data Label		Number Rows	4128	Segment	
Engine	BASE	Number Columns	11	Data Library	HUDATA

Node=HOUSING_20_TRAIN
Variables List

Name	Label	Role	Level	Type	Length	Format	Creator
dataobs	Observation Number	INPUT	INTERVAL	N	8		
households		INPUT	INTERVAL	N	8	BEST12.0	
housing_median_age		INPUT	INTERVAL	N	8	BEST12.0	
latitude		INPUT	INTERVAL	N	8	BEST12.0	
longitude		INPUT	INTERVAL	N	8	BEST12.0	
median_house_value		INPUT	INTERVAL	N	8	BEST12.0	
median_income		INPUT	INTERVAL	N	8	BEST12.0	
ocean_proximity		INPUT	NOMINAL	C	8	$8.	
population		INPUT	INTERVAL	N	8	BEST12.0	
total_bedrooms		INPUT	INTERVAL	N	8	BEST12.0	
total_rooms		INPUT	INTERVAL	N	8	BEST12.0	

SAS Enterprise Miner Report

Node=Predict Model Comparison Summary

Node id = MdlComp3
Node label = Predict Model Comparison
Meta path = Ids2 => Part => Neural2 => MdlComp4 => MdlComp3
Notes =

Node=Predict Model Comparison Properties

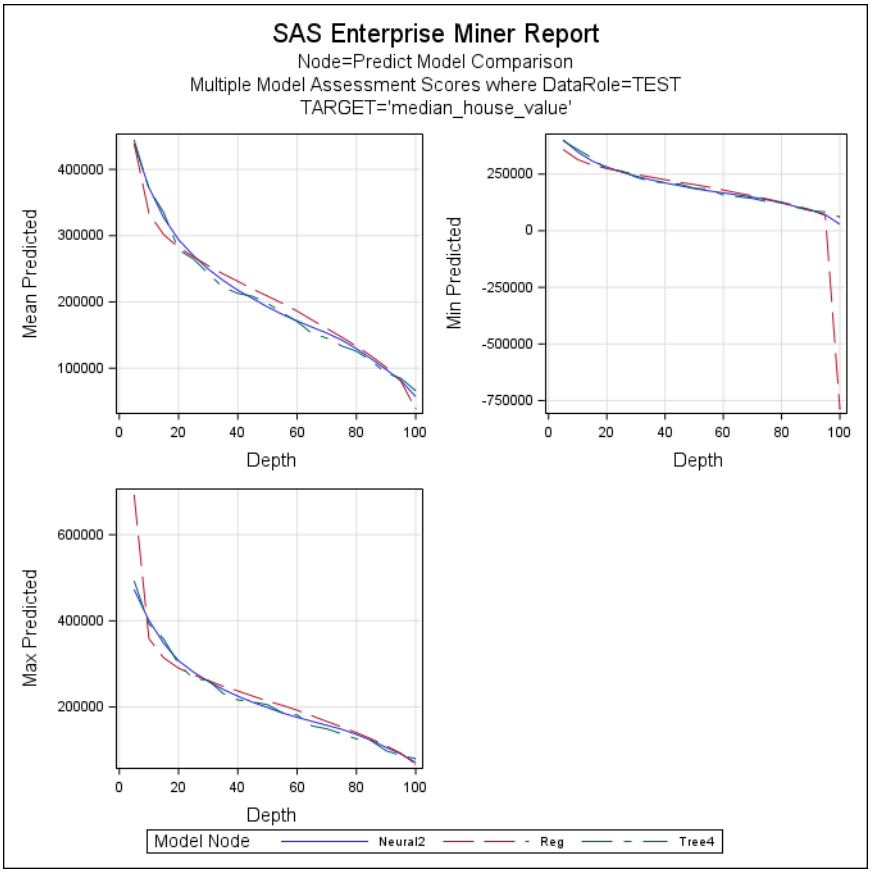
Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	ModelCompare		NumberOfReportedLevels	1E-6		SelectionData	DEFAULT	
AssessAllTargetLevels	N		NumberofBins	20		SelectionDepth	10	
DecileBin	20		ProfitEpsilon	1E-6		SelectionTable	TRAIN	TABLE
HPCriteria	DEFAULT		RecomputeAssess	N		StatisticUsed	_VASE_	
LiftEpsilon	1E-6		RocChart	Y		TargetLabel		
ModelCriteria	Valid: Average Squared Error		RocEpsilon	0.01		TargetName	median_house_value	
ModelDescription	5HU		RoiEpsilon	1E-6		classViyaCriteria	DEFAULT	
ModelId	Neural2		ScoreDistBin	20		intervalViyaCriteria	DEFAULT	
NormalizeReportingVariables	Y		SelectionCriteria	DEFAULT				

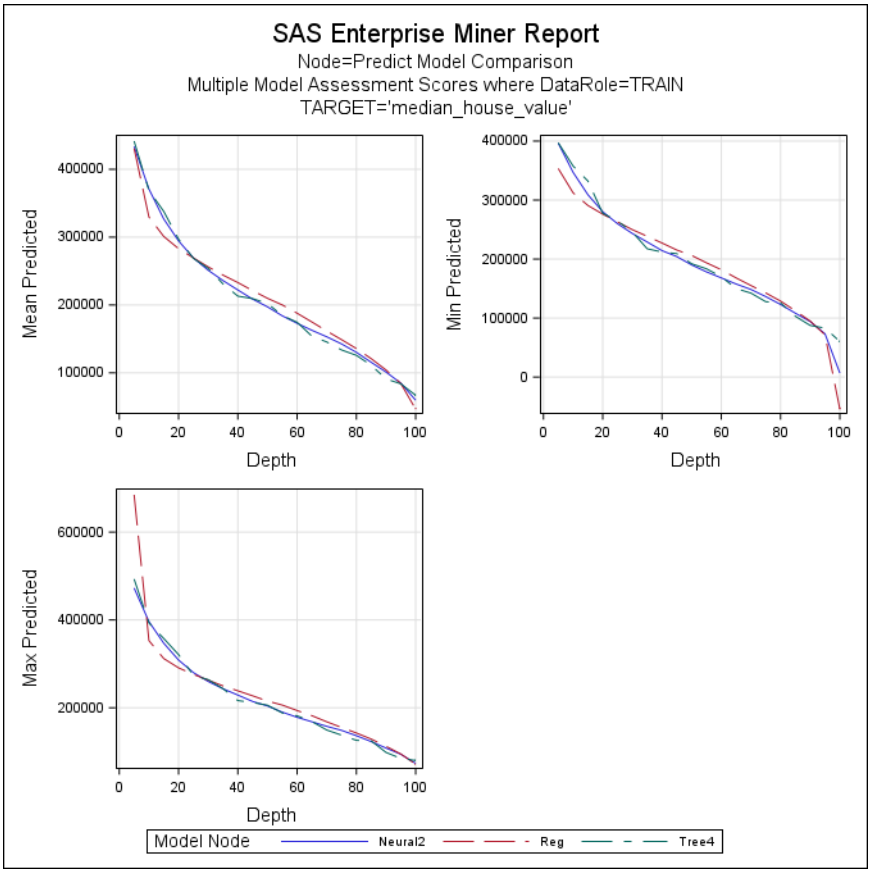
Node=Predict Model Comparison Variable Summary

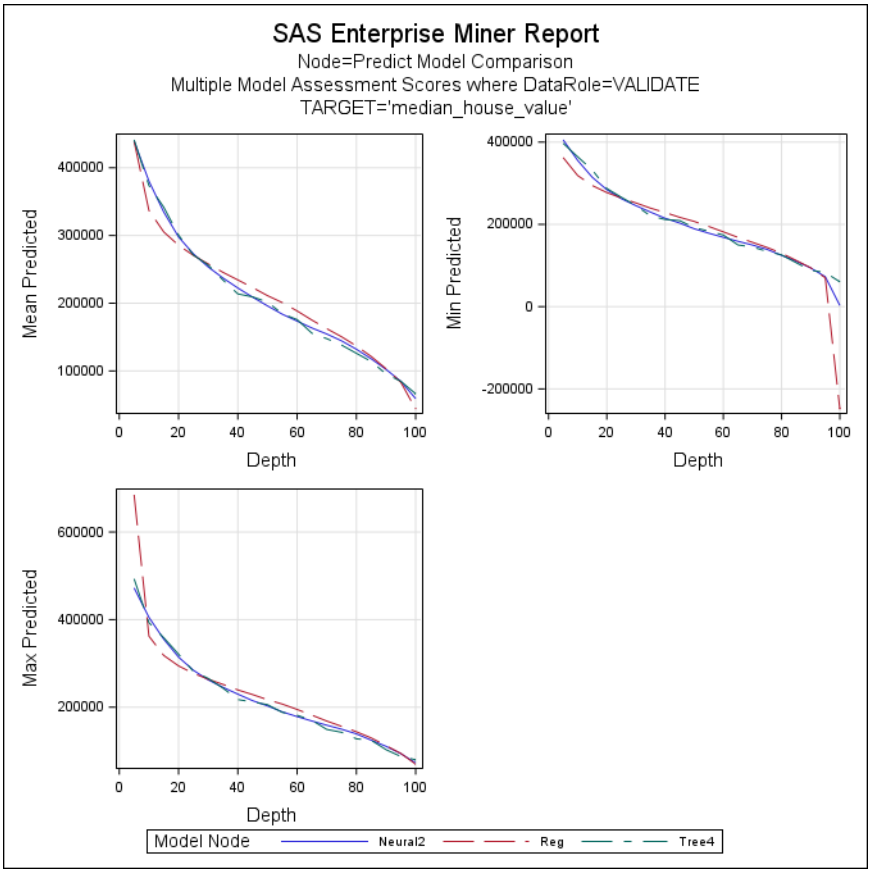
Role	Level	Frequency Count	Name
TARGET	INTERVAL	1	median_house_value

Node=Predict Model Comparison Fit Statistics Table

Selected Model	Predecessor Node	Model Node	Model Description	Target Variable	Target Label	Selection Criterion: Valid: Average Squared Error	Train: Average Squared Error
Y	MdlComp4	Neural2	5HU	median_house_value		3633790314.4	3727167485.1
	MdlComp	Tree4	RegTree B3D6	median_house_value		3703212490.8	3214560819.2
	MdlComp2	Reg	Exhaustive Regression	median_house_value		4798889268.6	4899002783.0


SAS Enterprise Miner Report
Node=Predict Model Comparison
Multiple Model Assessment Scores where DataRole=TEST
TARGET='median_house_value'
Mean Predicted
Min Predicted
Max Predicted
Depth
Model Node
Neural2
Reg
Tree4


SAS Enterprise Miner Report
Node=Predict Model Comparison
Multiple Model Assessment Scores where DataRole=TRAIN
TARGET='median_house_value'
Mean Predicted
Min Predicted
Max Predicted
Depth
Model Node
Neural2
Reg
Tree4


SAS Enterprise Miner Report
Node=Predict Model Comparison
Multiple Model Assessment Scores where DataRole=VALIDATE
TARGET='median_house_value'
Mean Predicted
Min Predicted
Max Predicted
Depth
Model Node
Neural2
Reg
Tree4

SAS Enterprise Miner Report

Node=Score
Summary

Node id = Score
Node label = Score
Meta path = Ids2 => Part => Neural2 => MdlComp4 => MdlComp3 => Score
Notes =

Node=Score
Properties

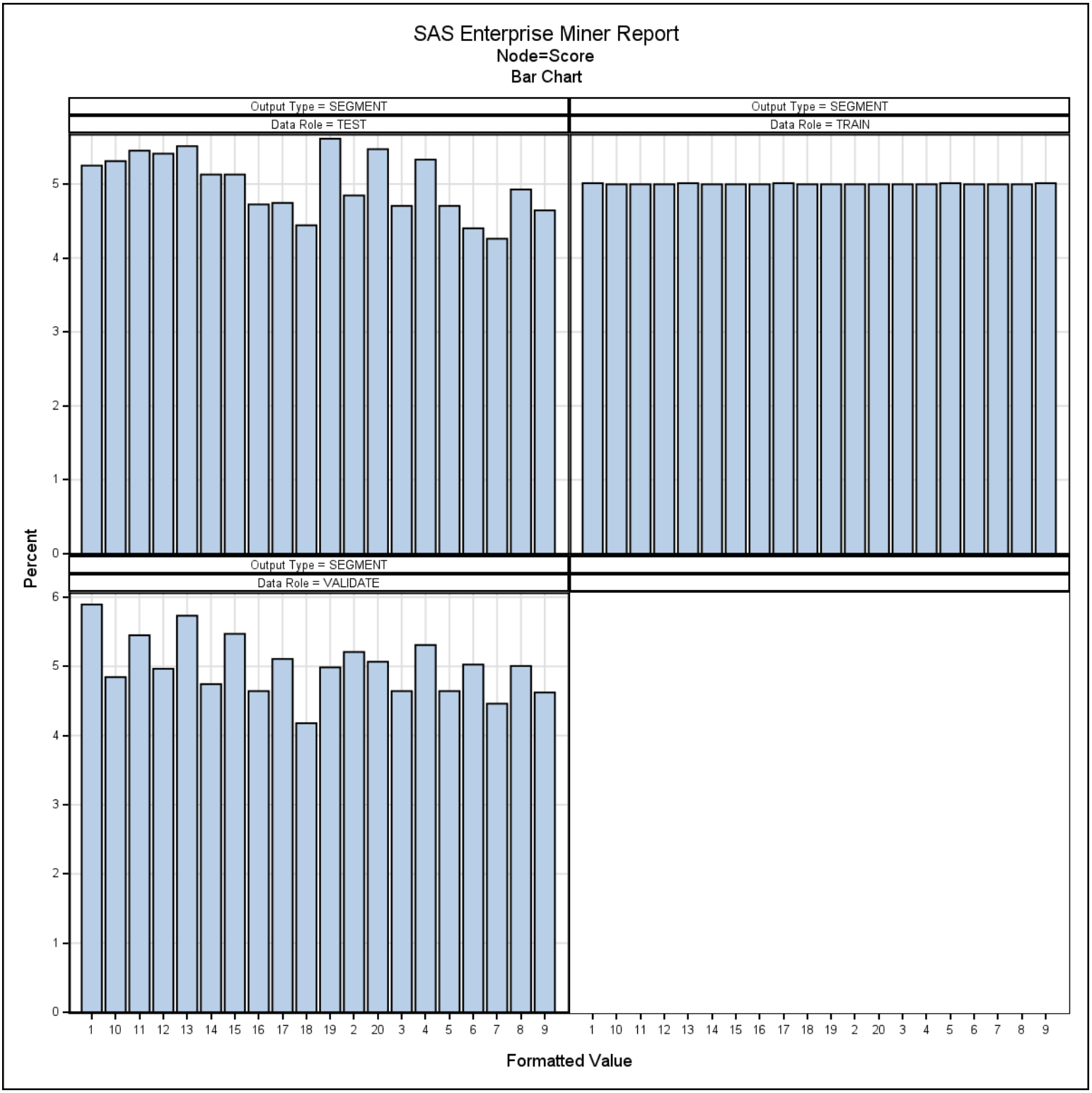
Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	Score		HideInput	Y		JScore	N	
CScore	N		HideOther	Y		OptimizedCode	Y	
FixedOutputNames	Y		HidePredict	Y		OutputType	VIEW	
GraphReports	Y		HideRejected	Y		PackageName	DEFAULT	
HideAssess	Y		HideResidual	Y		PreferenceName		
HideClassification	Y		HideTarget	Y		ScoreTest	N	
HideFreq	Y		HideVariables	N		ScoreValidate	N	

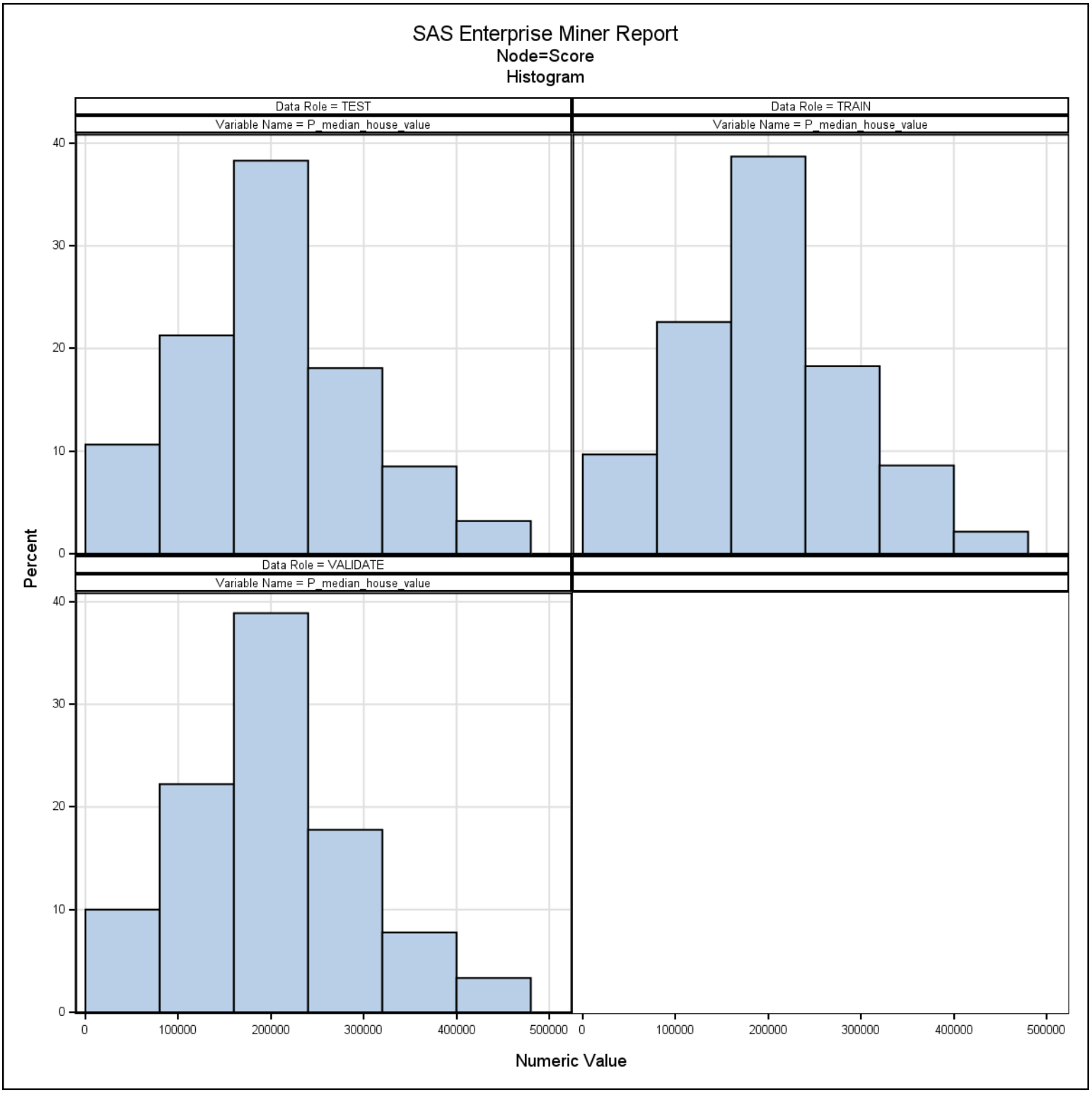
Node=Score
Variable Summary

Role	Level	Frequency Count	Name
TARGET	INTERVAL	1	median_house_value
SEGMENT	NOMINAL	1	b_median_house_val

Node=Score
Output Variables

Variable Name	Creator	Variable Label	Function	Type
EM_PREDICTION	Score	Prediction for median_house_value	PREDICT	N
EM_SEGMENT	Score	Segment	TRANSFORM	N
P_median_house_value	Neural2	Predicted: median_house_value	PREDICT	N
WARN	Neural2	Warnings	ASSESS	C
b_median_house_val	MdlComp4		TRANSFORM	N


SAS Enterprise Miner Report
Node=Score
Bar Chart
Output Type = SEGMENT
Data Role = TEST
Output Type = SEGMENT
Data Role = TRAIN
Output Type = SEGMENT
Data Role = VALIDATE
Percent
Formatted Value
1 10 11 12 13 14 15 16 17 18 19 2 20 3 4 5 6 7 8 9


SAS Enterprise Miner Report
Node=Score
Histogram
Data Role = TEST
Variable Name = P_median_house_value
Data Role = TRAIN
Variable Name = P_median_house_value
Data Role = VALIDATE
Variable Name = P_median_house_value
Percent
0
10
20
30
40
0
100000
200000
300000
400000
500000
Numeric Value

SAS Enterprise Miner Report

Node=Save Data
Summary

Node id = EMSave
Node label = Save Data
Meta path = Ids2 => Part => Neural2 => MdlComp4 => MdlComp3 => Score => EMSave
Notes =

Node=Save Data
Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	EMSave		Name	PredictiveAnalysis.Scored		Train	Y	
AllObs	Y		OutObs	1000		Transaction	Y	
AllRoles	Y		Replace	Y		Type	XLSX	SAS7DBAT
DirectorySelector	\\uisnutvdiprof1a\redirected$\smupp\Documents\HU.PREDICTIVEANALYSIS		Score	Y		Validate	Y	
Lib			Test	Y				

Node=Save Data
Variable Summary

Role	Level	Frequency Count	Name
INPUT	INTERVAL	8	households housing_median_age latitude longitude median_income population total_bedrooms total_rooms
INPUT	NOMINAL	1	ocean_proximity

Node=Save Data
Data

Output Location	Total Observations	Saved Observations	Number of Variables
\\uisnutvdiprof1a\redirected$\smupp\Documents\HU.PREDICTIVEANALYSIS\PredictiveAnalysis.Scored_TRAIN.xlsx	6605	MAX	18
\\uisnutvdiprof1a\redirected$\smupp\Documents\HU.PREDICTIVEANALYSIS\PredictiveAnalysis.Scored_VALIDATE.xlsx	4954	MAX	18
\\uisnutvdiprof1a\redirected$\smupp\Documents\HU.PREDICTIVEANALYSIS\PredictiveAnalysis.Scored_TEST.xlsx	4953	MAX	18

End of Report